

2025 CFA[®] CFA Level I



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CFA Program



LEVEL I BOOK 4

Premium Practice Pack

Portfolio Management &
Quantitative Methods

CFA L-1 Premium Practice Pack - 2025

Volume 4

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Portfolio Management: Practice Pack- Question

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Question

Which of the following statements about pension plans is *most* accurate?

- A. Defined benefit plans typically have a low risk tolerance.
- B. Defined contribution plans typically have a low risk tolerance.
- C. The sponsor of a defined benefit plan specifies the obligation owed to participants.

Question 2 of 100

Question

The process of risk management is *best* described as the set of decisions that maximizes a company's value while:

- A. minimizing the risk taken.
- B. bearing a tolerable level of risk.
- C. predicting the potential risk correctly.

Question 3 of 100

Question

Which of the following is *least* consistent with effective risk governance?

- A. Taking an enterprise-wide view
- B. Defining the enterprise's risk tolerance
- C. Following a bottom-up process to direct risk management activities

Question 4 of 100

Question

With regard to an investment policy statement, which of the following statements about return objectives is *most* accurate?

- A. A return objective cannot be a required rate of return.
- B. Return objectives must be set independent of risk objectives.
- C. When setting a relative return objective, a good benchmark should be investable.

Question 5 of 100

Question

The correlation between the risk-free asset and the optimal risky portfolio is expected to be:

- A. negative.
- B. zero.
- C. positive.

Question 6 of 100

Question

An investor who can lend and borrow at the risk-free rate builds a portfolio using the risk-free asset and the market portfolio. The risk-free rate is 3% and the expected market return is 15%. If the expected portfolio return is 18%, the investor's portfolio is:

- A. a lending portfolio.
- B. a leveraged portfolio.
- C. the optimal risky portfolio.

Question 7 of 100

Question

An analyst gathers the following information about a company:

Payables turnover	8
Inventory turnover	2
Receivables turnover	10

If all purchases and sales were made on credit, the cash conversion cycle (based on a 360-day year) is:

- A. less than the utility generated for a risk-averse investor.
- B. equal to the utility generated for a risk-averse investor.
- C. greater than the utility generated for a risk-averse investor.

Question 8 of 100

Question

A risk-neutral investor *most likely* seeks to maximize:

- A. both risk and return.
- B. return irrespective of risk.
- C. return for a given level of risk.

Question 9 of 100

Question

The market has a return of 8% and a standard deviation of returns of 12%. The risk-free rate is 2%. If a portfolio has a Sharpe ratio of 0.8, the portfolio's M² alpha is *closest* to:

- A. 3.6%.
- B. 5.6%.
- C. 11.6%.

Question 10 of 100

Question

Which of the following is *most likely* a consequence of overconfidence bias? Investors:

- A. holding poorly diversified portfolios.
- B. continuing to hold classes of assets with which they are familiar.
- C. holding investments in a loss position longer than justified, in the hope that they will return to breakeven.

Question 11 of 100

Question

Failing to act in pursuit of long-term goals in favor of short-term satisfaction *best* describes which of the following emotional biases?

- A. Self-control bias
- B. Endowment bias
- C. Loss-aversion bias

Question 12 of 100

Question

When creating a long-only portfolio, which of the following correlation coefficients between assets would be *most* effective at reducing portfolio risk?

- A. -0.5.
- B. 0.
- C. 0.5.

Question 13 of 100

Question

Which of the following can *best* be explained by overconfidence when predicting companies' earnings growth rates?

- A. Base-rate neglect
- B. The value anomaly
- C. The disposition effect

Question 14 of 100

Question

Which of the following is *most likely* developed by combining a client's investment objectives and constraints with long-term capital market expectations?

- A. Risk budget
- B. Strategic asset allocation
- C. Investment policy statement

Question 15 of 100

Question

The diversification ratio of a portfolio is *best* described as the ratio of the:

- A. standard deviation of the equally weighted portfolio's returns to the average standard deviation of the individual securities' returns.
- B. standard deviation of the market-capitalization-weighted portfolio's returns to the standard deviation of the equally weighted portfolio's returns.
- C. average standard deviation of the individual securities' returns to the standard deviation of the market-capitalization-weighted portfolio's returns.

Question 16 of 100

Question

In regard to the asset allocation process, a top-down analysis *most likely* begins with an examination of:

- A. macroeconomic growth.
- B. a company's board of directors.

the expected growth of a company's competitors

Question 17 of 100

Question

Which of the following is *most likely* a consequence of the illusion of control bias?

- A. An investor's portfolio turnover is too low.
- B. The investor's portfolio contains concentrated positions in companies.
- C. An investor uses a simple forecasting model for portfolio construction.

Question 18 of 100

Question

Which of the following is *best* classified as a non-financial risk?

- A. Credit risk
- B. Liquidity risk
- C. Accounting risk

Question 19 of 100

Question

Portfolio rebalancing is *best* described as a process aimed at:

- A. generating alpha.
- B. aligning portfolio weights with the tactical asset allocation decision.
- C. restoring the portfolio's original exposures to systematic risk factors.

Question

A. the risk-free asset is uncorrelated with the other assets in the portfolio.

- C. the correlations among the risky assets decrease, providing a diversification effect.

Question

A. Engaging in herding behaviour

- B. Borrowing excessively to finance present consumption
- C. Misidentifying risk tolerances because of how questions about risk tolerance were framed

Question

- A. stem from impulses and intuition.
- B. result in the same decision as assumed by traditional finance theory.
- C. can often be corrected or eliminated through better information, education, and advice.

Question

Exchange-traded funds (ETFs):

- A. are priced once a trading day.
- B. usually pay out dividends to shareholders.
- C. are generally structured as closed-end funds.

Question

Two assets have the following characteristics:

Variance of returns for Asset 1	0.05
Variance of returns for Asset 2	0.06
Correlation of returns between Asset 1 and Asset 2	0.75

The variance of returns for an equally weighted portfolio of the two assets is *closest* to: A. 0.038.

- | | |
|----|--------|
| 3. | 0.048. |
| 2. | 0.055. |

Question

With respect to return-generating models, statistical factor models:

- A. only include factors that have economic meaning.
- B. identify factors that explain the covariance in observed returns.
- C. only include factors that have a fundamental connection to returns.

Question 26 of 100

Question

Which of the following statements about asset allocation is *most* accurate?

- A. Investors should diversify their wealth between asset classes in order to eliminate systematic risk.
- B. Investors with a below-average risk tolerance should have an above-average weight in alternative investments.
- C. Adding asset classes with a low correlation to existing asset classes improves an investor's risk–return trade-off.

Question 27 of 100

Question

An analyst gathers the following information about a portfolio and the market:

Portfolio return	7.0%
Market return	5.0%
Risk-free return	1.0%
Portfolio beta	1.2

Jensen's alpha for the portfolio is:

- A. 0.0%.
- B. 1.2%.>
- C. 2.2%.

Question 28 of 100

Question

Which of the following behavioral biases is *most likely* the hardest to correct?

- A. Hindsight bias
- B. Loss-aversion bias
- C. Representativeness bias

Question 29 of 100

Question

Momentum, defined as relative past stock returns, is *most likely* a factor in:

- A. fundamental factor models.
- B. the Carhart four-factor model.
- C. the Fama–French three-factor model.

Question 30 of 100

Question

Which of the following measures uses only systematic risk to evaluate portfolio performance?

- A. M^2
- B. Sharpe ratio
- C. Jensen's alpha

Question 31 of 100

Question

When evaluating the return distribution of an asset class, the probability of extreme returns is *best* assessed by the distribution's:

- A. kurtosis.
- B. variance.
- C. skewness.

Question 32 of 100

Question

Robo-advisers *most likely*:

- A. face high barriers to entry.
- B. cater to the demand from investors with lower levels of investable assets.
- C. prefer actively managed funds to index funds when constructing client portfolios.

Question 33 of 100

Question

Which of the following is *most* closely associated with representativeness bias?

- A. Momentum
- B. The halo effect
- C. Bubbles and crashes

Question 34 of 100

Question

Which of the following statements about different types of investors is *most* accurate?

- A. For banks, the liquidity of their investments is a paramount concern.
- B. For endowments, investment horizons are short due to their short-term spending needs.
- C. For insurance companies, the risk tolerance of their general and surplus accounts is typically the same.

Question 35 of 100

Question

Which of the following is *most likely* undertaken during the risk budgeting process?

- A. Assessing risk appetite
- B. Setting a limit for value at risk (VaR)
- C. Establishing a reserve to cover potential future losses

Question 36 of 100

Question

The portfolio approach to investing *most likely*:

- A. prevents portfolio losses during market downturns.
- B. reduces the systematic risk of individual assets in a portfolio.
- C. helps avoid disastrous investment outcomes during normal market conditions.

Question 37 of 100

Question

A client's time horizon is *most* appropriately used by an investment adviser to determine the client's:

- A. risk attitude.
- B. ability to take risk.
- C. willingness to take risk.

Question 38 of 100

Question

Which of the following sections of an investment policy statement *most likely* provides guidance on obtaining feedback on investment results?

- A. Investment Guidelines
- B. Evaluation and Review
- C. Statement of Duties and Responsibilities

Question 39 of 100

Question

The security market line plots the expected return of a portfolio against a measure of the portfolio's:

- A. total risk.
- B. systematic risk.
- C. unsystematic risk.

Question 40 of 100

Question

An analyst estimates the standard deviation of returns for the market portfolio to be 15% and the standard deviation of returns for a stock to be 25%. If the correlation of returns between the stock and the market portfolio is 0.6, the stock has:

- A. less systematic risk than the market portfolio.
- B. the same systematic risk as the market portfolio.
- C. more systematic risk than the market portfolio.

Question 41 of 100

Question

Risk-averse investors make investment decisions that maximize:

- A. both return and risk.
- B. return irrespective of risk.
- C. return for the same amount of risk.

Question 42 of 100

Question

Sovereign wealth funds are *best* described as investment funds:

- A. owned by governments.
- B. traded as closed-end country funds.
- C. restricted from investing in foreign securities.

Question 43 of 100

Question

Which of the following lines is plotted on a graph with the excess return of a security on the y-axis and the excess return of the market on the x-axis?

- A. Capital market line
- B. Security market line
- C. Security characteristic line

Question 44 of 100

Question

According to capital market theory, the only type of risk that is priced is:

- A. systematic risk.
- B. diversifiable risk.
- C. idiosyncratic risk.

Question 45 of 100

Question

Which of the following is *most* accurate regarding an investment policy statement (IPS)?

- A. Policies on sustainable investing require a separate IPS.
- B. Investment constraints can be determined by the client or by the law.
- C. Clients can specify different spending goals, but each goal must have the same risk tolerance and return objective.

Question 46 of 100

Question

An investor's ability to take risk:

- A. is a function of risk tolerance.
- B. is typically assessed by a psychometric questionnaire.
- C. increases with the length of the investment horizon, all else being equal.

Question 47 of 100

Question

The capital market line *most likely* consists of portfolios that:

- A. are fully diversified.
- B. have zero systematic risk.
- C. have nonsystematic risk equal to beta.

Question 48 of 100

Question

The Markowitz efficient frontier is *best* described as a curve that:

- A. lies above and to the left of the minimum-variance frontier.
- B. connects the minimum-variance portfolios for all possible returns.
- C. contains all portfolios of risky assets that rational, risk-averse investors will choose.

Question 49 of 100

Question

Which of the following *most likely* affects a client's ability to take risk? The client's:

- A. utility function
- B. degree of risk aversion
- C. level of wealth relative to liabilities

Question 50 of 100

Question

The expected return for a security is equal to the market's risk premium. If the risk-free rate is positive and the CAPM holds, the beta of the security is:

- A. less than 1.
- B. equal to 1.
- C. greater than 1.

Question 51 of 100

Question

Which of the following characteristics is *most likely* used to determine an investor's ability to take risk? The investor's:

- A. risk attitude.
- B. self-confidence.
- C. years until retirement.

Question 52 of 100

Question

An analyst gathers the following information:

Risk-free rate	2%
Expected return of the market portfolio	10%
Standard deviation of the market portfolio	20%
Standard deviation of the security	35%
Correlation between the security and the market	0.8

According to the CAPM, the expected return of the security is *closest* to:

- A. 5.7%.
- B. 13.2%.
- C. 16.0%.

Question 53 of 100

Question

The correlation of returns between two securities with equal standard deviation of returns is 0.75. If the covariance of returns is 5.5%, the standard deviation of returns for each security is *closest* to:

- A. 2.7%.
- B. 3.7%.
- C. 7.3%.

Question 54 of 100

Question

An analyst gathers the following information about an asset and the market:

Risk-free rate 1%

Market risk premium 5%

Asset's expected return 5%

Based on the CAPM, the asset's beta is *closest* to:

- A. 0.80.
- B. 1.00.
- C. 1.25.

Question 55 of 100

Question

Long-term historical data on the risk–return trade-off of securities show that investors are *most likely*:

- A. risk averse.
- B. risk neutral.
- C. risk seeking.

Question 56 of 100

Question

Information regarding the permissible use of derivatives in a portfolio is *most likely* found in which of the following sections of an investment policy statement?

- A. Procedures
- B. Investment Guidelines
- C. Statement of Duties and Responsibilities

Question 57 of 100

Question

In the portfolio management process, the feedback step *most likely* involves:

- A. rebalancing the portfolio.
- B. deciding on an asset allocation.
- C. understanding the client's constraints.

Question 58 of 100

Question

Which of the following investment principles *best* explains the use of strategic asset allocation in portfolio construction?

- A. Returns on similar assets reflect exposures to certain sets of systematic factors.
- B. Nonsystematic risk accounts for most of the change in portfolio value over the long term.

- C. Deviating from policy exposures to systematic risk factors may add value to the portfolio.

Question 59 of 100

Question

In the investment policy statement of a pension fund, a countrywide limit on the proportion of high-risk assets that can be held in long-term pension portfolios is *most likely* a:

- A. liquidity constraint.
- B. legal and regulatory constraint.
- C. time horizon constraint.

Question 60 of 100

Question

The global minimum-variance portfolio is a portfolio that lies:

- A. anywhere along the minimum-variance frontier.
- B. at the left-most point of the minimum-variance frontier.
- C. at the upper right-most point of the minimum-variance frontier.

Question 61 of 100

Question

Which of the following principles *best* explains the focus on a strategic asset allocation when constructing a client's IPS?

- A. Adding assets with high correlation improves the risk–return trade-off.
- B. Nonsystematic risk accounts for most of a portfolio's change in value over the long term.
- C. The returns to groups of similar assets predictably reflect exposures to certain systematic factors.

Question 62 of 100

Question

Which of the following sections of an investment policy statement (IPS) *most likely* explains how and when the IPS should be reviewed?

- A. Procedures
- B. Investment Guidelines
- C. Statement of Duties and Responsibilities

Question 63 of 100

Question

Which of the following *best* describes a characteristic of defined contribution pension plans?

- A. The employee accepts the investment and inflation risk.
- B. The employer is responsible for adequately funding the plan.
- C. Defined contribution plans typically have a higher cost to the company than defined benefit plans.

Question 64 of 100

Question

The market model is *most likely* used to predict:

- A. market returns in a future period.
- B. economic growth in a future period.
- C. company-specific returns in a future period.

Question 65 of 100

Question

Open-end mutual funds typically:

- A. are priced intraday.
- B. have a fixed number of shares outstanding.
- C. have a larger required minimum investment than ETFs.

Question 66 of 100

Question

A portfolio consists of two securities with the following characteristics:

	<u>Expected Return</u>	<u>Standard Deviation of Returns</u>
Security 1	17%	24%
Security 2	6%	12%

If the portfolio has an expected return of 12.6% and the returns of the two securities are uncorrelated, the portfolio's standard deviation is *closest* to:

- A. 13.4%.
- B. 15.2%.
- C. 19.2%.

Question 67 of 100

Question

An equally weighted portfolio is composed of two risky assets. If the correlation of asset returns is equal to zero, the portfolio standard deviation is:

- A. equal to zero.
- B. equal to the weighted average of the assets' standard deviations.
- C. less than the weighted average of the assets' standard deviations.

Question 68 of 100

Question

If all investors have homogeneous expectations, the total risk and expected return of portfolios consisting of the risk-free asset and the optimal risky portfolio are plotted on the:

- A. capital market line.
- B. security market line.
- C. security characteristic line.

Question 69 of 100

Question

The risk–return profile of a portfolio's strategic asset allocation is *most likely* determined by the expected returns and risks of the individual asset classes and the:

- A. correlations between those asset classes.
- B. use of security selection for each of those asset classes.
- C. allowable deviation of portfolio weights from policy weights for those asset classes.

Question 70 of 100

Question

A portfolio has an annual return of 15.2% and a standard deviation of returns of 11.7%. If the risk-free rate is 3.1%, the portfolio's Sharpe ratio is *closest* to:

- A. 1.03.
- B. 1.30.
- C. 1.56.

Question 71 of 100

Question

Which of the following is *best* classified as a financial risk?

- A. Tax risk
- B. Credit risk
- C. Accounting risk

Question 72 of 100

Question

The risk management measure that captures the sensitivity of a derivative's delta to a change in the value of the underlying *best* describes:

- A. rho.
- B. vega.
- C. gamma.

Question 73 of 100

Question

The intercept on the y-axis of the security characteristic line is:

- A. beta.
- B. Jensen's alpha.
- C. the risk-free rate of return.

Question 74 of 100

Question

With respect to an investment policy statement, which of the following is *most* closely linked to the client's distinctive needs?

- A. The evaluation and review section
- B. The objectives and constraints sections
- C. The statement of duties and responsibilities

Question 75 of 100

Question

The slope of the security market line is *most likely* the:

- A. security's beta.
- B. market risk premium.
- C. market risk premium divided by the market standard deviation.

Question 76 of 100

Question

Which of the following performance measures is equal to the slope of the capital allocation line?

- A. M^2
- B. Sharpe ratio
- C. Treynor ratio

Question 77 of 100

Question

Which of the following sections of an investment policy statement for a pension plan *most likely* specifies the discretion that portfolio managers have with respect to executing the investment strategy?

- A. Procedures
- B. Investment Constraints
- C. Statement of Duties and Responsibilities

Question 78 of 100

Question

Which of the following measures is *most* appropriate to evaluate the performance of a portfolio that is **not** fully diversified?

- A. Sharpe ratio
- B. Treynor ratio
- C. Jensen's alpha

Question 79 of 100

Question

An analyst gathers the following information about three stocks:

	<u>Beta</u>	<u>Standard Deviation of Returns</u>
Stock 1	0.9	17%
Stock 2	1.1	18%
Stock 3	1.2	16%

If the standard deviation of market returns is equal to 10%, the stock with the highest nonsystematic risk is:

- A. Stock 1.
- B. Stock 2.
- C. Stock 3.

Question 80 of 100

Question

Which of the following metrics is *most* appropriate to estimate a bond's average extreme loss?

- A. VaR of loss
- B. Standard deviation of loss
- C. Expected loss given default

Question 81 of 100

Question

An analyst gathers the following information about the security market line (SML) and a stock:

Intercept of the SML	2%
Slope of the SML	5%
Stock's beta	1.3

According to capital market theory, if the analyst believes the stock will have a return of 7%, the stock is:

- A. undervalued.
- B. properly valued.
- C. overvalued.

Question 82 of 100

Question

Relative to passive market-cap-weighted strategies, smart beta strategies typically have:

- A. lower management fees and higher portfolio turnover.
- B. higher management fees and lower portfolio turnover.
- C. higher management fees and higher portfolio turnover.

Question 83 of 100

Question

When defining asset classes for a strategic asset allocation, which of the following pairwise correlations between asset class returns is *most* preferable?

- A. 0.0
- B. 0.5
- C. 1.0

Question 84 of 100

Question

According to capital market theory, an efficient market does **not** reward investors for taking on:

- A. market risk.
- B. systematic risk.
- C. idiosyncratic risk.

Question 85 of 100

Question

Which of the following is *most likely* a legal and regulatory constraint in an investment policy statement?

- A. A pension fund's decision to limit investments in real estate
- B. A taxable investor's requirement to avoid investments in securities generating interest income
- C. A public company director's restriction on trading the company's stock shortly before the publication of financial results

Question 86 of 100

Question

When defining asset classes, the paired correlations of assets within an asset class should be:

- A. negative.
- B. zero.
- C. positive.

Question 87 of 100

Question

An investor gathers the following information about a security and the market:

Security's beta	0.35
Standard deviation of the security's returns	12%
Standard deviation of market returns	18%

The correlation between the security's returns and the market's returns is *closest* to:

- A. 0.2.
- B. 0.5.
- C. 0.8.

Question 88 of 100

Question

A bank determines that its value at risk (VaR) is £5 million at 5% for one day. The bank is expecting a minimum loss of £5 million once every:

- A. 5 business days.
- B. 13 business days.
- C. 20 business days.

Question 89 of 100

Question

A good risk management process should:

- A. predict when a crisis will occur.
- B. consider the balance between expected returns and losses.
- C. only consider losses occurring from events that have a high likelihood.

Question 90 of 100

Question

A portfolio consisting of two securities has the following characteristics:

<u>Security</u>	<u>Portfolio Weight</u>	<u>Standard Deviation</u>
1	40%	15%
2	60%	18%

If the correlation of returns between the two securities is 0.20, the portfolio's standard deviation of returns is *closest* to:

- A. 1.8%.
- B. 9.1%.
- C. 13.4%.

Question 91 of 100

Question

An investor has a 15-year time horizon but needs to withdraw funds from her portfolio in one year's time to pay for tuition fees. Which of the following investments is *most* suitable to cover the investor's liquidity requirement due to the tuition fees?

- A. Commercial paper
- B. Private equity securities
- C. Large-capitalization stocks

Question 92 of 100

Question

With respect to capital market theory, which of the following statements is *most* accurate?

- A. The optimal risky portfolio is dependent on the risk-free rate.
- B. The optimal risky portfolio is dependent on the investor's risk profile.
- C. The investor's optimal portfolio must lie on the Markowitz efficient frontier.

Question 93 of 100

Question

An analyst gathers the following information:

	<u>Standard Deviation of Returns</u>	<u>Beta</u>
Asset	70%	1.8
Market portfolio	35%	1.0

The covariance between the returns of the asset and the market is *closest* to:

- A. 0.22.
- B. 0.40.
- C. 0.90.

Question 94 of 100

Question

A security with a beta of 1.5 has an expected return of 11% according to the CAPM. If the risk-free rate is 2%, the market risk premium is *closest* to:

- A. 4.0%.
- B. 6.0%.
- C. 7.3%.

Question 95 of 100

Question

An analyst gathers the following information about a portfolio and the market:

Portfolio's Sharpe ratio	0.8
Volatility of portfolio returns	19%
Volatility of market returns	10%
Correlation between portfolio returns and market returns	0.7

The portfolio's Treynor ratio is *closest* to:

- A. 0.060.
- B. 0.114.
- C. 0.413.

Question 96 of 100

Question

For an equally weighted portfolio, an increase in the correlations between asset returns *most likely* decreases the:

- A. portfolio's expected return.
- B. portfolio's standard deviation of returns.
- C. level of risk reduction provided by the portfolio.

Question 97 of 100

Question

An adviser gathers the following information about a client's retirement income needs:

- The current balance of the client's retirement fund is \$150,000.
- The client plans to retire in 15 years' time.
- \$500,000 in today's money is needed to fund retirement.
- No further retirement fund contributions will be made.

Inflation is expected to average 2% per year over the next 15 years. Based only on the above information and ignoring taxes, the minimum annual rate of return required to meet the client's retirement income objective is *closest* to:

- A. 6.2%.
- B. 8.4%.
- C. 10.5%.

Question 98 of 100

Question

As the number of assets in an equally weighted portfolio becomes large, the portfolio's variance of returns *most likely* approaches:

- A. zero.
- B. the average variance of the individual assets' returns.
- C. the average covariance between the individual assets' returns.

Question 99 of 100

Question

In risk management, which of the following should be taken into account when determining an enterprise's risk tolerance?

- A. Management compensation
- B. The enterprise's value at risk (VaR)
- C. The government and regulatory landscape

Question 100 of 100

Question

Two investors have indifference curves that are tangent to the same capital allocation line (CAL). If Investor 1 is more risk averse than Investor 2, Investor 1's optimal portfolio is:

- A. to the left of Investor 2's optimal portfolio on the CAL.
- B. at the same point on the CAL as Investor 2's optimal portfolio.
- C. to the right of Investor 2's optimal portfolio on the CAL.

Portfolio Management: Practice Pack- Answers

Answer 1 of 100

Answer

Solution

- A. Incorrect because the risk tolerance for defined benefit pension plans will vary with the maturity of the plan, but is typically quite high.
- B. Incorrect because defined contribution pension plans are managed for individual investors, therefore the risk tolerance of each plan varies with the individual. Some individuals will be investing for growth and will therefore seek assets that have the potential for capital gains. Others, such as retirees, may need to draw an income from their assets and may therefore choose to invest more in fixed-income and dividend-paying shares. The investment needs of individuals will depend in part on their broader financial circumstances, such as their employment prospects and whether or not they own their own residence. They may also need to consider such issues as building up a cash reserve and the purchase of appropriate insurance policies before undertaking longer-term investments. The risk tolerance of an individual investor varies by individual.
- C. **Correct** because **defined benefit pension plans** (DB plans) are company-sponsored plans that offer employees a predefined benefit on retirement. The future benefit is defined because the DB plan requires the plan sponsor to specify the obligation stated in terms of the retirement income benefits owed to participants.

Answer 2 of 100

Answer

Solution

- A. Incorrect because risk management is *not* about minimizing risk; it is about actively understanding and embracing those risks that best balance the achievement of goals with an acceptable chance of failure, quantifying the exposure, and continually monitoring and modifying it.
- B. **Correct** because risk management is the process by which an organization or individual defines the level of risk to be taken, measures the level of risk being taken, and adjusts the latter toward the former, with the goal of maximizing the company's or portfolio's value. Said differently, risk management comprises all the decisions and actions needed to *best* achieve organizational or personal objectives while bearing a tolerable level of risk.
- C. Incorrect because risk management is not even about predicting risks. 'The Doctrine of No Surprises' is a key mantra among many risk managers, but it does not mean they are expected to predict what will happen.

Answer 3 of 100

Answer

Solution

- A. Incorrect because to achieve the best results for an organization, risk governance should take an enterprise-wide view.
- B. Incorrect because good governance should include defining an organization's risk tolerance and providing risk oversight.
- C. **Correct** because risk governance is the top-down (not bottom-up) process and guidance that directs risk management activities to align with and support the overall enterprise.

Answer 4 of 100

Answer

Solution

- A. Incorrect because a return objective can be a required rate of return. The return objective could be a required return—that is, the amount the investor needs to earn to meet a particular future goal—such as a certain level of retirement income.
- B. Incorrect because the return objective must be consistent with the client's risk objective (high expected returns are unlikely to be possible without high levels of risk).
- C. **Correct** because a benchmark is used as a relative return objective and a good benchmark should be investable.

Answer 5 of 100

Answer

Solution

- A. Incorrect because a combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The optimal risky portfolio is a risky asset, and thus has zero correlation with the risk-free asset.
- B. **Correct** because a combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The optimal risky portfolio is a risky asset, and thus has zero correlation with the risk-free asset.
- C. Incorrect because a combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The optimal risky portfolio is a risky asset, and thus has zero correlation with the risk-free asset.

Answer 6 of 100

Answer

Solution

- A. Incorrect because a lending portfolio is a portfolio that has a positive investment in the risk-free asset.

A portfolio's expected return, $E(R_p)$, is calculated as: $E(R_p) = w_1 R_f + (1 - w_1) E(R_m)$, where w_1 is the proportion invested in the risk-free asset, returning R_f , and $E(R_m)$ is the expected return on the market portfolio.

Thus, $0.18 = w_1 \times 0.03 + (1 - w_1) \times 0.15$

$0.18 - 0.15 = w_1 \times (0.03 - 0.15)$

$0.03 = w_1 \times (-0.12)$

$w_1 = 0.03 / (-0.12)$

$w_1 = -0.25$

A negative proportion invested in the risk-free rate implies a leveraged, not a lending, portfolio.

- B. **Correct** because a leveraged portfolio is a portfolio that has a negative investment in the risk-free asset.

A portfolio's expected return, $E(R_p)$, is calculated as: $E(R_p) = w_1 R_f + (1 - w_1) E(R_m)$, where w_1 is the proportion invested in the risk-free asset, returning R_f , and $E(R_m)$ is the expected return on the market portfolio.

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$0.03 = w_1 \times (-0.12)$

$w_1 = 0.03 / (-0.12)$

$w_1 = -0.25$

A negative proportion invested in the risk-free rate implies a leveraged portfolio.

C. Incorrect because the expected market return is 15%, while the expected portfolio return is 18%. Since the two returns are different, the investor's portfolio is different from the optimal risky (market) portfolio.

Answer 7 of 100

Answer

Solution

A. Incorrect because a risk-free asset ($\sigma^2 = 0$) generates the same utility for all individuals.

B. **Correct** because a risk-free asset ($\sigma^2 = 0$) generates the same utility for all individuals. If $\sigma^2 = 0$, then $U = E(r) - 1/2(A)(\sigma^2) = E(r)$ for all individuals.

C. Incorrect because a risk-free asset ($\sigma^2 = 0$) generates the same utility for all individuals.

Answer 8 of 100

Answer

Solution

A. Incorrect because a risk-seeking investor would maximize both risk and return. A risk-neutral investor maximizes return irrespective of risk. That is, a risk-neutral investor deliberately seeks to maximize only return, not risk. It is possible that a risk-neutral investor could end up maximizing risk by maximizing return, but he only seeks to maximize return and risk is not a consideration.

B. **Correct** because a risk-neutral investor would maximize return irrespective of risk. This is because such an investor cares only about return and not about risk, so higher return investments are more desirable even if they come with higher risk.

C. Incorrect because only risk-averse investors want to minimize their risk for the same amount of return, and maximize their return for the same amount of risk. A risk-neutral investor seeks to maximize return irrespective of risk, not for a given level of risk.

Answer 9 of 100

Answer

Solution

A. **Correct** because M^2 provides a measure of portfolio return that is adjusted for the total risk of the portfolio and is computed as $M^2 = [E(R_p) - R_f](\sigma_m/\sigma_p) + R_f = SR \times \sigma_m + R_f$, where SR = Sharpe ratio, σ_m = market standard deviation of returns, and R_f = risk-free rate. Thus, $M^2 = 0.8 \times 0.12 + 0.02 = 0.116$. The difference between the risk-adjusted performance of the portfolio and the performance of the market is frequently referred to as M^2 alpha. Thus, M^2 alpha = $0.116 - 0.08 = 0.036 = 3.6\%$.

B. Incorrect because it computes $M^2 = 0.02 + 0.8 \times 0.12 = 0.116$. M^2 alpha is wrongly calculated as $0.116 - (0.08 - 0.02) = 0.056 = 5.6\%$.

C. Incorrect because it computes M^2 instead of M^2 alpha, which is an intermediate step in the correct calculation; $M^2 = 0.02 + 0.8 \times 0.12 = 0.116 = 11.6\%$.

Answer 10 of 100

Answer

Solution

A. **Correct** because **overconfidence bias** is a bias in which people demonstrate unwarranted faith in their own abilities. As a result of overconfidence bias, FMPs [financial market participants] may hold poorly diversified portfolios, which may result in significant downside risk.

B. Incorrect because this is a potential consequence of endowment bias, not overconfidence bias. **Endowment bias** is an emotional bias in which people value an asset more when they own it than when they do not. Endowment bias may lead FMPs [financial market participants] to continue to hold classes of assets with which they are familiar. FMPs may believe they understand the characteristics of

the investments they already own and may be reluctant to purchase assets with which they have less experience. Familiarity adds to owners' perceived value of a security.

- C. Incorrect because this is a potential consequence of loss-aversion bias, not overconfidence. As a result of loss-aversion bias, FMPs [financial market participants] may hold investments in a loss position longer than justified by fundamental analysis, in the hope that they will return to breakeven.

Answer 11 of 100

Answer

Solution

- A. **Correct** because **self-control bias** is a bias in which people fail to act in pursuit of their long-term, overarching goals in favor of short-term satisfaction.

B. Incorrect because **endowment bias** is an emotional bias in which people value an asset more when they own it than when they do not, and is not about acting in favor of short-term satisfaction the cost of long-term goals.

- C. Incorrect because **loss-aversion bias** refers to the tendency to strongly prefer avoiding losses to achieving gains, and is not about acting in favor of short-term satisfaction at the cost of long-term goals.

Answer 12 of 100

Answer

Solution

- A. **Correct** because -0.5 is the smallest of the three correlations and the closer the correlation coefficient is to -1 , the greater the reduction in portfolio risk. The correlation coefficient between two assets determines the effect on portfolio risk when the two assets are combined. You will find that portfolio risk is unaffected when the two assets are perfectly correlated ($\rho_{12} = +1$). In other words, the portfolio's standard deviation is simply a weighted average of the standard deviations of the two assets and as such a portfolio's risk is unchanged with the addition of assets with the same risk parameters. Portfolio risk falls, however, when the two assets are not perfectly correlated ($\rho_{12} < +1$). Sufficiently low values of the correlation coefficient can make the portfolio riskless under certain conditions. For an extreme case in which $\rho_{12} = -1$ (that is, the two asset returns move in opposite directions), the portfolio can be made risk free. Analytically, the standard deviation risk of a two asset portfolio is given by the square root of the portfolio's variance: $\sigma_p = (w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2)^{1/2}$, which is a strictly increasing function of ρ_{12} . Hence, -0.5 , the smallest of the three correlations, has the lowest risk.

B. Incorrect because the correlation coefficient is not the lowest of the three options. The closer the correlation coefficient is to -1 , the greater the reduction in portfolio risk. Portfolio risk falls when the two assets are not perfectly correlated ($\rho_{12} < +1$). Sufficiently low values of the correlation coefficient can make the portfolio riskless under certain conditions. For an extreme case in which $\rho_{12} = -1$ (that is, the two asset returns move in opposite directions), the portfolio can be made risk free.

- C. Incorrect because the correlation coefficient is not the lowest of the three options. The closer the correlation coefficient is to -1 , the greater the reduction in portfolio risk. "Portfolio risk falls when the two assets are not perfectly correlated ($\rho_{12} < +1$). Sufficiently low values of the correlation coefficient can make the portfolio riskless under certain conditions. For an extreme case in which $\rho_{12} = -1$ (that is, the two asset returns move in opposite directions), the portfolio can be made risk free.

Answer 13 of 100

Answer

Solution

A. Incorrect because, in **base-rate neglect**, a phenomenon's rate of incidence in a larger population—its base rate—is neglected in favor of specific information. The specific, individual information may be misleading relative to the more appropriate base rate or general information. Base-rate neglect is connected to representativeness bias and not overconfidence, nor does it relate to earnings growth rates.

B. **Correct** because a number of other studies have offered behavioral explanations for value anomalies, presenting the anomalies as mispricing rather than compensation for increased risk. These studies recognize the emotional factors involved in appraising stocks. Overconfidence can also be involved in predicting growth rates, potentially leading growth stocks to be overvalued, leading to the value anomaly.

C. Incorrect because an important concept embedded in this utility representation is what has been termed the **disposition effect**: the holding of investments that have experienced losses too long, and the selling of investments that have experienced gains too quickly (i.e., holding on to losers and selling winners). The disposition effect does not relate to prediction errors, nor to earnings growth rates.

Answer 14 of 100

Answer

Solution

A. Incorrect because **risk budgeting** is the process of deciding on the amount of risk to assume in a portfolio (the overall risk budget), and subdividing that risk over the sources of investment return (e.g., strategic asset allocation, tactical asset allocation, and security selection). Because the decision about the amount of risk to be taken is made in constructing the IPS, at this stage we are concerned about the subdivision of that risk.

B. **Correct** because a strategic asset allocation results from combining the constraints and objectives articulated in the IPS [investment policy statement] and long-term capital market expectations regarding the asset classes.

C. Incorrect because long-term capital market expectations are not commonly stated in the IPS, which in addition to Investment Objectives and Investment Constraints typically includes the following sections: Introduction, Statement of Purpose, Statement of Duties and Responsibilities, Procedures, Investment Guidelines, Evaluation and Review, Appendices. Therefore, combining investment objectives and constraints with long-term capital market expectations does not result in a client's IPS.

Answer 15 of 100

Answer

Solution

A. **Correct** because a simple measure of the value of diversification is calculated as the ratio of the standard deviation of the equally weighted portfolio to the standard deviation of the randomly selected security. This ratio may be referred to as the **diversification ratio**. In the example of the 5-stock portfolio given, the equally weighted portfolio's standard deviation is approximately 71 percent of the average standard deviation of the 5 stocks (24.9%); i.e., the denominator is the average standard deviation of all individual securities in the portfolio and the numerator is the standard deviation of the equally weighted portfolio.

B. Incorrect because the standard deviation of the equally weighted portfolio should be the numerator (not the denominator) and the denominator should be the average standard deviation of all individual securities in the portfolio.

C. Incorrect because the average standard deviation of all individual securities in the portfolio should be the denominator (not the numerator) and the numerator should be the standard deviation of the equally weighted portfolio.

Answer 16 of 100

Answer

Solution

A. **Correct** because a **top-down analysis** begins with consideration of macroeconomic conditions. Based on the current and forecasted economic environment, analysts evaluate markets and industries with the purpose of investing in those that are expected to perform well. Finally, specific companies within these industries are considered for investment.

B. **Incorrect** because a **top-down analysis** begins with consideration of macroeconomic conditions. Bottom-up analysis focuses on company specifics. Rather than emphasizing economic cycles or industry analysis, a **bottom-up analysis** focuses on company-specific circumstances, such as management quality and business prospects.

C. **Incorrect** because a **top-down analysis** begins with consideration of macroeconomic conditions. Bottom-up analysis focuses on company specifics. Rather than emphasizing economic cycles or industry analysis, a **bottom-up analysis** focuses on company-specific circumstances, such as management quality and business prospects.

Answer 17 of 100

Answer

Solution

A. **Incorrect** because as a result of illusion of control bias, FMPs [financial market participants] may trade more than is prudent. Researchers have found that portfolio turnover is negatively correlated with investment returns.

B. **Correct** because as a result of illusion of control bias, FMPs [financial market participants] may inadequately diversify portfolios. Research has found that some investors prefer to invest in companies that they feel they have control over, such as the companies they work for, leading them to hold concentrated positions.

C. **Incorrect** because as a result of illusion of control bias, FMPs [financial market participants] may construct financial models and forecasts that are overly detailed. FMPs may require detailed models before making an investment decision, believing that forecasts from these models control uncertainty. Although a greater understanding of an investment, issuer, or industry is often useful, increased model complexity does not control the inherent risk and uncertainty of investment outcomes.

Answer 18 of 100

Answer

Solution

A. **Incorrect** because the second primary financial risk is credit risk. Credit risk is the risk of loss if one party fails to pay an amount owed on an obligation, such as a bond, loan, or derivative, to another party.

B. **Incorrect** because liquidity risk is a financial risk and defined as the risk of a significant downward valuation adjustment when selling a financial asset.

C. **Correct** because although most risks have monetary consequences, there are a number of risks that are typically classified as non-financial in nature. These risks arise from a variety of sources, such as the relationship between the entity and counterparties, regulators, governments, the environment, suppliers, customers, and employees. The following three non-financial risks are related: regulatory risk, accounting risk, and tax risk. They could even be collectively referred to as compliance risk because they all deal with the matter of conforming to policies, laws, rules, and regulations as set forth by governments and authoritative bodies, such as accounting governing boards.

Answer 19 of 100

Answer

Solution

A. **Incorrect** because periodically, or when a certain threshold deviation from the policy weight (the bandwidth) has been breached, the portfolio should be rebalanced back

to the policy weights. The risk–return profile of the strategic asset allocation depends on the expected returns and risks of the individual asset classes, as well as the correlation between those asset classes. On the other hand, **security selection** is an attempt to generate higher returns than the asset class benchmark by selecting securities with a higher expected return. An "alpha process is a process for adding value through active investing.

- B. Incorrect because periodically, or when a certain threshold deviation from the policy weight (the bandwidth) has been breached, the portfolio should be rebalanced back to the policy weights. On the other hand, **tactical asset allocation** is the decision to deliberately deviate from the policy exposures to systematic risk factors (i.e., the policy weights of asset classes) with the intent to add value based on forecasts of the near-term returns of those asset classes.
- C. **Correct** because as the portfolio is constructed and its value changes with the returns of the asset classes and securities in which it is invested, the weights of the asset classes will gradually deviate from the policy weights in the strategic asset allocation. This process is referred to as drift. Periodically, or when a certain threshold deviation from the policy weight (the bandwidth) has been breached, the portfolio should be rebalanced back to the policy weights. The set of rules that guide the process of restoring the portfolio's original exposures to systematic risk factors is known as the **rebalancing policy**.

Answer 20 of 100

Answer

Solution

- A. **Correct** because an investor's portfolio improves if a risk-free asset is added to the mix. In other words, a combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset.
- B. Incorrect because the risk–return trade-off is improved by lower risk not a lower return. It is the zero correlation of the risk-free asset with the risky assets in the portfolio that lowers the risk level.
- C. Incorrect because the correlations among the risky assets are unaffected by the addition of a risk-free asset to the portfolio. The risk–return trade-off is improved by the zero correlation of the risk-free asset with the risky assets in the portfolio, not a change in correlations among the risky assets.

Answer 21 of 100

Answer

Solution

- A. **Correct** because as a result of regret-aversion bias, FMPs [financial market participants] may engage in herding behavior. FMPs may feel safer in popular investments in order to limit potential future regret. **Regret-aversion bias** is an emotional bias in which people tend to avoid making decisions out of fear that the decision will turn out poorly.
- B. Incorrect because it is a likely consequence of the self-control bias, and not the regret-aversion bias. As a result of self-control bias, FMPs [financial market participants] may borrow excessively to finance present consumption.
- C. Incorrect because it is a likely consequence of the framing bias, and not the regret-aversion bias. As a result of framing bias, FMPs [financial market participants] may misidentify risk tolerances because of how Answers about risk tolerance were framed, becoming more risk-averse when presented with a gain frame of reference and more risk-seeking when presented with a loss frame of reference.

Answer 22 of 100

Answer

Solution

- A. Incorrect because emotional biases [not cognitive errors], on the other hand, are harder to correct because they stem from impulses and intuitions.
- B. Incorrect because both forms of bias [cognitive and emotional], regardless of their source, may cause decisions to deviate from what is assumed by traditional finance theory.
- C. **Correct** because cognitive errors can often be **Correct** ed or eliminated through better information, education, and advice.

Answer 23 of 100

Answer

Solution

- A. Incorrect because ETFs can be transacted (and are priced) intraday.
- B. **Correct** because dividends on ETFs are paid out to the shareholders whereas mutual funds usually reinvest the dividends.
- C. Incorrect because exchange-traded funds (ETFs) are generally structured as open-end funds.

Answer 24 of 100

Answer

Solution

- A. Incorrect because the last term has not been doubled, leading to:

$$\sigma^2p = (0.5)^2(0.05) + (0.5)^2(0.06) + (0.5)(0.5)(0.75)(0.05)^{0.5}(0.06)^{0.5} = 0.0125 + 0.015 + 0.0102698 = 0.03777 \approx 0.038.$$

- B. **Correct** because for a two asset portfolio, the expression for portfolio variance simplifies to the following using correlation:

$\sigma^2p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$, where w_i is the weight of asset i in the portfolio, σ_i^2 is the variance of asset i , and ρ_{ij} is the correlation between the returns of assets i and j . Here,

$$\sigma^2p = (0.5)^2(0.05) + (0.5)^2(0.06) + 2(0.5)(0.5)(0.75)(0.05)^{0.5}(0.06)^{0.5} = 0.0125 + 0.015 + 0.0205396 = 0.04804 \approx 0.048.$$

- C. Incorrect because it is the weighted average of the asset variances: $(0.5)(0.05) + (0.5)(0.06) = 0.025 + 0.03 = 0.055$. However, a weighted average calculation is only appropriate for the portfolio standard deviation (not variance) when the correlation among assets is 1.

Answer 25 of 100

Answer

Solution

- A. Incorrect because in statistical factor models, the statistical factors, however, may or may not have an economic or fundamental connection to returns. Furthermore, in statistical factor models data mining may generate many spurious factors that are devoid of any economic meaning.
- B. **Correct** because in a statistical factor model, historical and cross-sectional return data are analyzed to identify factors that explain variance or covariance in observed returns.
- C. Incorrect because in statistical factor models, the statistical factors, however, may or may not have an economic or fundamental connection to returns. Furthermore, in statistical factor models data mining may generate many spurious factors that are devoid of any economic meaning.

Answer 26 of 100

Answer

Solution

- A. Incorrect because **systematic risk** is risk related to the economic system (e.g., risk related to business cycle) that cannot be eliminated by holding a diversified portfolio. Thus, dividing wealth between asset classes cannot eliminate the systematic risk.
- B. Incorrect because investors with below-average risk tolerance should have below-average weights in alternative investments. Typically, the strategic asset allocation for risk-averse investors will have a large weight in government bonds and cash, whereas those with more willingness and ability to take risk will have more of their assets in risky asset classes, such as equities and many types of alternative investments.
- C. **Correct** because in general, adding assets classes with low correlation improves the risk–return trade-off (more return for similar risk).

Answer 27 of 100

Answer

Solution

- A. Incorrect because the market return is used instead of the market risk premium. Therefore, Jensen's alpha = $R_p - [R_f + (\beta p \times R_m)] = 7\% - (1\% + 1.2 \times 5\%) = 0.0\%$.
- B. **Correct** because Jensen's alpha is defined as the portfolio return less (the risk-free rate plus the portfolio beta times (the market return minus the risk-free rate)). Therefore, Jensen's alpha = $R_p - [R_f + \beta p(R_m - R_f)] = 7\% - (1\% + 1.2 \times (5\% - 1\%)) = 7.0\% - 5.8\% = 1.2\%$.
- C. Incorrect because this omits the initial risk-free rate in the equation. Therefore, Jensen's alpha = $R_p - \beta p[R_m - R_f] = 7\% - (1.2 \times (5\% - 1\%)) = 7.0\% - 4.8\% = 2.2\%$.

Answer 28 of 100

Answer

Solution

- A. Incorrect because hindsight bias is a cognitive error and cognitive errors can often be corrected or eliminated through better information, education, and advice. Emotional biases, on the other hand, are harder to correct because they stem from impulses and intuitions. Thus, it is often possible only to recognize an emotional bias and adapt to it. **Hindsight bias** refers to believing past events as having been predictable and reasonable to expect. This behavior results from the obvious fact that outcomes that did occur are more readily evident than outcomes that did not.
- B. **Correct** because loss-aversion bias is an emotional bias, as opposed to a cognitive error, and cognitive errors can often be **Correct** ed or eliminated through better information, education, and advice. Emotional biases, on the other hand, are harder to **Correct** because they stem from impulses and intuitions. Thus, it is often possible only to recognize an emotional bias and adapt to it. **Loss-aversion bias** refers to the tendency to strongly prefer avoiding losses to achieving gains.
- C. Incorrect because representativeness bias is a cognitive error and cognitive errors can often be corrected or eliminated through better information, education, and advice. Emotional biases, on the other hand, are harder to correct because they stem from impulses and intuitions. Thus, it is often possible only to recognize an emotional bias and adapt to it. **Representativeness bias** refers to the tendency to classify new information based on past experiences and classifications.

Answer 29 of 100

Answer

Solution

- A. Incorrect because fundamental factor models analyze and use relationships between security returns and the company's underlying fundamentals, for example, earnings, earnings growth, cash flow generation, investment in research, advertising, and number of patents.
- B. **Correct** because Mark Carhart (1997) extended the Fama and French model by adding another factor: momentum, defined as relative past stock returns. The best example of a practical model is the four-factor model proposed by Fama and French (1992) and Carhart (1997).
- C. Incorrect because Fama and French (1992) proposed that three factors seem to explain asset returns better than just systematic risk. Those three factors are relative size, relative book-to-market value, and beta of the asset. Mark Carhart (1997) extended the Fama and French model by adding another factor: momentum, defined as relative past stock returns.

Answer 30 of 100

Answer

Solution

- A. Incorrect because M^2 provides a measure of portfolio return that is adjusted for the total risk of the portfolio relative to that of some benchmark.
- B. Incorrect because the Sharpe ratio measure is based on total risk. Total risk is relevant for an investor when he or she holds a portfolio that is not fully diversified, which is not a desirable portfolio. In such cases, the Sharpe ratio and M^2 are appropriate performance measures.
- C. **Correct** because Jensen's alpha is based on systematic risk. The difference between the actual portfolio return and the calculated risk-adjusted return is a measure of the portfolio's performance relative to the market portfolio and is called Jensen's alpha.

Answer 31 of 100

Answer

Solution

- A. **Correct** because in evaluating investments using only the mean (expected return) and variance (risk), we are implicitly making two important assumptions: 1) that the returns are normally distributed and can be fully characterized by their means and variances and 2) that markets are not only informationally efficient but that they are also operationally efficient. To the extent that these assumptions are violated, we need to consider additional investment characteristics. One of them is kurtosis. **Kurtosis** refers to fat tails or higher than normal probabilities for extreme returns and has the effect of increasing an asset's risk that is not captured in a mean-variance framework. This applies to asset returns as several market participants note that the probability and the magnitude of extreme events is underappreciated and was a primary contributing factor to the financial crisis of 2008.
- B. Incorrect because using only mean and variance would be appropriate to evaluate investments if returns were distributed normally. Returns, however, are not normally distributed...because the probability of extreme events is significantly greater than what a normal distribution would suggest. The latter deviation is referred to as kurtosis.
- C. Incorrect because **skewness** refers to asymmetry of the return distribution, that is, returns are not symmetric around the mean. Skewness does not refer specifically to the probability of extreme events (and does not measure it) even though a negatively skewed distribution would have a longer left tail relative to the normal distribution.

Answer 32 of 100

Answer

Solution

- A. Incorrect because reflecting low barriers to entry, large wealth management firms have introduced robo-adviser solutions to service certain customer segments and appeal to a new generation of investors.
- B. **Correct** because rapid growth in robo-advisory assets is based on several industry trends including **growing demand from 'mass affluent' and younger investors**. Traditional investment advice has often underserved younger and 'mass affluent' investors with lower relative levels of investable assets.
- C. Incorrect because robo-advisers often rely on lower fee underlying portfolio investment options, such as index funds or ETFs, when constructing portfolios for clients.

Answer 33 of 100

Answer

Solution

- A. Incorrect because representativeness bias pertains to sample-size neglect, where a small sample is believed to represent a population, or base-rate neglect, where specific information is seen to supercede an established trend. Momentum is attributable to the fear of missing out or recent trends in the market, not a specific piece of information. Momentum can be partly explained by availability, hindsight, and loss aversion biases.
- B. **Correct** because representativeness bias attributes one positive trait as being representative of an overall positive investment. The **halo effect** extends a favorable evaluation of some characteristics to other characteristics. A company with a good growth record and good previous share price performance might be seen as a good investment, with higher expected returns than its risk characteristics merit. This view is a form of representativeness that can lead investors to extrapolate recent past performance into expected returns.
- C. Incorrect because bubbles and crashes are most closely identified with overconfidence, not representativeness bias. In bubbles, investors often exhibit symptoms of *overconfidence*; overtrading, underestimation of risks, failure to diversify, and rejection of contradictory information. The overconfidence and excessive trading that contribute to a bubble are linked to *confirmation bias* and *self-attribution bias*.

Answer 34 of 100

Answer

Solution

- A. **Correct** because liquidity is a paramount concern for banks that stand ready to meet depositor requests for withdrawals.
- B. Incorrect because most foundations and endowments are established with the intent of having perpetual lives. The large allocation to alternative investments primarily reflects the typically long time horizon of endowments and foundations.
- C. Incorrect because an insurance company's general account is typically invested conservatively in a diverse allocation of fixed-income securities, while an insurer's surplus account typically targets a higher return than the general account and thus often invests in less-conservative asset classes.

Answer 35 of 100

Answer

Solution

- A. Incorrect because it describes risk tolerance, not risk budgeting. Risk tolerance identifies the extent to which the organization is willing to experience losses or opportunity costs and to fail in meeting its objectives. Whereas risk tolerance focuses on the appetite for risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken.
- B. **Correct** because risk budgeting quantifies and allocates the tolerable risk by specific metrics. Four well-known single-dimension measures that are often used are standard deviation, beta, value at risk (VaR), and scenario loss.
- C. Incorrect because it describes self-insurance, an example of risk acceptance, not risk budgeting. Self-insurance is the notion of bearing a risk that is considered undesirable but too costly to eliminate by external means. It may involve the establishment of a reserve to cover losses.

Answer 36 of 100

Answer

Solution

- A. Incorrect because, although the portfolio approach helps avoid disasters by way of diversification, it does not necessarily provide downside protection (prevent losses) during market downturns. A major reason that portfolios can effectively reduce risk is that combining securities whose returns do not move together provides diversification. However, an important issue is that the co-movement or correlation pattern of the securities' returns in the portfolio can change in a manner unfavorable to the investor. Despite diversification, a portfolio is unlikely to preserve its value during market downturns.
- B. Incorrect because an asset's risk should be measured in relation to the remaining systematic or non-diversifiable risk, which should be the only risk that affects the asset's price. Systematic risk is risk that cannot be avoided and is inherent in the overall market. It is non-diversifiable because it includes risk factors that are innate within the market and affect the market as a whole. Therefore, portfolio diversification does not reduce the systematic risk of individual assets in a portfolio.
- C. **Correct** because portfolio diversification helps investors avoid disastrous investment outcomes. A main tenet of the portfolio approach to investing is diversification. A disastrous outcome can result from 'putting all your eggs into one basket' or investing everything into one stock whose value could then go to zero. A diversified portfolio holding many securities is likely to avoid this outcome. Although diversification may not prevent losses during market downturns, it does help avoid disastrous investment outcomes during normal market conditions.

Answer 37 of 100

Answer

Solution

- A. Incorrect because the *willingness* to take risk, or risk attitude, is a more subjective factor based on the client's psychology and perhaps also his or her current circumstances.
- B. **Correct** because the *ability* to bear risk is measured mainly in terms of objective factors, such as time horizon, expected income, and the level of wealth relative to liabilities.
- C. Incorrect because the *willingness* to take risk, or risk attitude, is a more subjective factor based on the client's psychology and perhaps also his or her current circumstances.

Answer 38 of 100

Answer

Solution

- A. Incorrect because the *Investment Guidelines* section provides information about how policy should be executed (e.g., on the permissible use of leverage and derivatives) and on specific types of assets excluded from investment, if any.
- B. **Correct** because the *Evaluation and Review* section provides guidance on obtaining feedback on investment results.
- C. Incorrect because the *Statement of Duties and Responsibilities* details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers.

Answer 39 of 100

Answer

Solution

- A. Incorrect because the **security market line (SML)** is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk (not total risk), on the x-axis and expected return on the y-axis. The security market line applies to any security, efficient or not. Total risk and systematic risk are equal only for efficient portfolios because those portfolios have no diversifiable risk remaining.
- B. **Correct** because the **security market line (SML)** is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis.
- C. Incorrect because the **security market line (SML)** is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk (not unsystematic risk), on the x-axis and expected return on the y-axis.

Answer 40 of 100

Answer

Solution

- A. Incorrect because the company has a beta equal to one, which is the same as the beta of the market portfolio.
- B. **Correct** because the amount of systematic risk for a company is measured by the stock's beta. $\beta = \rho_{i,m} \times \sigma_i / \sigma_m = 0.60 \times 0.25 / 0.15 = 0.15 / 0.15 = 1$. Since the company's beta equals the market portfolio's beta (the market portfolio's beta with itself equals 1), the company has the same level of systematic risk as the market portfolio.
- C. Incorrect because the company has a beta equal to one, which is the same as the beta of the market portfolio.

Answer 41 of 100

Answer

Solution

- A. Incorrect because a risk-seeking investor would maximize both risk and return.
- B. Incorrect because a risk-neutral investor would maximize return irrespective of risk.
- C. **Correct** because risk-averse investors make investment decisions based on the risk–return trade-off, maximizing return for the same risk, and minimizing risk for the same return.

Answer 42 of 100

Answer

Solution

A. **Correct** because sovereign wealth funds (SWFs) are government-owned investment funds.

B. Incorrect because sovereign wealth funds (SWFs) are government-owned investment funds. Closed-end funds are mutual funds in which no new investment money is accepted into the fund. New investors invest by buying existing shares, and investors in the fund liquidate by selling their shares to other investors. A mutual fund is a comingled investment pool in which investors in the fund each have a pro-rata claim on the income and value of the fund. SWFs do not pool investment money from individual investors and institutions, and they are not traded.

C. Incorrect because sovereign wealth funds (SWFs) are government-owned investment funds or entities that invest in financial or real assets. In order to meet their long-term return objectives, SWFs invest in a wide range of real and financial assets or in alternative investments. The name "sovereign" wealth funds refers to the funds' ownership rather than their permitted investments.

Answer 43 of 100

Answer

Solution

A. Incorrect because the **capital market line** is a special case of the capital allocation line, where the risky portfolio is the market portfolio. The capital market line is shown in Exhibit 3, where the standard deviation (σ_p), or total risk, is on the x-axis and expected portfolio return, $E(R_p)$, is on the y-axis.

B. Incorrect because the security market line represents a graph of the capital asset pricing model. The **security market line** (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis. Using the same concept as the capital market line, the SML intersects the y-axis at the risk-free rate of return, and the slope of this line is the market risk premium, $R_m - R_f$.

C. **Correct** because similar to the SML (security market line), we can draw a **security characteristic line** (SCL) for a security. The SCL is a plot of the excess return of the security on the excess return of the market. The security characteristic line can also be estimated by regressing the excess security return, $R_i - R_f$, on the excess market return, $R_m - R_f$.

Answer 44 of 100

Answer

Solution

A. **Correct** because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk. Pricing or valuing an asset is equivalent to estimating its expected rate of return.

B. Incorrect because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk.

C. Incorrect because systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk. Nonsystematic risk is the risk that pertains to a single company or industry and is also known as company-specific, industry-specific, diversifiable, or idiosyncratic risk.

Answer 45 of 100

Answer

Solution

A. Incorrect because although the IPS is sometimes complemented by a document outlining policy on **sustainable investing** policies on sustainable investing may also be integrated within the IPS itself. In the remainder of this reading, the integration of sustainable investing within the IPS will be our working assumption.

B. **Correct** because the constraints may be internal (i.e., set by the client), or external (i.e., set by law or regulation).

C. Incorrect because as part of their financial planning, clients may specify specific spending goals, each of which could have different risk tolerance and return objectives.

Answer 46 of 100

Answer

Solution

A. Incorrect because a client's overall risk tolerance is a function of the client's ability to bear (accept) risk and his or her 'risk attitude', which might be considered as the client's willingness to take risk," not the other way around.

B. Incorrect because a willingness (not ability) to take risk may be gauged by discussing risk with the client or by asking the client to complete a psychometric Answersnaire.

C. **Correct** because the *ability* to bear risk is measured mainly in terms of objective factors, such as time horizon, expected income, and the level of wealth relative to liabilities. For example, an investor with a 20-year time horizon can be considered to have a greater ability to bear risk, other things being equal, than an investor with a 2-year horizon. This difference is because over 20 years there is more scope for losses to be recovered or other adjustments to circumstances to be made than there is over two years.

Answer 47 of 100

Answer

Solution

A. **Correct** because the capital market line (CML) does not apply to all securities or assets but only to portfolios on the efficient frontier. The efficient frontier gives optimal combinations of expected return and total risk. Total risk and systematic risk are equal only for efficient portfolios because those portfolios have no diversifiable risk remaining. Thus, the CML holds only for well-diversified portfolios.

B. Incorrect because the nonsystematic risk of the market portfolio is zero, but not its systematic risk. Because nonsystematic risk is zero for well-diversified portfolios, such as the market portfolio, the total risk of a market portfolio and other similar portfolios is only systematic risk, which is $\beta\sigma_m$. Therefore, portfolios on the CML have zero nonsystematic (not systematic) risk.

C. Incorrect because the total risk of a market portfolio and other similar portfolios is only systematic risk, which is $\beta\sigma_m$. Therefore, portfolios on the CML have nonsystematic risk equal to zero, not beta.

Answer 48 of 100

Answer

Solution

A. Incorrect because the curve that lies above and to the right of the global minimum-variance portfolio is referred to as the **Markowitz efficient frontier** because it contains all portfolios of risky assets that rational, risk-averse investors will choose. The efficient frontier is the upper part of the minimum-variance frontier, it does not lie above and to the left of it.

B. Incorrect because the entire collection of minimum-variance portfolios for all possible returns is referred to as the minimum-variance frontier, not the efficient frontier (which is only the upper arm of the minimum-variance frontier).

Compared to the investment opportunity set, the minimum-variance frontier defines the smaller set of portfolios in which investors would want to invest. Note that no risk-averse investor will choose to invest in a portfolio to the right of the minimum-variance frontier because a portfolio on the minimum-variance frontier can give the same return but at a lower risk.

- C. **Correct** because the **Markowitz efficient frontier** contains all portfolios of risky assets that rational, risk-averse investors will choose.

Answer 49 of 100

Answer

Solution

- A. Incorrect because the client's utility function affects their willingness, not ability, to take risk. Utility is a measure of relative satisfaction from consumption of various goods and services or in the case of investments, the satisfaction that an investor derives from different portfolios.
- B. Incorrect because the client's degree of risk aversion affects their willingness, not ability, to take risk. Risk attitude, or willingness to take risk, is a more subjective factor based on the client's psychology.
- C. **Correct** because the ability to bear risk is measured mainly in terms of objective factors, such as time horizon, expected income, and the level of wealth relative to liabilities.

Answer 50 of 100

Answer

Solution

- A. **Correct** because the CAPM equation is $E(R_i) = R_f + \beta[E(R_m) - R_f]$. Therefore, the beta of the security has to be less than 1 in order for the security's return to be the same as the market risk premium, given that the risk-free rate is positive.
- B. Incorrect because the CAPM equation is $E(R_i) = R_f + \beta[E(R_m) - R_f]$. Therefore, if beta is equal to 1, the security's return will be higher than the market risk premium.
- C. Incorrect because the CAPM equation is $E(R_i) = R_f + \beta[E(R_m) - R_f]$. Therefore, if beta is greater than 1, the security's return will be higher than the market risk premium.

Answer 51 of 100

Answer

Solution

- A. Incorrect because risk attitude, or willingness to take risk, is a more subjective factor based on the client's psychology and perhaps also his or her current circumstances. Risk attitude is an investor's willingness to take risk, not the investor's ability to take risk.
- B. Incorrect because risk attitude, or willingness to take risk, is a more subjective factor based on the client's psychology and perhaps also his or her current circumstances. Although the list of factors that are related to an individual's risk attitude remains open to debate, it is believed that some psychological factors, such as personality type, self-esteem, and inclination to independent thinking, are correlated with risk attitude. Self-confidence is therefore associated with an investor's willingness to take risk, not the investor's ability to take risk.
- C. **Correct** because the ability to bear risk is measured mainly in terms of objective factors, such as time horizon. For example, an investor with a 20-year time horizon can be considered to have a greater ability to bear risk, other things being equal, than an investor with a 2-year horizon.

Answer 52 of 100

Answer

Solution

A. Incorrect because the security beta is wrongly calculated due to the reversal of the standard deviations of the security and the market portfolio. Therefore, the beta is calculated as $\beta = (0.8 \times 20\%) / 35\% = 0.457$. Next, using the CAPM equation, $E(R_i) = R_f + \beta[E(R_m) - R_f]$, the expected return of the security is calculated as $E(R_i) = 2\% + 0.457 \times (10\% - 2\%) = 5.66\% \approx 5.7\%$.

B. **Correct** because the expected return of the security can be calculated using the CAPM equation: $E(R_i) = R_f + \beta[E(R_m) - R_f]$. The beta of the security can be calculated using the equation $\beta = (\rho_{i,m} \times \sigma_i) / \sigma_m$. Therefore, $\beta = (0.8 \times 35\%) / 20\% = 1.4$. Beta is the product of the asset's correlation with the market with a ratio of standard deviations of return (i.e., the ratio of the asset's standard deviation to the market's). Therefore, using the CAPM equation: $E(R_i) = 2\% + 1.4 \times (10\% - 2\%) = 13.2\%$.

C. Incorrect because the risk-free rate is not included in the calculation of the market risk premium. Beta is calculated as $\beta = (0.8 \times 35\%) / 20\% = 1.4$. Therefore, using the CAPM equation: $E(R_i) = 2\% + 1.4 \times (10\%) = 16.0\%$.

Answer 53 of 100

Answer

Solution

A. **Correct** because the correlation between the returns of the two securities is

$$r_{ij} = \frac{\text{Cov}_{ij}}{\sigma_i \sigma_j} ; \sigma_i \sigma_j = \frac{\text{Cov}_{ij}}{r_{ij}} ; \sigma_i \sigma_j = \frac{5.5}{0.75} = 7.3$$

. Since the two securities have equal risk,
 $\sigma_i = \sigma_j, \sigma_i^2 = 7.3, \sigma_i = 2.7\%$.

B. Incorrect because it is a miscalculation and mistakes $\sigma_i \sigma_j$ as $2 \times \sigma_i$;

$$\sigma_i \sigma_j = \frac{5.5}{0.75} = 7.3 ; 2 \times \sigma_i = 7.33\% ; \sigma_i = 3.67\% \approx 3.7\%.$$

C. Incorrect because it is an intermediate step in the correct calculation;

$$\sigma_i \sigma_j = \frac{5.5}{0.75} = 7.3 \%$$

Answer 54 of 100

Answer

Solution

A. **Correct** because the expected return of an asset is $E(R_i) = R_f + \beta_i[E(R_m) - R_f]$, where R_i , R_m , and R_f denote the return on the asset, the market, and the risk-free asset, respectively, β_i is the asset's beta, and $[E(R_m) - R_f]$ is the market risk premium. Thus, $\beta_i = [E(R_i) - R_f] / [E(R_m) - R_f] = [5\% - 1\%] / 5\% = 4\% / 5\% = 0.8$.

B. Incorrect because it does not subtract the risk-free rate in the numerator and calculates $\beta_i = E(R_i) / [E(R_m) - R_f] = 5\% / 5\% = 1.0$. It is also the answer if the market risk premium is confused with the market return, thus $\beta_i = [E(R_i) - R_f] / [\text{market risk premium} - R_f] = [5\% - 1\%] / [5\% - 1\%] = 4\% / 4\% = 1.0$.

C. Incorrect because it is the inverse of beta, $1/\beta_i = [E(R_m) - R_f] / [E(R_i) - R_f] = \text{market risk premium} / [E(R_i) - R_f] = 5\% / [5\% - 1\%] = 5\% / 4\% = 1.25$. It is also the answer if two mistakes are made: The risk-free rate is not subtracted from the expected return in the numerator and the market risk premium is confused with the market return. Thus, $\beta_i = E(R_i) / [\text{market risk premium} - R_f] = 5\% / [5\% - 1\%] = 5\% / 4\% = 1.25$.

Answer 55 of 100

Answer

Solution

- A. **Correct** because the expression 'risk–return trade-off' refers to the positive relationship between expected risk and return. In other words, a higher return is not possible to attain in **efficient markets** and over long periods of time without accepting higher risk. Expected returns should be greater for assets with greater risk. Over long periods of time, we observe that higher risk does result in higher mean returns. Thus, it is reasonable to claim that, over the long term, market prices reward higher risk with higher returns, which is a characteristic of a risk-averse investor.
- B. **Incorrect** because long-term historical data on securities suggest that investors are risk averse, not risk neutral.
- C. **Incorrect** because because long-term historical data on securities suggest that investors are risk averse, not risk seeking.

Answer 56 of 100

Answer

Solution

- A. **Incorrect** because the Procedures section of an Investment Policy Statement (IPS) explains the steps to take to keep the IPS current and the procedures to follow to respond to various contingencies. Information regarding the permissible use of derivatives is included under Investment Guidelines.
- B. **Correct** because the Investment Guidelines section of an Investment Policy Statement (IPS) provides information about how policy should be executed (e.g., on the permissible use of leverage and derivatives) and on specific types of assets excluded from investment, if any.
- C. **Incorrect** because the Statement of Duties and Responsibilities section of an Investment Policy Statement (IPS) details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers. Information regarding the permissible use of derivatives is included under Investment Guidelines.

Answer 57 of 100

Answer

Solution

- A. **Correct** because the feedback step assists the portfolio manager in rebalancing the portfolio due to a change in, for example, market conditions or the circumstances of the client.
- B. **Incorrect** because it is part of the execution step, not the feedback step. The portfolio execution step consists of first deciding on a target asset allocation, which determines the weighting of asset classes to be included in the portfolio.
- C. **Incorrect** because it is part of the planning step, not the feedback step. The first step (the planning step) in the investment process is to understand the client's needs (objectives and constraints) and develop an investment policy statement (IPS).

Answer 58 of 100

Answer

Solution

- A. **Correct** because the focus on the SAA [strategic asset allocation] is the result of a number of important investment principles. One such principle is that the returns to groups of similar assets (e.g., long-term debt claims) predictably reflect exposures to certain sets of systematic factors (e.g., for the debt claims, unexpected changes in the inflation rate).
- B. **Incorrect** because the focus on the SAA [strategic asset allocation] is the result of a number of important investment principles. One principle is that a

portfolio's systematic risk accounts for most of its change in value over the long term.

C. Incorrect because the decision to deliberately deviate from the policy exposures to systematic risk factors (i.e., the policy weights of asset classes) with the intent to add value based on forecasts of the near-term returns of those asset classes is called tactical asset allocation. It is not one of the investment principles which strategic asset allocation is based upon.

Answer 59 of 100

Answer

Solution

A. Incorrect because the countrywide constraint for pension funds that limits the proportion of high-risk assets in the portfolio is a regulatory constraint. The fact that the constraint is formulated as a limit to invest in high-risk assets does not indicate it is a liquidity constraint. Liquidity constraints state what the likely requirements are to withdraw funds from the portfolio.

B. **Correct** because the IPS should state any legal and regulatory restrictions that constrain how the portfolio is invested. In some countries, such institutional investors as pension funds are subject to restrictions on the composition of the portfolio. For example, there may be a limit on the proportion of equities or other risky assets in the portfolio, or on the proportion of the portfolio that may be invested overseas.

C. Incorrect because the countrywide constraint for pension funds that limits the proportion of high-risk assets in the portfolio is a regulatory constraint. The constraint does not change the broad investment time horizon of the pension fund and is therefore not a time horizon constraint. Time horizon constraints state the time horizon over which the investor is investing.

Answer 60 of 100

Answer

Solution

A. Incorrect because the entire collection of these minimum-variance portfolios is referred to as the minimum-variance frontier. This describes the minimum variance portfolio, but not the global minimum-variance portfolio.

B. **Correct** because the left-most point on the minimum-variance frontier is the portfolio with the minimum variance among all portfolios of risky assets, and is referred to as the global minimum-variance portfolio.

C. Incorrect because the upper right-most point of the minimum-variance frontier is the opposite position of the global minimum-variance portfolio, it is actually the maximum variance on the curve of risky portfolios.

Answer 61 of 100

Answer

Solution

A. Incorrect because this is not one of the two principles which resulted in the focus on a strategic asset allocation in portfolio construction. Furthermore, in general, adding assets classes with low (not high) correlation improves the risk-return trade-off (more return for similar risk).

B. Incorrect because a portfolio's systematic (not nonsystematic) risk accounts for most of its change in value over the long term.

C. **Correct** because the focus on the SAA (strategic asset allocation) is the result of a number of important investment principles. A second principle is that the returns to groups of similar assets (e.g., long-term debt claims) predictably reflect exposures to certain sets of systematic factors (e.g., for the debt claims, unexpected changes in the interest rate).

Answer 62 of 100

Answer

Solution

- A. **Correct** because the IPS should be reviewed on a regular basis to ensure that it remains consistent with the client's circumstances and requirements. The IPS should also be reviewed if the manager becomes aware of a material change in the client's circumstances, or on the initiative of the client when his or her objectives, time horizon, or liquidity needs change. The major components of an IPS include the following section: *Procedures*. This section explains the steps to take to keep the IPS current and the procedures to follow to respond to various contingencies.
- B. Incorrect because the Investment Guidelines section of the investment policy statement (IPS) provides information about how policy should be executed (e.g., on the permissible use of leverage and derivatives) and on specific types of assets excluded from investment, if any. Information about how and when the IPS should be reviewed can be found in the Procedures section.
- C. Incorrect because the Statement of Duties and Responsibilities section of the investment policy statement (IPS) details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers. While this section may outline who is responsible for reviewing the IPS, information about how and when the IPS should be reviewed is more likely to be found in the Procedures section.

Answer 63 of 100

Answer

Solution

- A. **Correct** because the key to a defined contribution (DC) plan is that the employee accepts the investment and inflation risk and is responsible for ensuring that there are enough assets in the plan to meet their needs upon retirement.
- B. Incorrect because this applies for defined benefit (DB) plans, as generally, employers are responsible for the contributions made to a DB plan and bear the risk associated with adequately funding the benefits offered to employees.
- C. Incorrect because defined contribution (DC) plans typically have lower costs/risk to the company.

Answer 64 of 100

Answer

Solution

- A. Incorrect because the intercept, α_i , and slope coefficient, β_i , of the market model can be estimated by using historical security and market returns. These parameter estimates are then used to predict company-specific returns that a security may earn in a future period.
- B. Incorrect because the market model is a return generating model which can be used to predict company-specific returns based on market returns. Macroeconomic factor models (not the market model) use economic factors that are correlated with security returns. These factors may include economic growth.
- C. **Correct** because the intercept, α_i , and slope coefficient, β_i , of the market model can be estimated by using historical security and market returns. These parameter estimates are then used to predict company-specific returns that a security may earn in a future period.

Answer 65 of 100

Answer

Solution

- A. Incorrect because it is an ETF that is priced intraday, not a mutual fund. The value of a mutual fund is referred to as the 'net asset value'. It is computed daily based on the closing price of the securities in the portfolio. Because they are traded on exchanges, ETFs

can be transacted (and are priced) intraday. In contrast, mutual funds typically can be purchased or sold only once a day.

B. Incorrect because closed-end, not open-end, mutual funds have a fixed number of shares outstanding. An alternative to setting the fund up as an open-end fund would be to create a **closed-end fund** in which no new investment money is accepted into the fund. New investors invest by buying existing shares, and investors in the fund liquidate by selling their shares to other investors. Hence, the number of outstanding shares does not change. One consequence of this fixed share base is that, unlike open-end funds in which new shares are created and sold at the current net asset value per share, closed-end funds can sell for a premium or discount to net asset value depending on the demand for the shares.

C. **Correct** because the minimum required investment in ETFs is usually smaller than that of mutual funds.

Answer 66 of 100

Answer

Solution

A. Incorrect because it uses an equally weighted portfolio of securities instead of deriving the securities' weights from the portfolio's expected return. Given that the two securities' returns are uncorrelated, it follows that the correlation between the two securities' returns is 0. Thus, portfolio standard deviation is calculated as:

$$\sigma_p^2 = 0.5^2 \times 0.24^2 + 0.5^2 \times 0.12^2$$

$$\sigma_p^2 = 0.0144 + 0.0036$$

$$\sigma_p^2 = 0.018$$

$$\sigma_p = \sqrt{0.018} = 0.1342 \approx 13.4\%$$

B. **Correct** because the portfolio's weights are calculated by setting the portfolio return equal to 12.6%. The portfolio return of a two-security portfolio is:

$$R_p = w_1 R_1 + w_2 R_2$$

$$R_p = w_1 R_1 + (1 - w_1) R_2$$

$$12.6\% = w_1 \times 17\% + (1 - w_1) \times 6\%$$

$$w_1 = 0.6 \text{ and } w_2 = 1 - 0.6 = 0.4$$

The variance of a two-security portfolio is:

$$\sigma_p^2 = (w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)) >$$

Since the two securities' returns are uncorrelated, it follows that $\text{Cov}(R_1, R_2) = 0$, such that:

$$\sigma_p^2 = (0.6^2 \times 0.24^2 + 0.4^2 \times 0.12^2)$$

$$\sigma_p^2 = (0.0207 + 0.0023)$$

$$\sigma_p = \sqrt{0.0230} = 0.1518 \approx 15.2\%$$

C. Incorrect because it assumes that, since the securities' returns are uncorrelated, the portfolio standard deviation is the weighted mean of the individual securities' standard deviations.

The portfolio's weights are calculated by setting the portfolio return equal to 12.6%. The portfolio return of a two-security portfolio is:

$$R_p = w_1 R_1 + (1 - w_1) R_2$$

$$12.6\% = w_1 \times 17\% + (1 - w_1) \times 6\%$$

$$w_1 = 0.6 \text{ and } w_2 = 1 - 0.6 = 0.4$$

Then, σ_p is incorrectly calculated as:

$$\sigma_p = 0.6 \times 0.24 + 0.4 \times 0.12$$

$\sigma_p = 0.192 = 19.2\%$. This would be the correct portfolio standard deviation if the correlation was equal to 1.

Answer 67 of 100

Answer

Solution

A. Incorrect because the portfolio risk is less than the weighted average of risks when the correlation of asset returns is less than one. For an equally weighted portfolio, the standard deviation can only be equal to zero when the two assets are perfectly negatively correlated ($\rho = -1$) and the two assets have the same standard deviation. For an extreme case in which $\rho_{12} = -1$ (that is, the two asset returns move in opposite directions), the portfolio can be made risk free.

B. Incorrect because the portfolio's standard deviation is simply a weighted average of the standard deviations of the two assets when the correlation of asset returns is equal to one.

C. **Correct** because the portfolio risk is less than the weighted average of risks when the correlation of asset returns is less than one.

Answer 68 of 100

Answer

Solution

A. **Correct** because the risk-free asset could be combined with a risky portfolio to create a capital allocation line (CAL). A specific CAL that uses the market portfolio as the optimal risky portfolio is known as the capital market line. When assuming homogeneous expectations, only one optimal portfolio exists. The capital market line is shown in Exhibit 3, where the standard deviation (σ_p), or total risk, is on the x-axis and expected portfolio return, $E(R_p)$, is on the y-axis.

B. Incorrect because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis. Using the same concept as the capital market line, the SML intersects the y-axis at the risk-free rate of return, and the slope of this line is the market risk premium, $R_m - R_f$.

C. Incorrect because a security characteristic line (SCL) for a security is a plot of the excess return of the security on the excess return of the market.

Answer 69 of 100

Answer

Solution

A. **Correct** because the risk–return profile of the strategic asset allocation depends on the expected returns and risks of the individual asset classes, as well as the correlation between those asset classes.

B. Incorrect because the risk–return profile of the strategic asset allocation depends on the expected returns and risks of the individual asset classes, as well as the correlation between those asset classes. Deciding to deviate from policy weights or to select securities aiming to beat the benchmark creates additional uncertainty about returns. This risk is over and above the risk inherent in the policy portfolio. Hence, an investment policy should set risk limits and desired payoffs for each of these activities. Risk budgeting implies that the portfolio manager has to choose, for every asset class, whether to deploy security selection as a return generator. Although security selection could impact the returns of a portfolio, the risk–return profile of a portfolio's strategic asset allocation is based on expected returns for the asset classes and the correlation between those asset classes rather than the additional risk associated with the use of security selection.

C. Incorrect because the risk–return profile of the strategic asset allocation depends on the expected returns and risks of the individual asset classes, as well as the correlation between those asset classes. As the portfolio is constructed and its value changes with the returns of the asset classes and securities in which it is invested, the weights of the asset classes will gradually deviate from the policy weights in the strategic asset allocation. This

process is referred to as drift. Periodically, or when a certain threshold deviation from the policy weight (the bandwidth) has been breached, the portfolio should be rebalanced back to the policy weights. The set of rules that guide the process of restoring the portfolio's original exposures to systematic risk factors is known as the **rebalancing policy**. Deciding to deviate from policy weights or to select securities aiming to beat the benchmark creates additional uncertainty about returns. This risk is over and above the risk inherent in the policy portfolio. Although the weights of asset classes will deviate over time, the risk– return profile of a portfolio's strategic asset allocation is not determined by the rebalancing policy.

Answer 70 of 100

Answer

Solution

- A. **Correct** because the Sharpe ratio is defined as the portfolio's risk premium divided by its risk. Hence, the Sharpe ratio = $(R_p - R_f) / \sigma_p = (15.2\% - 3.1\%) / 11.7\% = 12.1\% / 11.7\% = 1.0342 \approx 1.03$.
- B. Incorrect because it omits the risk-free rate when determining the portfolio's risk premium. Hence, the Sharpe ratio = Mean Annual Return/Standard Deviation of Return = $15.2\% / 11.7\% = 1.2991 \approx 1.30$.
- C. Incorrect because the risk-free rate is added to (not subtracted from) the portfolio's return to determine the risk premium. Hence, the Sharpe ratio = $(R_p + R_f) / \sigma_p = (15.2\% + 3.1\%) / 11.7\% = 18.3\% / 11.7\% = 1.5641 \approx 1.56$.

Answer 71 of 100

Answer

Solution

- A. Incorrect because tax risk is a non-financial risk as the following three non-financial risks are related: regulatory risk, accounting risk, and tax risk.
- B. **Correct** because the risk management industry has come to classify three types of risks as primarily financial in nature and the second primary financial risk is credit risk.
- C. Incorrect because accounting risk is a non-financial risk as the following three non-financial risks are related: regulatory risk, accounting risk, and tax risk.

Answer 72 of 100

Answer

Solution

- A. Incorrect because derivatives are also sensitive to changes in interest rates, which are reflected in a measure called rho.
- B. Incorrect because vega is a first-order measure of the change in the derivative price for a change in the volatility of the underlying.
- C. **Correct** because the sensitivity of the derivative price to a small change in the value of the underlying asset is called the delta. Large changes are captured by the concept of gamma. Whereas delta is a first-order risk, gamma is considered a second-order risk because it reflects the risk of changes in delta. Gamma is a numerical measure of how sensitive an option's delta is to a change in the value of the underlying.

Answer 73 of 100

Answer

Solution

- A. Incorrect because beta is the slope of the security characteristic line (SCL), not the intercept.

- B. **Correct** because the SCL is a plot of the excess return of the security on the excess return of the market. Jensen's alpha is the intercept and the beta is the slope.
- C. Incorrect because the risk-free rate of return is the intercept on the y-axis of the security market line, whereas Jensen's alpha is the intercept on the y-axis of the security characteristic line.

Answer 74 of 100

Answer

Solution

- A. Incorrect because the *Evaluation and Review* section provides guidance on obtaining feedback on investment results.
- B. **Correct** because the sections that are most closely linked to the client's distinctive needs, and probably the most important from a planning perspective, are those dealing with investment objectives and constraints. An IPS [investment policy statement] focusing on these two elements has been called an IPS in an 'objectives and constraints' format.
- C. Incorrect because the *Statement of Duties and Responsibilities* details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers.

Answer 75 of 100

Answer

Solution

- A. Incorrect because the security's beta is the slope of the security characteristic line (SCL), not the security market line. The SCL is a plot of the excess return of the security on the excess return of the market. The beta is the slope.
- B. **Correct** because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis. The slope of this line is the market risk premium, $R_m - R_f$.
- C. Incorrect because it is the slope of the capital market line (CML), $E(R_m) - R_f / \sigma_m$ not the security market line. Since the market standard deviation is unlikely to be 1, the slope of the CML is most likely not the slope of the SML.

Answer 76 of 100

Answer

Solution

- A. Incorrect because M^2 provides a measure of portfolio return that is adjusted for the total risk of the portfolio relative to that of some benchmark. While M^2 is related to the Sharpe ratio, it is calculated in a different way and is measured in different units, therefore it does not represent the slope of the capital allocation line.
- B. **Correct** because the Sharpe ratio, also called the reward-to-variability ratio, is simply the slope of the capital allocation line.
- C. Incorrect because the Treynor ratio corresponds to the slope of the security market line, not the capital allocation line. The **Treynor ratio** is a simple extension of the Sharpe ratio and resolves the Sharpe ratio's first limitation by substituting beta (systematic risk) for total risk.

Answer 77 of 100

Answer

Solution

- A. Incorrect because the Procedures section explains the steps to take to keep the IPS current and the procedures to follow to respond to various contingencies.
- B. Incorrect because the Investment Constraints section presents the factors that constrain the client in seeking to achieve the investment objectives. The constraints section covers factors that need to be taken into account when constructing a portfolio for the client that meets the objectives. The typical categories are liquidity requirements, time horizon, regulatory requirements, tax status, and unique needs. The constraints may be internal (i.e., set by the client), or external (i.e., set by law or regulation).
- C. **Correct** because the Statement of Duties and Responsibilities section details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers. In the case of an institution, such as a pension plan or university endowment, the IPS may set out the governance arrangements that apply to the investment funds. For example, this information could cover the investment committee's approach to appointing and reviewing investment managers for the portfolio, and the discretion that those managers have.

Answer 78 of 100

Answer

Solution

- A. **Correct** because total risk is relevant for an investor when he or she holds a portfolio that is not fully diversified, which is not a desirable portfolio. In such cases, the Sharpe ratio and M^2 are appropriate performance measures. The Sharpe ratio uses total risk as a measure of risk.
- B. Incorrect because performance measures relative to beta risk—Treynor ratio and Jensen's alpha—are relevant when the investor holds a well-diversified portfolio with negligible diversifiable risk.
- C. Incorrect because performance measures relative to beta risk—Treynor ratio and Jensen's alpha—are relevant when the investor holds a well-diversified portfolio with negligible diversifiable risk.

Answer 79 of 100

Answer

Solution

- A. **Correct** because Total variance = Systematic variance + Nonsystematic variance, which can be written as $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{e_i}^2$. Thus, $\sigma_{e_i}^2 = \sigma_i^2 - \beta_i^2 \sigma_m^2$. Thus, nonsystematic variance for Stock 1 is equal to $(0.17)^2 - (0.9)^2(0.1)^2 = 0.0208$. For Stock 2, this is equal to $(0.18)^2 - (1.1)^2(0.1)^2 = 0.0203$. For Stock 3, this is equal to $(0.16)^2 - (1.2)^2(0.1)^2 = 0.0112$. Therefore, Stock 1 has the highest nonsystematic variance and the highest idiosyncratic risk. Nonsystematic risk is the risk that pertains to a single company or industry and is also known as company-specific, industry-specific, diversifiable, or idiosyncratic risk.
- B. Incorrect because Stock 2 has the highest total risk, not the highest nonsystematic risk. Total risk is calculated as the square root of total variance.
- C. Incorrect because Stock 3 has the highest beta and thus highest systematic risk, not the highest nonsystematic risk. Systematic risk is calculated as $\beta \sigma_m$.

Answer 80 of 100

Answer

Solution

- A. Incorrect because in a VaR measure, there is no ultimate maximum that one can state. VaR is thus a minimum extreme loss metric. Thus, the average loss cannot be provided by the VaR.
- B. Incorrect because a standard deviation may not exist for return distributions with fat tails. Thus, the use of standard deviation is not appropriate when estimating extreme losses.
- C. **Correct** because the statistics used to estimate VaR can be used to gauge average extreme losses. Conditional VaR or **CVaR** is a common tail loss measure, defined as the weighted average of all loss outcomes in the statistical distribution that exceed the VaR loss. Another tail risk metric in the credit risk space that is analogous to CVaR is expected loss given default, which answers the Answer for a debt security, If the underlying company or asset defaults, how much do we lose on average?

Answer 81 of 100

Answer

Solution

- A. Incorrect because the analyst forecasts the return of the asset to be 7%, which is lower than the expected return of 8.5% according to the CAPM. Thus, the security plots below the SML and should be considered overvalued. This response would be chosen if the expected return of the asset was wrongly computed as $0.02 + 1.3 \times (0.05 - 0.02) = 0.059 = 5.9\%$, which would imply that the asset plots above the SML.
- B. Incorrect because the analyst forecasts the return of the asset to be 7%, which is lower than the expected return of 8.5% according to the CAPM. Thus, the security does not plot on the SML and it is not properly valued according to the CAPM. This response would be chosen if beta was omitted in the CAPM equation and the expected return of the asset was wrongly computed as $0.02 + 0.05 = 0.07 = 7.0\%$, which would imply that the asset plots on the SML.
- C. **Correct** because the **security market line** (SML) is a graphical representation of the capital asset pricing model with beta, reflecting systematic risk, on the x-axis and expected return on the y-axis. Using the same concept as the capital market line, the SML intersects the y-axis at the risk-free rate of return, and the slope of this line is the market risk premium, $R_m - R_f$. Potential investors can plot a security's expected return and beta against the SML and use this relationship to decide whether the security is overvalued or undervalued in the market. All securities that reflect the consensus market view are points directly on the SML (i.e., properly valued). If a point representing the estimated return of an asset is above the SML, the asset has a low level of risk relative to the amount of expected return and would be a good choice for investment. In contrast, if the point representing a particular asset is below the SML, the stock is considered overvalued. The asset will be on the SML if the forecasted return equals the expected return of $0.02 + 1.3 \times 0.05 = 0.085 = 8.5\%$. Since the analyst forecasts the return of the asset to be 7%, which is lower than the expected return according to the CAPM, the security plots below the SML and should be considered overvalued.

Answer 82 of 100

Answer

Solution

- A. Incorrect because typically, smart beta strategies feature somewhat higher (not lower) management fees and higher portfolio turnover relative to passive market-cap weighted strategies.

- B. Incorrect because typically, smart beta strategies feature somewhat higher management fees and higher (not lower) portfolio turnover relative to passive market-cap weighted strategies.
- C. **Correct** because typically, smart beta strategies feature somewhat higher management fees and higher portfolio turnover relative to passive market-cap weighted strategies.

Answer 83 of 100

Answer

Solution

- A. **Correct** because when defining asset classes, a number of criteria apply. Intuitively, an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of assets should be relatively high within an asset class but should be lower versus assets in other asset classes. A between asset class correlation of zero would indicate better defined asset classes than higher correlations would.
- B. Incorrect because when defining asset classes, a number of criteria apply. Intuitively, an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of assets should be relatively high within an asset class but should be lower versus assets in other asset classes. A between asset class correlation of 0.5 would indicate less well defined asset classes than a zero correlation would.
- C. Incorrect because when defining asset classes, a number of criteria apply. Intuitively, an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of assets should be relatively high within an asset class but should be lower versus assets in other asset classes. A between asset class correlation of 1.0 would indicate less well defined asset classes than a zero correlation would. A candidate may choose this response if they confuse the criteria for correlation across asset classes with the criteria for correlation within each asset class (which should be high).

Answer 84 of 100

Answer

Solution

- A. Incorrect because systematic risk, also known as non-diversifiable or market risk, is the risk that affects the entire market or economy. Systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk.
- B. Incorrect because systematic risk, also known as non-diversifiable or market risk, is the risk that affects the entire market or economy. Systematic or non-diversifiable risk is priced and investors are compensated for holding assets or portfolios based only on that investment's systematic risk. Investors do not receive any return for accepting nonsystematic or diversifiable risk.
- C. **Correct** because we can assume that in an efficient market, no incremental reward can be earned for taking on diversifiable risk. Nonsystematic risk is the risk that pertains to a single company or industry and is also known as company-specific, industry-specific, diversifiable, or idiosyncratic risk.

Answer 85 of 100

Answer

Solution

A. Incorrect because there is no indication that the pension fund's decision to constrain investments in real estate is driven by legal and regulatory factors. Moreover, the word 'decision' highlights that it is not an external requirement, which would be the case if it was required by law or regulation.

B. Incorrect because this constraint is driven by economic considerations of the client, and thus would be reflected in the tax concerns section of the IPS, and not in the legal and regulatory constraints. In many cases, the portfolio should reflect the tax status of the client. For example, a taxable investor may wish to hold a portfolio that emphasizes capital gains and receives little income.

C. **Correct** because when an individual has access to material nonpublic information about a particular security, this situation may also form a [legal and regulatory] constraint. For example, the directors of a public company may need to refrain from trading the company's stock at certain points of the year before financial results are published. The IPS should note this constraint so that the portfolio manager does not inadvertently trade the stock on the client's behalf.

Answer 86 of 100

Answer

Solution

A. Incorrect because when defining asset classes paired correlations of assets should be relatively high within an asset class.

B. Incorrect because when defining asset classes paired correlations of assets should be relatively high within an asset class.

C. **Correct** because when defining asset classes an asset class should contain relatively homogeneous assets and paired correlations of assets should be relatively high within an asset class.

Answer 87 of 100

Answer

Solution

A. Incorrect because it confuses σ_i and σ_m and calculates $\rho_{im} = \beta_i \times \sigma_i / \sigma_m = 0.35 \times 0.12 / 0.18 = 0.233 \approx 0.2$. This answer is also obtained if, as an additional mistake, variances are used; $\rho_{im} = \beta_i \times \sigma_i^2 / \sigma_m^2 = 0.35 \times (0.12)^2 / (0.18)^2 = 0.156 \approx 0.2$.

B. **Correct** because $\beta_i = \rho_{im} \times \sigma_i / \sigma_m$, where ρ_{im} denotes the correlation between the asset returns and the market returns, and σ_i and σ_m denote the standard deviation of the asset returns and the market returns, respectively. Thus, correlation $\rho_{im} = \beta_i \times \sigma_m / \sigma_i = 0.35 \times 0.18 / 0.12 = 0.525 \approx 0.5$.

Incorrect because it uses variances instead of standard deviations in the correct formula; $\rho_{im} = \beta_i \times \sigma_m^2 / \sigma_i^2 = 0.35 \times (0.18)^2 / (0.12)^2 = 0.788 \approx 0.8$.

Answer 88 of 100

Answer

Solution

- A. Incorrect because with a probability of 5% and a measurement period of one day, we can interpret the bank's VaR as expecting a minimum loss of £5 million once every 20 business days.
- B. Incorrect because with a probability of 5% and a measurement period of one day, we can interpret the bank's VaR as expecting a minimum loss of £5 million once every 20 business days. Based on a year with 250 to 260 business days, this should occur 12.5 to 13 times per year (roughly once per month).
- C. **Correct because with a probability of 5% and a measurement period of one day, we can interpret the bank's VaR as expecting a minimum loss of £5 million once every 20 business days.**

Answer 89 of 100

Answer

Solution

- A. Incorrect because risk management is not even about predicting risks. 'The Doctrine of No Surprises' is a key mantra among many risk managers, but it does not mean they are expected to predict what will happen. For example, a risk manager of a bank would not have been expected to know that a real estate crisis was going to occur and cause significant defaults on the bank's real estate securities.
- B. **Correct** because a good risk management process would include a deep discussion at the governance level about the balance between the likely returns and the unlikely—but sizable—losses and whether such losses are tolerable.
- C. Incorrect because a poor risk management process would have ignored the possibility, though small, of such a significant market event and not quantified the potential loss.

Answer 90 of 100

Answer

Solution

- A. Incorrect because it is portfolio variance instead of standard deviation:
- $$w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2$$

$$\begin{aligned} &= (0.4)^2(0.15)^2 + (0.6)^2(0.18)^2 + 2(0.4)(0.6)(0.2)(0.15)(0.18) \\ &= (0.16)(0.0225) + (0.36)(0.0324) + (0.002592) = 0.017856 \approx 1.8\%. \end{aligned}$$

- B. Incorrect because it uses standard deviations instead of variances in the formula for the portfolio standard deviation:

$$\begin{aligned} &w_1^2 \sigma_1 + w_2^2 \sigma_2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{1,2} \\ &= (0.4)^2(0.15) + (0.6)^2(0.18) + 2(0.4)(0.6)(0.20)(0.15)(0.18) \\ &= (0.024) + (0.0648) + (0.002592) \\ &= 0.091392 \approx 9.1\% \end{aligned}$$

- C. **Correct** because $\sigma_{\text{port}} = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{1,2} \sigma_1 \sigma_2}$

$$\begin{aligned}
&= \sqrt{(0.4)^2 (0.15)^2 + (0.6)^2 (0.18)^2 + 2(0.4)(0.6)(0.2)(0.15)(0.18)} \\
&= \sqrt{(0.16) (0.0225) + (0.36) (0.0324) + 0.002592} \\
&= \sqrt{0.017856} = 0.133626 \approx 13.4\%.
\end{aligned}$$

Answer 91 of 100

Answer

Solution

- A. **Correct** because, although the investor has a time horizon of 15 years, she has liquidity needs in one year. When the client does have such a requirement, the manager should allocate part of the portfolio to cover the liability. This part of the portfolio will be invested in assets that are liquid—that is, easily converted to cash—and low risk at the point in time the liquidity need is actually present (e.g., a bond maturing at the time when private education expenses will be incurred), so that their value is known with reasonable certainty. Commercial paper is a short-term, negotiable, unsecured promissory note that represents a debt obligation of the issuer. Thus, commercial paper being a short-term investment, it is more suitable compared to private equity and large-capitalization stocks to be included in the portion of the investor's portfolio that has the short-term liquidity need.
- B. **Incorrect** because, although the investor has a time horizon of 15 years, she has liquidity needs in one year. When the client does have such a requirement, the manager should allocate part of the portfolio to cover the liability. This part of the portfolio will be invested in assets that are liquid—that is, easily converted to cash—and low risk at the point in time the liquidity need is actually present (e.g., a bond maturing at the time when private education expenses will be incurred), so that their value is known with reasonable certainty. Private equity securities are not listed on public exchanges and have no active secondary market. They are issued primarily to institutional investors via non-public offerings, such as private placements. Being illiquid, they are not suitable to be included in the portion of the client's portfolio that has the short-term liquidity need.
- C. **Incorrect** because, although the investor has a time horizon of 15 years, she has liquidity needs in one year. When the client does have such a requirement, the manager should allocate part of the portfolio to cover the liability. This part of the portfolio will be invested in assets that are liquid—that is, easily converted to cash—and low risk at the point in time the liquidity need is actually present (e.g., a bond maturing at the time when private education expenses will be incurred), so that their value is known with reasonable certainty. Thus, large-capitalization stocks, being a more volatile investment than commercial paper, would be less suitable to be included in the portion of the investor's portfolio that has the short-term liquidity need.

Answer 92 of 100

Answer

Solution

- A. **Correct** because, for a given Markowitz efficient frontier, a different risk-free rate will result in a different tangent to the frontier, hence a different optimal risky portfolio. CAL(P) is the optimal capital allocation line and Portfolio P is the optimal risky portfolio. Thus, with the addition of the risk-free asset, we are able to narrow our selection of risky portfolios to a single optimal risky portfolio, P, which is at the tangent of CAL(P) and the efficient frontier of risky assets.

- B. Incorrect because the optimal risky portfolio is selected from numerous risky portfolios without considering the investor's preferences.
- C. Incorrect because all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk-free asset and an optimal portfolio of risky assets. The investor's optimal portfolio thus lies on the capital allocation line (CAL), not on the Markowitz efficient frontier. If it is optimal for an investor to allocate no wealth to the risk-free asset, the investor's optimal portfolio will lie on the Markowitz efficient frontier. However, this is not 'always' the case.

Answer 93 of 100

Answer

Solution

- A. **Correct** because, given the beta of an asset (β_i), the covariance between the asset returns (R_i) and the market portfolio returns (R_m) is given by: $\text{Cov}(R_i, R_m) = \beta_i \times \sigma_m^2 = 1.8 \times 0.35^2 = 0.22$.
- B. Incorrect because it is the systematic variance of the portfolio; $\beta_i^2 \times \sigma_m^2 = 1.8^2 \times 0.35^2 = 0.40$.
- C. Incorrect because it is the correlation coefficient, not the covariance, between the asset returns and the market portfolio returns; $\rho_{im} = \beta_i \times \sigma_m / \sigma_i = 1.8 \times 0.35 / 0.70 = 0.90$. It is also the closest answer if the asset return variance is used in the correct formula instead of the market return variance; $1.8 \times 0.70^2 = 0.88$.

Answer 94 of 100

Answer

Solution

- A. Incorrect because it uses an incorrect CAPM equation to calculate the expected return of a security as $E(R_i) = R_f + \beta_i \times E(R_m)$, such that $E(R_m) = [E(R_i) - R_f] / \beta_i = (0.11 - 0.02) / 1.5 = 0.06 = 6\%$. Since the market risk premium is $R_m - R_f$, it follows that the market risk premium is calculated as $6\% - 2\% = 4\%$.
- B. **Correct** because, according to the CAPM, the expected return of a security is $E(R_i) = R_f + \beta_i [E(R_m) - R_f]$, such that $[E(R_m) - R_f] = [E(R_i) - R_f] / \beta_i = (0.11 - 0.02) / 1.5 = 0.06 = 6\%$. The market risk premium is $E(R_m) - R_f = 6\%$.
- C. Incorrect because it uses an incorrect CAPM equation to calculate the expected return of a security as $E(R_i) = \beta_i \times [E(R_m) - R_f]$, such that $[E(R_m) - R_f] = E(R_i) / \beta_i = 0.11 / 1.5 = 0.0733 \approx 7.3\%$. The market risk premium is incorrectly calculated as 7.3%.
This is also closest to the market return of 8% calculated according to the CAPM equation; $0.11 = 0.02 + 1.5 \times [E(R_m) - 0.02]$; $E(R_m) = 0.08 = 8\%$.

Answer 95 of 100

Answer

Solution

- A. Incorrect because

$$0.060 = \text{SR} \times (\sigma_p / \beta_p') = 0.8 \times (0.19 / 2.527),$$

where

$$\beta_p' = \text{incorrectly calculated portfolio beta} = \rho_{pm} (\sigma_p^2 / \sigma_m^2) = 0.7 \times (0.19^2 / 0.1^2) = 2.527.$$

- B. **Correct** because

$$\text{TR} = (R_p - R_f) / \beta_p = [(R_p - R_f) / \sigma_p] \times (\sigma_p / \beta_p) = \text{SR} \times (\sigma_p / \beta_p) = 0.8 \times (0.19 / 1.33) = 0.114,$$

where

$$\sigma_m = \text{standard deviation of market returns} = 0.10 \text{ (given);}$$

σ_p = standard deviation of portfolio returns = 0.19 (given);
 $\rho_{p,m}$ = correlation between market and portfolio = 0.7 (given);
 β_p = portfolio beta = $\rho_{p,m}(\sigma_p/\sigma_m) = 0.7 \times (0.19/0.1) = 1.33$ (per $\beta_i = \rho_{i,m}(\sigma_i/\sigma_m)$);
 R_p = portfolio's return;
 R_f = risk-free rate of interest;

TR = Treynor ratio;

SR = Sharpe ratio, where $SR = (R_p - R_f)/\sigma_p$.

C. Incorrect because

$$0.413 = SR \times (\sigma_p/\beta_p^*) = 0.8 \times (0.19/0.368),$$

where

$$\beta_p^* = \text{incorrectly calculated portfolio beta} = \rho_{p,m}(\sigma_m/\sigma_p) = 0.7 \times (0.1/0.19) = 0.368.$$

Answer 96 of 100

Answer

Solution

A. Incorrect because the equally weighted portfolio's return is the same as the return on the randomly selected security. Thus, a portfolio's expected return is not influenced by the correlations between assets. The portfolio return varies with weights but is unaffected by the correlation coefficient.

B. Incorrect because the standard deviation of an equally weighted portfolio is not simply the average of the standard deviations of the individual shares. Such a portfolio offers a lower standard deviation of return than the average of its individual components due to the correlations or interactions between the individual securities. Thus, a portfolio's standard deviation of returns is influenced by the correlations between assets.

For example, the standard deviation of a two-asset portfolio simplifies to the following:

$\sigma_p = \sqrt{(w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2)}$. It follows that an increase in the correlation among assets will result in an increase in the portfolio's standard deviation, not a decrease.

C. **Correct** because, as the correlations between asset returns increase, the diversification benefit provided by portfolios decreases. A major reason that portfolios can effectively reduce risk is that combining securities whose returns do not move together provides diversification. However, an important issue is that the co-movement or correlation pattern of the securities' returns in the portfolio can change in a manner unfavorable to the investor. When we examine the returns of a set of global equity indexes over the last 15 years, we observe a reduction in the diversification benefit due to a change in the pattern of co-movements of returns. The degree to which these global equity indexes move together has increased over time.

The lesson is that although portfolio diversification generally does reduce risk, it does not necessarily provide the same level of risk reduction during times of severe market turmoil as it does when the economy and markets are operating 'normally'.

Answer 97 of 100

Answer

Solution

A. Incorrect because it adjusts for inflation in the wrong direction, i.e. it assumes that, in order to reach \$500,000 in 15 years' time, we must equate \$500,000 to $\$150,000 \times 1.02^{15} \times (1 + r)^{15}$, which yields: $r = [\$500,000/(\$150,000 \times 1.02^{15})]^{1/15} - 1 = 0.062 = 6.2\%$.

Alternatively, after calculating the nominal return via $\$150,000(1 + r)^{15} = \$500,000$, it then adjusts for inflation by subtracting, rather than adding, the expected inflation rate from the nominal rate; $[\$500,000/\$150,000]^{1/15} - 1 - 0.02 = 8.4\% - 2.0\% = 6.4\%$, which is closest to this response.

B. Incorrect because it does not take into account the effect of inflation on the individual's retirement income needs. In other words, it assumes that the portfolio needs to grow to at least \$500,000 in 15 years' time, rather than to this amount adjusted upward for inflation. Hence, the minimum required rate of return, r , satisfies $\$150,000(1 + r)^{15} = \$500,000$, giving $r = [\$500,000/\$150,000]^{1/15} - 1 = 0.084 = 8.4\%$. Calculator solution: $N = 15$; $PV = -150,000$; $FV = 500,000$; compute $I/Y = 8.4\%$.

C. **Correct** because, at 2% annual inflation, \$500,000 in today's money equates to $\$500,000(1.02)^{15} = \$672,934$ in 15 years' time. Since no further contributions to the retirement fund will be made, the current savings of \$150,000 must grow to at least \$672,934 in 15 years' time to meet the individual's retirement income objective. Hence, the minimum required rate of return, r , satisfies $\$150,000(1 + r)^{15} = \$672,934$, giving $r = [\$672,934/\$150,000]^{1/15} - 1 = 0.105 = 10.5\%$. Calculator solution: (1) $N = 15$; $I/Y = 2\%$; $PV = -150,000$; compute $FV = 672,934$. (2) $N = 15$; $PV = -150,000$; $FV = 672,934$; compute $I/Y = 10.5\%$.

Alternatively, a candidate could compute the nominal required rate of return via $\$150,000(1 + r)^{15} = \$500,000$, giving $r = [\$500,000/\$150,000]^{1/15} - 1 = 0.084 = 8.4\%$, and then adjust for inflation as follows: $(1 + 0.084) \times (1 + 0.02) - 1 = 0.105 = 10.5\%$. If a candidate adjusted for inflation by adding the expected inflation rate, they would also arrive at (approximately) the **Correct** answer; $8.4\% + 2.0\% = 10.4\%$.

Answer 98 of 100

Answer

Solution

A. Incorrect because for portfolios with a large number of assets, covariance among the assets accounts for almost all of the portfolio's risk.

B. Incorrect because the contribution of the average variance of the individual assets to the total portfolio variance becomes negligible as the number of assets becomes large. Furthermore, for portfolios with a large number of assets, covariance among the assets accounts for almost all of the portfolio's risk.

C. **Correct** because, given equal weights and average variance/covariance (denoted σ^2 and Cov , respectively), we can rewrite the portfolio variance as $\sigma_p^2 = \sigma^2/N + [(N - 1)/N]\text{Cov}$. The equation shows that as N becomes large, the first term on the right side with the denominator of N becomes smaller and smaller, implying that the contribution of one asset's variance to portfolio variance gradually becomes negligible. The second term, however, approaches the average covariance as N increases. It is reasonable to say that for portfolios with a large number of assets, covariance among the assets accounts for almost all of the portfolio's risk.

Answer 99 of 100

Answer

Solution

A. Incorrect because, in the context of risk management, certain factors should *not* determine risk tolerance, but in many cases they do. Personal motivations, beliefs, and agendas of board members (the agency problem); company size; whether the market environment seems stable; short-term pressures; and management compensation often affect risk tolerance in ways that might not be in line with the owners' best interests.

B. Incorrect because, in the context of risk management, risk budgeting picks up where risk tolerance leaves off. Whereas risk tolerance focuses on the appetite for

risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken. Risk budgeting quantifies and allocates the tolerable risk by specific metrics; it extends and guides implementation of the risk tolerance decision.

C. **Correct** because, in the context of risk management, factors such as a company's goals, its expertise in certain areas, and its strategies will help a board determine which risks the company may pursue and with how much intensity. The government and regulatory landscape is important too, both in their *ex ante* demands on how companies approach risk and in the likely *ex post* reaction in the event of disasters.

Answer 100 of 100

Answer

Solution

A. **Correct** because the optimal portfolio maximizes the return per unit of risk (as it is on the capital allocation line), and it simultaneously supplies the investor with the most satisfaction (utility).

B. Incorrect because, as the investors have different degrees of risk aversion, they will not have the same optimal portfolio. The optimal portfolio maximizes the return per unit of risk (as it is on the capital allocation line), and it simultaneously supplies the investor with the most satisfaction (utility).

C. Incorrect because the less risk-averse investor is willing to accept the additional risk.

Therefore, the more risk-averse Investor 1 will have an optimal portfolio with less risk and a lower expected return than Investor 2. The optimal portfolio maximizes the return per unit of risk (as it is on the capital allocation line), and it simultaneously supplies the investor with the most satisfaction (utility).

Quantitative Methods- Practice Pack Ques

Question 1 of 90

Question

Which of the following factors is **not** used in the calculation of a confidence interval?

- A. Point estimate
- B. Sampling error
- C. Reliability factor

Question 2 of 90

Question

An analyst performs a simple linear regression of a stock's monthly return on the monthly return of a market index (both in %) and gathers the following information:

Estimated slope	1.0
Estimated intercept	1.2%
Standard error of the forecast	1.4%
Critical t -values at a 5% significance level	± 2.032

The 95% prediction interval for the stock's monthly return, given that the forecasted monthly return on the index is 3.5%, is *closest* to:

- A. 0.7% to 6.3%.
- B. 1.9% to 7.5%.
- C. 3.3% to 6.1%.

Question 3 of 90

Question

An investor purchases a stock for \$100. Immediately after receiving a dividend of \$7, the investor sells the stock for \$107. The holding period return of the investment is *closest* to:

- A. 0%.
- B. 7%.
- C. 14%.

Question 4 of 90

Question

For a sample of 50 observations, in which of the following situations is a nonparametric test *least likely* to be appropriate? The data:

- A. contain outliers.
- B. are given in ranks.
- C. come from a population with a lognormal distribution.

Question 5 of 90

Question

Which of the following test statistics is *most* appropriate for a hypothesis test concerning the mean difference between two normally distributed populations?

- A. t -statistic
- B. F -statistic
- C. Chi-square statistic

Question 6 of 90

Question

An asset earns 13.1% over a 16-month period. The asset's annualized compound rate of return is *closest* to:

- A. 9.3%.
- B. 9.7%.
- C. 9.8%.

Question 7 of 90

Question

A company estimates its revenue will be 50% higher than today in four years' time. The compound annual growth rate is *closest* to:

- A. 10.7%.
- B. 11.8%.
- C. 12.5%.

Question 8 of 90

Question

An investor purchased a stock for \$450 and then sold the stock immediately after receiving a dividend of \$2. If the holding period return is a loss of 10.2%, the investor sold the stock at a price *closest* to:

- A. \$402.
- B. \$404.
- C. \$406.

Question 9 of 90

Question

An investor considers the following certificates of deposit (CDs) available for purchase at face value:

CD Interest Rate

1 2.2%

2 3.3%

3 4.4%

If each CD has the same maturity and default risk, the opportunity cost of investing in CD 1 is *closest* to:

- A. 0.0%.
- B. 1.1%.
- C. 2.2%.

Question 10 of 90

Which of the following statements is *most* accurate with respect to the widespread adoption of algorithmic trading in financial markets?

- A. The need for low-latency networks has grown.
- B. Markets have become less fragmented in terms of trading venues.
- C. Average trade size has increased as algorithmic trading is used to execute large institutional orders.

Question 11 of 90

Question

The failure of machine learning models to accurately predict outcomes can be the result of:

- A. overfitting, but not underfitting.
- B. underfitting, but not overfitting.
- C. either overfitting or underfitting.

Question 12 of 90

Question

An investment pays \$1,000 annually for five years, with the first payment occurring three years from today. If the discount rate is 6% compounded annually, the present value of the investment today is *closest* to:

- A. \$3,537.
- B. \$3,749.
- C. \$4,212.

Question 13 of 90

Question

Which of the following is *most likely* an advantage of traditional financial advisers over fully automated digital wealth managers?

- A. Lower account minimums
- B. Dividend reinvestment options
- C. Solutions that better address the needs of complex portfolios

Question 14 of 90

To test whether a population's mean, μ , is greater than zero, the alternative hypothesis should be formulated as:

- A. $\mu \leq 0$.
- B. $\mu \geq 0$.
- C. $\mu > 0$.

Question 15 of 90

Question

A test of independence is based on the data in a contingency table with 5 rows and 4 columns. Using a nonparametric test statistic that is chi-square distributed, the number of degrees of freedom is:

- A. 7.
- B. 12.
- C. 20.

Question 16 of 90

Question

In a parametric test of the correlation between two variables with a sample size of 51 and sample correlation of 0.6, the *t*-statistic is *closest* to:

- A. 0.07.
- B. 5.25.
- C. 6.64.

Question 17 of 90

Question

If a unimodal return distribution is negatively skewed, which of the following *most likely* has the highest value?

- A. Mean
- B. Mode
- C. Median

Question 18 of 90

Question

A portfolio manager will invest €100,000 and is presented with the following information about three portfolios with normally distributed returns:

	Expected Annual Return	Standard Deviation of Returns
Portfolio 1	23%	15%
Portfolio 2	12%	6%
Portfolio 3	15%	8%

If the manager wants to withdraw €5,000 in one year without invading initial capital, the safety-first optimal portfolio is:

- A. Portfolio 1.
- B. Portfolio 2.
- C. Portfolio 3.

Question 19 of 90

Question

Which of the following *best* describes when a transformation of the data may be needed to enable the use of a simple linear regression model? When the:

- A. dependent variable is non-normally distributed
- B. pairs of the dependent and independent variables are uncorrelated with one another
- C. relationship between the independent variable and the dependent variable is non-linear

Question 20 of 90

Question

For a continuous positively skewed unimodal distribution:

- A. both the mode and the median are less than the mean.
- B. both the mode and the median are greater than the mean.
- C. the mode is less than the mean and the median is greater than the mean.

Question 21 of 90
Question

In its broadest sense, fintech is *best* described as:

- A. the vast amount of data being generated by the financial services industry.
- B. the execution of investment strategies through computer-generated algorithms.
- C. technological innovation in the design and delivery of financial services and products.

Question 22 of 90
Question

An analyst estimates the following information from a simple linear regression:

Sum of squares error	280
Sum of squares regression	25
Number of paired observations	30

The standard error of the estimate is *closest* to:

- A. 2.5.
- B. 3.2.
- C. 10.0.

Question 23 of 90
Question

An analyst draws samples from an original sample to estimate the standard error of a population mean. Which of the following *best* describes this sampling procedure?

- A. Bootstrap method
- B. Cluster sampling method
- C. Convenience sampling method

Question 24 of 90
Question

An investor is considering buying a US T-bill. If the real risk-free rate is 1% and the inflation premium is 2%, the investor's opportunity cost of failing to make this investment is *closest* to:

- A. 1%.
- B. 2%.
- C. 3%.

Question 25 of 90
Question

An investor records the following information and transactions for a stock:

Year	Price per Share at Beginning of Year	Cash Flow at Beginning of Year
1	\$100	\$100 used to purchase one share
2	\$100	\$9 received in dividends and not reinvested
3	\$100	\$100 received from selling one share

The investor's money-weighted rate of return is *closest* to:

- A. 4.4%.
- B. 4.5%.

C. 4,6%.

Question 26 of 90

Question

The null hypothesis for the F -distributed test statistic in a simple linear regression model tests whether the:

- A. slope is equal to zero.
- B. intercept is equal to zero.
- C. slope is not equal to zero.

Question 27 of 90

Question

An analyst estimates the probabilities of three possible economic scenarios and the probabilities of a stock having a positive or a negative return in each scenario. These scenarios are *best* represented by a:

- A. tree-map.
- B. tree diagram.
- C. probability density function.

Question 28 of 90

Question

Which of the following is *most likely* used to detect sentiment shifts in an analyst's commentary?

- A. Tokenization
- B. Data curation
- C. Natural language processing

Question 29 of 90

Question

Which of the following visualizations is *most* appropriate for interpreting the correlation between two variables?

- A. Tree-map
- B. Scatter plot
- C. Clustered bar chart

Question 30 of 90

An investor has three options for receiving payments from an investment:

- Option 1: a single payment of \$136,000 today;
- Option 2: 30 annual payments of \$12,000, beginning one year from today;
- Option 3: 20 annual payments of \$13,000, beginning today.

If the annual discount rate is 8%, the option with the highest present value is:

- A. Option 1.
- B. Option 2.
- C. Option 3.

Question 31 of 90

Question

Roy's safety-first criterion:

- A. evaluates only downside risk.
- B. uses semideviation as a risk measure.
- C. assumes asset prices are normally distributed.

Question 32 of 90

Question

All else being equal, when compared to non-probability sampling, probability sampling *most likely* yields:

- A. a less representative sample.
- B. an equally representative sample.
- C. a more representative sample.

Question 33 of 90

Question

An analyst examines 30 paired monthly returns for two stock indexes. To determine if the mean difference of the returns is zero, the number of degrees of freedom of the *t*-test is: A. 28.

- B. 29.
- C. 58.

Question 34 of 90

An analyst calculates the following statistics for a sample with 100 observations:

	Value
First quartile	11
Second quartile	62
Third quartile	93
Fourth quartile	359

The interquartile range of the sample is equal to:

- A. 31.
- B. 82.
- C. 348.

Question 35 of 90

Question

The probability of correctly rejecting a null hypothesis is *best* defined as the:

- A. *p*-value.
- B. power of the test.
- C. level of significance.

Question 36 of 90

Question

An investor summarizes end-of-year cash outlays and proceeds from a two-year investment in a company's shares below:

Year	Outlays	Proceeds
0	\$100 to purchase the first share	---
1	\$110 to purchase the second share	\$10 dividend received from first share (not reinvested)
2	---	\$230 received from selling two shares at \$115 per share

The annualized time-weighted rate of return of the investment over the two-year period is
closest to:

- A. 9.7%.
- B. 12.0%.
- C. 12.3%.

Question 37 of 90

Question

Ranked in ascending order, the 19th observation in a sample of 75 is in the second:

- A. decile.
- B. quintile.
- C. quartile.

Question 38 of 90

Question

For a given dataset with different non-negative observations, which of the following will have the largest value?

- A. Harmonic mean
- B. Arithmetic mean
- C. Geometric mean

Question 39 of 90

Question

Sampling error is the difference between the observed value of a:

- A. random variable and the respective statistic.
- B. random variable and its hypothesized value.
- C. statistic and the quantity it is intended to estimate.

Question 40 of 90

Question

The central limit theorem:

- A. requires that the population be approximately normally distributed.
- B. implies that the sample mean is a consistent estimator of the population mean.
- C. states that the product of independent random variables is normally distributed.

Question 41 of 90

Question

An analyst assumes that a company's future EPS will be either \$2.00, \$2.20, or \$2.40. If each scenario is equally likely, the variance [in \$²] of the company's future EPS is *closest* to:

- A. 0.03.
- B. 0.16.
- C. 0.20.

Question 42 of 90
Question

An analyst observes the following EPS for four companies: -£0.50, £0.50, £2.50, and £5.50. The 50th percentile of the EPS values is *closest* to:

- A. £1.50.
- B. £2.00.
- C. £2.50.

Question 43 of 90
Question

The standard error of the estimate in a simple linear regression is *best* described as:

- A. a relative measure of fit for the regression.
- B. the percentage of the variation of the dependent variable that is explained by the independent variable.
- C. a measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression.

Question 44 of 90
Question

An analyst gathers the following information about a portfolio's returns:

**Year Return
(%)**

- | | |
|---|----|
| 1 | 6% |
| 2 | 7% |
| 3 | 3% |
| 4 | 2% |
| 5 | 4% |

If the target return is 5%, the target downside deviation is *closest* to:

- A. 1.7%.
- B. 1.9%.
- C. 2.2%.

Question 45 of 90
Question

Which of the following measures *best* quantifies the amount of risk per unit of mean return?

- A. Sharpe ratio
- B. Standard deviation
- C. Coefficient of variation

Question 46 of 90
Question

An analyst gathers the following returns for seven funds:

12% 7% 5% 4% 8% 3% 3%

The second quartile return is:

- B. 4%.
- C. 5%.
- 6%.

Question 47 of 90

Question

The correlation coefficient:

- A. ranges from 0 to 1.
- B. is not affected by outliers.
- C. indicates the strength of the linear relationship between two random variables.

Question 48 of 90

Question

If the covariance between two positively correlated random variables remains the same but the variance of both variables increases, the correlation between the two variables:

- A. decreases.
- B. stays the same.
- C. increases.

Question 49 of 90

Question

The correlation between two variables measures:

- A. only their linear relationship.
- B. only their non-linear relationship.
- C. both their linear and non-linear relationships.

Question 50 of 90

Question

Which of the following is required to compute the standard error of a sample mean using the bootstrap resampling method?

- A. The mean of each resample
- B. The mean of the original sample
- C. The standard deviation of the original sample

Question 51 of 90

Question

A discrete random variable X has the following probability distribution:

Probability Outcome	
y	
0.20	35
0.30	50
0.50	80

The standard deviation of X is closest to:

- A. 18.73.
- B. 20.00.
- C. 22.91.

Question 52 of 90

Question

A tree diagram contains the following information about the dividend per share payable by a company under two scenarios:

Scenario	Probability of Scenario	Dividend per Share	Probability of Dividend
		\$2.00	0.80
Favorable	0.60	\$1.50	0.20
		\$0.75	0.30
Unfavorable	0.40	\$0.50	0.70

The expected dividend per share under the favorable scenario is *closest* to: A.

- A. \$1.14.
- B. \$1.37.
- C. \$1.90.

Question 53 of 90

Question

An investor is considering two term deposits with the following characteristics:

	Term Deposit 1	Term Deposit 2
Compounding frequency	Quarterly	Continuous
Stated annual rate	4%	?

The stated annual rate for Term Deposit 2 that should make the investor indifferent between the two term deposits is *closest* to:

- A. 3.92%.
- B. 3.98%.
- C. 4.06%.

Question 54 of 90

Question

An investor invests a fixed amount of money into a fund each year for three years as follows:

Year	Price per Share
1	€14.00
2	€12.00
3	€17.00

The investor's average cost per share is *closest* to:

- A. €14.05.
- B. €14.33.
- C. €14.63.

Question 55 of 90

Question

The simple linear regression model in which only the independent variable is in logarithmic

form is *best* described as the:

- A. log-lin model.
- B. lin-log model.
- C. log-log model.

Question 56 of 90

Question

An analyst produces the following joint probability function for the returns on two companies, X and Y :

	Return		
	Return of $Y = 15\%$	of $Y = 10\%$	Return of $Y = 5\%$
Return of $X = 20\%$	0.2	0	0
Return of $X = 15\%$	0	0.4	0
Return of $X = 10\%$	0	0	0.4

The expected returns of companies X and Y are 14% and 9%, respectively. The covariance of returns between X and Y (in percent squared) is *closest* to:

- A. 0.
- B. 5.
- C. 14.

Question 57 of 90

Question

An analyst gathers the following sample returns for a security:

Return

-2%

-1%

1%

2%

The mean absolute deviation of the sample returns is:

- A. less than the sample standard deviation.
- B. equal to the sample standard deviation.
- C. greater than the sample standard deviation.

Question 58 of 90

Question

An investor gathers the following information about a stock:

Stock price at $t = 0$ \$20

Dividend paid at $t = 1$ \$3

Stock price at $t = 1$ \$12

Dividend paid at $t = 2$ \$1

Stock price at $t = 2$ \$20

The investor purchased one unit of the stock at $t = 0$ and sold it at $t = 2$. If the dividends were not reinvested, the money-weighted rate of return is *closest* to:

- A. 10%.
- B. 15%.
- C. 20%.

Question 59 of 90

Which of the following statements is *most* accurate? The money-weighted return:

- A. ignores cash withdrawals and additional cash investments.
- B. measures what the investor actually earned on the funds invested.
- C. should be used to compare the performance of different investment managers.

Question 60 of 90

Question

In evaluating portfolio performance, the return measure *most* affected by an addition of funds to the portfolio just before a market downturn is the:

- A. time-weighted return.
- B. arithmetic mean return.
- C. money-weighted return.

Question 61 of 90

Question

An equally weighted portfolio consists of two securities, each with a standard deviation of 3%. If the two securities' returns are uncorrelated, the portfolio's standard deviation is *closest* to:

- A. 0.0%.
- B. 2.1%.
- C. 3.0%.

Question 62 of 90

Question

An investment requires 10 equal annual payments, starting today, and will pay out a lumpsum of \$500,000 15 years from today. If the interest rate is 4% per year compounded annually, the required annual payment is *closest* to:

- A. \$32,913.
- B. \$34,230.
- C. \$40,044.

Question 63 of 90

Question

The probability of correctly rejecting a false null hypothesis is *best* described as one minus the:

- A. test statistic's *p*-value.
- B. probability of a Type I error.
- C. probability of a Type II error.

Question 64 of 90

Question

All else being equal, which of the following would *most likely* lead to a wider prediction interval for the dependent variable when re-estimating a linear regression model? An increase in the:

- A. sample size
- B. level of significance
- C. standard error of the estimate

Question 65 of 90

Question

An investor needs to make the following payments to cover college tuition fees, starting 10 years from today:

Annual fee (payable at the beginning of each year) \$50,000

Number of years of fee payments 4

If the investor's annual discount rate is 3%, the minimum investment amount required today to fund all four years of college tuition is *closest* to:

- A. \$138,294.
- B. \$142,442.
- C. \$146,716.

Question 66 of 90

Question

An annuity makes seven annual payments of \$10,000 each, with the first payment occurring five years from today. If the discount rate is 6% per year, the value of the annuity today is *closest* to:

- A. \$41,715.
- B. \$44,218.
- C. \$55,824.

Question 67 of 90

Question

With respect to simple linear regression, a residual is *best* described as the difference between the observed value of a dependent variable and:

- A. its mean.
- B. its estimated value using a fitted regression line based on the sample.
- C. its expected value based on the true underlying population relationship.

Question 68 of 90

Question

The time preference of individuals for current instead of future real consumption is captured by the:

- A. liquidity premium.
- B. maturity premium.
- C. real risk-free interest rate.

Question 69 of 90

Question

A portfolio has a mean return of 1.0% and a standard deviation of returns of 2.7%. If the specified minimum target return is 1.0%, the sample target semideviation is:

- A. less than 2.7%.
- B. equal to 2.7%.
- C. greater than 2.7%.

Question 70 of 90

Question

An analyst discards the lowest 2.5% and the highest 2.5% of values in a sample, and computes the mean of the remaining 95% of values. The resulting mean is *best* described as

- A. trimmed mean.
- B. harmonic mean.
- C. winsorized mean.

Question 71 of 90
Question

If a stock's continuously compounded return is normally distributed, the future stock price is *most likely*:

- A. normally distributed.
- B. uniformly distributed.
- C. lognormally distributed.

Question 72 of 90
Question

An analyst considers the population of all existing stocks and selects those where the company name starts with the letter P. This sampling procedure is *most likely* an example of:

- A. systematic sampling.
- B. non-probability sampling.
- C. two-stage cluster sampling.

Question 73 of 90
Question

A bank offers a savings account with a stated annual rate of 3% in the first year and 5% in the second year. If returns are compounded quarterly and €90,000 is deposited in the account at the beginning of the first year, the account's value at the end of the second year is *closest* to:

- A. €97,200.
- B. €97,335.
- C. €97,455.

Question 74 of 90
Question

The lognormal distribution:

- A. is unbounded.
- B. is asymmetrical.
- C. has the same mean as that of its associated normal distribution.

Question 75 of 90
Question

In simple linear regression analysis, the total sum of squares *best* describes:

- A. a scatter plot.
- B. the variation of the dependent variable.
- C. a paired observation between variables.

Question 76 of 90

Question

A pension fund needs to pay a lump sum \$10,000,000 to its participants in 15 years. If the fund is expected to earn 5% per year compounded semi-annually, the amount needed today to meet its liability in 15 years is *closest* to:

- A. \$4,767,427.
- B. \$4,810,171.
- C. \$4,892,771.

Question 77 of 90

Question

Which of the following is an underlying assumption of the simple linear regression model? The regression residuals:

- A. are normally distributed.
- B. have high correlations across observations.
- C. have different variances across observations.

Question 78 of 90

Question

For a set of return observations, the coefficient of variation is *best* described as a measure of:

- A. risk per unit of mean return.
- B. mean excess return earned per unit of risk.
- C. average absolute deviation around the mean return.

Question 79 of 90

Question

An analyst performs a hypothesis test concerning the difference between the mean returns of two portfolios, assuming normally distributed populations with unknown but equal variances. If the analyst decides to change the hypothesized difference in mean returns from 0% to 1%, which of the following will change?

- A. The value of the test statistic
- B. The degrees of freedom used in the test
- C. The pooled estimate of the common population variance

Question 80 of 90

An analyst tabulates the ranks of four paired observations of random variables X and Y as follows:

Observation	Rank of X	Rank of Y
1	1	2
2	2	3
3	3	4
4	4	1

The Spearman rank correlation coefficient between X and Y is *closest* to:

- A. 0.2.
- B. 0.8.
- C. 1.0.

Question 81 of 90

Question

An analyst runs a simple linear regression to test whether the variation in the demand for corn explains the variation in the supply of wheat. In this model, the supply of wheat is a(n):

- A. indicator variable.
- B. explained variable.
- C. independent variable.

Question 82 of 90

Question

A return distribution with negative skew and a mean of zero *most likely* has:

- A. frequent small gains and a few extreme losses.
- B. frequent small losses and a few extreme gains.
- C. frequent extreme losses and a few small gains.

Question 83 of 90

Question

If the relationship between the dependent variable and independent variable is linear, the regression residuals when plotted against the independent value should appear to:

- A. be linear.
- B. be random.
- C. follow a pattern.

Question 84 of 90

Question

In which of the following cases is cluster sampling *most likely* used? When:

- A. conducting a market survey
- B. auditing financial statements
- C. creating a bond portfolio to mirror the performance of a specified index

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Question

Samples are drawn from a population that follows a binomial distribution with a probability of success on a trial of 0.3. According to the central limit theorem, as the sample size increases, the distribution of the sample mean approaches a:

- A. negatively skewed distribution.
- B. symmetric distribution.
- C. positively skewed distribution.

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Question

A graphical depiction of a continuous distribution shows the left tail to be longer than the right tail. The distribution is *best* described as having:

- A. negative skewness.
- B. leptokurtosis.
- C. positive skewness.

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Question

A nonparametric test is *most* appropriate when:

- A. comparing differences between means.

- B. data are given in ranks.
- C. data meet distributional assumptions.

Question 88 of 90

Question

If a client controls the timing of cash flows into and out of a portfolio, which of the following is *most* appropriate when evaluating the performance of the portfolio manager?

- A. Time-weighted rate of return
- B. Arithmetic mean rate of return
- C. Money-weighted rate of return

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Question

Grouping all publicly traded US firms by sector and then randomly selecting subsamples of firms from each sector according to the sector's proportion in the total population is an example of:

- A. cluster sampling.
- B. simple random sampling.
- C. stratified random sampling.

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Question

In hypothesis testing, which of the following *best* describes a Type II error?

- A. Rejecting a true null hypothesis
- B. Rejecting a false null hypothesis
- C. Failure to reject a false null hypothesis

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Answer 1 of 90

Answer

Solution

- A. Incorrect because a point estimate is used to calculate a confidence interval; Point estimate $\pm$ Reliability factor $\times$ Standard error = confidence interval.
- B. **Correct** because a confidence interval for a parameter is calculated as: Point estimate $\pm$ Reliability factor $\times$ Standard error, where standard error is the standard error of the sample statistic providing the point estimate. Thus, sampling error is not part of the calculation. Sampling error is the difference between the observed value of a statistic and the quantity it is intended to estimate. It is because of sampling error that confidence intervals are used.
- C. Incorrect because a reliability factor is used to calculate a confidence interval; Point estimate $\pm$ Reliability factor $\times$ Standard error = confidence interval.

Answer 2 of 90

Answer

Solution

- A. Incorrect because it uses the forecasted value of the independent variable to construct the prediction interval rather than the predicted value of the dependent variable. In other words, it assumes that the interval is given by $Xf \pm t_{\text{critical for } \alpha/2, S_f} \times 3.5\% \pm 2.032 \times 1.4\% \approx (0.7\%, 6.3\%)$.
- B. **Correct** because a forecasted value of the dependent variable, Yf , is determined using the estimated intercept and slope, as well as the expected or forecasted independent variable, X_f : $Yf = b_0 + b_1 X_f$, where b_0 and b_1 are the estimated intercept and slope coefficients, respectively. Hence, $Yf = 1.2\% + 1.0 \times 3.5\% = 4.7\%$.
Next, the prediction interval is $Yf \pm t_{\text{critical for } \alpha/2, S_f} \times s_f$, where s_f denotes the standard error of the forecast. Hence, the prediction interval is given by: $4.7\% \pm 1.4\% \times 2.032 \approx (1.9\%, 7.5\%)$.
- C. Incorrect because it neglects the critical t -values when constructing the prediction interval. In other words, it assumes that the interval is given by $Yf \pm s_f \times 1.4\% = (3.3\%, 6.1\%)$.

Answer 3 of 90

Answer

Solution

- A. Incorrect because it incorrectly subtracts the dividend instead of adding it, and thus calculates $HPR = (\$107 - \$100 - \$7)/\$100 = \$0/\$100 = 0\%$. The same result is obtained if the holding period return is calculated as $R = (\$100 - \$107 + \$7)/\$107 = \$0/\$107 = 0\%$.
- B. Incorrect because it omits the dividend and calculates $R = (\$107 - \$100)/\$100 = \$7/\$100 = 7\%$. It is also the ratio of the dividend received over the initial investment; $\$7/\$100 = 7\%$.
- C. **Correct** because a holding period return is the return earned from holding an asset for a single specified period of time. This return can be generalized and shown as a mathematical expression in which P is the price and I is the income: $R = (P_1 - P_0 + D_1)/P_0$. Thus, $R = (\$107 - \$100 + \$7)/\$100 = \$14/\$100 = 14\%$.

Answer 4 of 90

Answer

Solution

- A. Incorrect because we primarily use nonparametric procedures in four situations:

(1) when the data we use do not meet distributional assumptions, (2) when there are outliers, (3) when the data are given in ranks or use an ordinal scale, or (4) when the hypotheses we are addressing do not concern a parameter. This is one of the situations

(situation 2) in which a nonparametric test would be appropriate.

B. Incorrect because we primarily use nonparametric procedures in four situations: (1) when the data we use do not meet distributional assumptions, (2) when there are outliers, (3) when the data are given in ranks or use an ordinal scale, or (4) when the hypotheses we are addressing do not concern a parameter. This is one of the situations (situation 3) in which a nonparametric test would be appropriate.

C. Correct because a nonparametric test would be less appropriate compared to other answers as in this case a parametric test can be used. We may want to test a hypothesis concerning the mean of a population but believe that neither t - nor z -distributed tests are appropriate because the sample is small and may come from a markedly non-normally distributed population. In that case, we may use a nonparametric test. In our case, the data sample is large, thus a parametric test can be used instead.

Answer 5 of 90

Answer

Solution

A. Correct because for a Test of Mean Differences (Normally Distributed Populations, Unknown Population Variances), when we have data consisting of paired observations from samples generated by normally distributed populations with unknown variances, a t -test is based on $t = (d - \mu d_0)/s_d$, with $n - 1$ degrees of freedom, where n is the number of paired observations, d is the sample mean difference, and s_d is the standard error of d .

B. Incorrect because for a Test of Mean Differences (Normally Distributed Populations, Unknown Population Variances), when we have data consisting of paired observations from samples generated by normally distributed populations with unknown variances, a t -test is based on $t = (d - \mu d_0)/s_d$, with $n - 1$ degrees of freedom, where n is the number of paired observations, d is the sample mean difference, and s_d is the standard error of d . An F -test can be appropriate for Tests Concerning Differences between the Variances of Two Populations.

C. Incorrect because for a Test of Mean Differences (Normally Distributed Populations, Unknown Population Variances), when we have data consisting of paired observations from samples generated by normally distributed populations with unknown variances, a t -test is based on $t = (d - \mu d_0)/s_d$, with $n - 1$ degrees of freedom, where n is the number of paired observations, d is the sample mean difference, and s_d is the standard error of d . In tests concerning the variance of a single normally distributed population (not the mean difference between two populations), we make use of a chi-square test statistic.

Answer 6 of 90

Answer

Solution

A. Incorrect because it is the compound rate of return per month times 12; $[(1 + 0.13100)^{(1/16)} - 1] \times 12 = 0.0077236 \times 12 = 0.09268 \approx 9.3\%$. This is also the geometric mean return per month times 12.

B. Correct because a general equation to annualize returns is given, where c is the number of periods in a year. For a quarter, $c = 4$ and for a month, $c = 12$: $r_{\text{annual}} = (1 + r_{\text{period}})^c - 1$. That is, for 16 months, $c = 12/16 = 0.75$ and the annualized return is $(1 + 0.13100)^{0.75} - 1 = 1.09672 - 1 = 0.09672 \approx 9.7\%$.

C. Incorrect because it is the arithmetic mean return per month times 12; $(0.13100/16) \times 12 = 0.0081875 \times 12 = 0.09825 \approx 9.8\%$.

Answer 7 of 90

Answer

Solution

A. Correct because a growth rate (g) is calculated as $g = (FV_N/PV)^{1/N} - 1$, where FV is the future value, PV is the present value and N is the number of periods. Here, $g = (1.5/1)^{1/4} - 1 = 0.10668 \approx 10.7\%$.

- B. Incorrect because it is calculated as $\ln(1 + 0.5/4) - 1 = \ln(1.125) - 1 = 0.11778 \approx 11.8\%$.
- C. Incorrect because it is calculated as: $50\%/4 = 12.5\%$.

Answer 8 of 90

Answer

Solution

A. **Correct** because a holding period return is the return earned from holding an asset for a single specified period of time.....This return can be generalized and shown as a mathematical expression in which P is the price and I is the income: $R = [(P_1 - P_0) + I_1]/P_0$. The subscript indicates the time of the price or income, ($t = 0$), is the beginning of the period and ($t = 1$) is the end of the period. Hence, $P_1 = R \times P_0 + P_0 - I_1 = -10.2\% \times \$450 + \$450 - \$2 = \$402.1 \approx \402 .

B. Incorrect because if the stock was sold for \$404, the holding period return would be $(\$404 - \$450 + \$2)/\$450 = -9.8\%$. The incorrect price of \$404 ignores the impact of the dividend; holding period return (HPR) $= (P_1 - P_0)/P_0$; $P_1 = \text{HPR} \times P_0 + P_0 = -10.2\% \times \$450 + \$450 = \$404.1 \approx \$404$.

C. Incorrect because if the stock was sold for \$406, the holding period return would be $(\$406 - \$450 + \$2)/\$450 = -9.3\%$. The incorrect price of \$406 incorrectly subtracts the dividend; holding period return (HPR) $= [(P_1 - P_0) - I_1]/P_0$; $P_1 = \text{HPR} \times P_0 + P_0 + I_1 = -10.2\% \times \$450 + \$450 + \$2 = \$406.1 \approx \406 .D.

Answer 9 of 90

Answer

Solution

A. Incorrect because the investor is foregoing higher rates of return by investing in the security with the lowest return.

B. Incorrect because 1.1% is not the most the investor is foregoing; however, it is the difference if incorrectly using the average return of the three securities.

C. **Correct** because all three securities have the same maturity and default risk so the investor is forgoing 2.2% ($4.4\% - 2.2\%$) by investing in CD 1 rather than investing in CD 3.

Answer 10 of 90

Solution

A. **Correct** because algorithmic trading requires access to low-latency networks, and with the wide-spread adoption of algorithmic trading, the need for low-latency networks has grown. Low-latency systems—systems that operate on networks that communicate high volumes of data with minimal delay (latency)—are essential for automated trading applications that make decisions based on real-time prices and market events. In contrast, high-latency systems do not require access to real-time data and calculations. High-frequency trading is a form of algorithmic trading that makes use of vast quantities of granular financial data (tick data, for example) to automatically place trades when certain conditions are met. Trades are executed on ultra-high-speed, low-latency networks in fractions of a second.

B. Incorrect because global financial markets have undergone substantial change as markets have fragmented into multiple trading destinations consisting of electronic exchanges, alternative trading systems, and so-called dark pools, and average trade sizes have fallen.

C. Incorrect because, although algorithmic trading is increasingly used to execute large institutional orders, it is slicing orders into smaller pieces and executing across different exchanges and trading venues. Global financial markets have undergone substantial change as markets have fragmented into multiple trading destinations consisting of electronic exchanges, alternative trading systems, and so-called dark pools, and average trade sizes have fallen.

Answer 11 of 90

Answer

Solution

A. Incorrect because underfitting also results in this problem. Underfitted models will typically fail to fully discover patterns that underlie the data" and thus may not be able to accurately predict outcomes. ML involves splitting the dataset into three distinct subsets: a training dataset, a validation dataset, and a test dataset. Once an algorithm has been

trained, validated, and tested, the ML model can be used to predict outcomes based on other datasets. ML models also require sufficiently large amounts of data and may not

perform well where there may not be enough available data to train and validate the model.

B. Incorrect because overfitting also results in this problem. An ML model that has been overfitted is not able to accurately predict outcomes using a different dataset and may be too complex.

C. **Correct** because an ML model that has been overfitted is not able to accurately predict outcomes using a different dataset and may be too complex. Also, underfitted models will typically fail to fully discover patterns that underlie the data and thus may not be able to accurately predict outcomes.

Answer 12 of 90

Answer

Solution

A. Incorrect because it recognizes the investment as a delayed annuity, but it wrongly assumes that the first annuity payment for an ordinary annuity starts at $t = 4$ instead of $t = 3$. Thus, it computes the present value of an ordinary annuity at $t = 3$ as $PV_3 = A[1 - 1/(1 + r)^n] / r = \$1,000 \times [1 - 1/(1 + 0.06)^5] / 0.06 = \$4,212.36$.

Using the present value formula for a lump sum to bring the single cash flow from $t = 3$ to $t = 0$, $PV_0 = FV_N(1 + r)^{-N} = \$4,212.36 (1 + 0.06)^{-3} = \$3,536.78 \sim \$3,537$.

Calculator solution:

(1) END mode; $N = 5$; $I = 6\%$; $PMT = -1,000$; $FV = 0$; solve for $PV = 4,212.36$.

(1) END mode; $N = 3$; $I = 6\%$; $PMT = 0$; $FV = 4,212.36$; solve for $PV = 3,536.78 \sim 3,537$.

B. **Correct** because by drawing a time line, the investment is recognized as a delayed annuity with the first payment starting at $t = 3$.

The first step is to compute the present value of an ordinary annuity at $t = 2$ because the first annuity payment is then one period away, as $PV_2 = A[1 - 1/(1 + r)^n] / r = \$1,000 \times [1 - 1/(1 + 0.06)^5] / 0.06 = \$4,212.36$.

Using the present value formula for a lump sum to bring the single cash flow from $t = 2$ to $t = 0$, $PV_0 = FV_N(1 + r)^{-N} = \$4,212.36 (1 + 0.06)^{-2} = \$3,748.99 \sim \$3,749$.

Calculator solution:

(1) END mode; $N = 5$; $I = 6\%$; $PMT = -1,000$; $FV = 0$; solve for $PV = 4,212.36$.

(2) END mode; $N = 2$; $I = 6\%$; $PMT = 0$; $FV = 4,212.36$; solve for $PV = 3,748.99 \sim 3,749$.

A second method to compute the present value of the investment is to recognize it as an annuity due with first payment at $t = 3$, and then discount back three periods using the present value formula for a lump sum. $PV_3 = \{A[1 - 1/(1 + r)^n] / r\}(1 + r) = \$1,000 \times \{[1 - 1/(1 + 0.06)^5] / 0.06\} \times (1 + 0.06) = \$4,465.11$.

Using the present value formula for a lump sum to bring the single cash flow from $t = 3$ to $t = 0$, $PV_0 = FV_N(1 + r)^{-N} = \$4,465.11 (1 + 0.06)^{-3} = \$3,748.99 \sim \$3,749$.

Calculator solution:

(1) BGN mode; $N = 5$; $I = 6\%$; $PMT = -1,000$; $FV = 0$; solve for $PV = 4,465.11$.

(2) END mode; $N = 3$; $I = 6\%$; $PMT = 0$; $FV = 4,465.11$; solve for $PV = 3,748.99 \sim 3,749$.

Another method to compute the **Correct** answer is to calculate the present value of a series of equal cash flows, with the first cash flow in the third year. Using a calculator with $CF_0 = 0$, $CF_1 = 0$, $CF_2 = 0$, $CF_3 = 1000$, $CF_4 = 1000$, $CF_5 = 1000$, $CF_6 = 1000$, $CF_7 = 1000$; $I = 6\%$; solve for $NPV = 3,748.99 \sim 3,749$.

C. Incorrect because it is the present value of the investment as $t = 2$ instead of $t = 0$, and it is an intermediate step in the correct calculation. It computes present value of an

ordinary annuity at $t = 3$ as $PV_3 = A[1 - 1/(1 + r)^n] / r = \$1,000 \times [1 - 1/(1 + 0.06)^5] / 0.06 = \$4,212.36$.

Calculator solution:

END mode; $N = 5$; $I = 6\%$; $PMT = -1,000$; $FV = 0$; solve for $PV = 4,212.36 \sim 4,212$.

Answer 13 of 90

Answer

Solution

A. Incorrect because robo-advisers typically have low fees and low account minimums, implementing their recommendations with low-cost, diversified index mutual funds or exchange-traded funds (ETFs). Fully automated digital wealth managers are one type of service in the robo-advice sector. The fully automated model does not rely on assistance from a human financial adviser. These services seek to offer a low-cost solution to investing and recommend an investment portfolio, which is often composed of ETFs.

B. Incorrect because the service package offered by fully automated digital wealth managers may include direct deposits, periodic rebalancing, and dividend reinvestment options.

C. **Correct** because as the complexity and size of an investor's portfolio grows, robo-advisers may not be able to sufficiently address the particular preferences and needs of the investor. In the case of extremely affluent investors who may own a greater number of asset types—including alternative investments (e.g., venture capital, private equity, hedge funds, and real estate)—in addition to global stocks and bonds and have greater demands for customization, the need for a team of human advisers, each with particular areas of investment or wealth-management expertise, is likely to endure.

Answer 14 of 90

Solution

A. Incorrect because $\mu \leq 0$ is the null and not the alternative hypothesis in this test. We state the null and alternative hypotheses such that they account for all possible values of the parameter. Since the alternative hypothesis is formulated using the "suspected" condition $\mu > 0$, the null hypothesis should be formulated using the condition $\mu \leq 0$.

B. Incorrect because the "suspected" condition in this test is $\mu > 0$, not $\mu \geq 0$. The best formulation of the alternative hypothesis uses the 'suspected' or 'hoped for' condition.

C. **Correct** because despite the different ways to formulate hypotheses, we always conduct a test of the null hypothesis at the point of equality, $\theta = \theta_0$. We may have a 'suspected' or 'hoped for' condition for which we want to find supportive evidence. In that case, we can formulate the alternative hypothesis as the statement that this condition is true; the null hypothesis that we test is the statement that this condition is not true. Here, the "suspected" condition is that the population's mean is greater than zero ($\mu > 0$).

Answer 15 of 90

Answer

Solution

A. Incorrect because this is the result if the candidate uses $r + c - 2 = 5 + 4 - 2 = 7$, instead of the correct $(r - 1)(c - 1)$. The incorrect formula is similar to the degrees of freedom for a test of the difference in means, which is $n_1 + n_2 - 2$.

B. **Correct** because for a **contingency table** we can perform a test of independence using a nonparametric test statistic that is chi-square distributed. This test statistic has $(r - 1)(c - 1)$ degrees of freedom, where r is the number of rows and c is the number of columns. Here, $r = 5$ and $c = 4$, so degrees of freedom = $(5 - 1)(4 - 1) = 4 \times 3 = 12$.

C. Incorrect because this is the result if the candidate omits '-1' in both factors, i.e. $rc = 5 \times 4 = 20$, instead of the correct $(r - 1)(c - 1)$.

Answer 16 of 90

Answer

Solution

- A. Incorrect because this is the result if the positions of $\sqrt{(1 - r^2)}$ and $\sqrt{(n - 2)}$ are swapped, i.e. the formula used is $r\sqrt{(1 - r^2)} / \sqrt{(n - 2)} = (0.6)\sqrt{(1 - 0.36)} / \sqrt{(51 - 2)} = (0.6)(0.8)/7 = 0.06857 \approx 0.07$.
- B. **Correct** because for a **Parametric Test of a Correlation** if the two variables are normally distributed, we can test to determine whether the null hypothesis ($H_0: \rho = 0$) should be rejected using the sample correlation, r . The formula for the t -test is $r\sqrt{(n - 2)} / \sqrt{(1 - r^2)} = (0.6)\sqrt{(51 - 2)} / \sqrt{(1 - 0.36)} = (0.6)(7)/0.8 = 5.25$.
- C. Incorrect because this is the result if the r in the denominator is not squared, i.e. $r\sqrt{(n - 2)} / \sqrt{(1 - r)} = (0.6)\sqrt{(51 - 2)} / \sqrt{(1 - 0.6)} = (0.6)(7)/0.63246 = 6.6408 \approx 6.64$.

Answer 17 of 90

Answer

Solution

- A. Incorrect because the mode, not the mean, is the measure with the highest value for a continuous negatively skewed unimodal distribution.
- B. **Correct** because for the continuous negatively skewed unimodal distribution, the mean is less than the median, which is less than the mode. Therefore, the mode has the highest value.
- C. Incorrect because the mode, not the median, is the measure with the highest value for a continuous negatively skewed unimodal distribution.

Answer 18 of 90

Answer

Solution

- A. Incorrect because Portfolio 1 has the highest expected return and the highest difference between expected return and standard deviation, but not the highest safety-first ratio.
- B. Incorrect because Portfolio 2 has the lowest standard deviation of returns, but not the highest safety-first ratio. Portfolio 2 also has the highest ratio of expected return to standard deviation (i.e., when the return threshold is omitted); Portfolio 1: $23\%/15\% \approx 1.53$; Portfolio 2: $12\%/6\% = 2.00$; Portfolio 3: $15\%/8\% \approx 1.88$.
- C. **Correct** because if returns are normally distributed, the safety-first optimal portfolio maximizes the safety-first ratio. $SFRatio = [E(R_p) - R_i] / \sigma_p$, where $E(R_p)$ is the expected portfolio return, R_i is the investor's minimum acceptable return, and σ_p is the standard deviation of portfolio returns. The minimum acceptable return is 5% (= €5,000 / €100,000) as the investor needs to withdraw €5,000 without invading initial capital; $SF_{P1} = (23\% - 5\%) / 15\% = 1.20$; $SF_{P2} = (12\% - 5\%) / 6\% \approx 1.17$; $SF_{P3} = (15\% - 5\%) / 8\% = 1.25$. Therefore, Portfolio 3 is the safety-first optimal portfolio. "The portfolio for which $E(R_p) - R_i$ is largest relative to standard deviation minimizes $P(R_p < R_i)$."

Answer 19 of 90

Answer

Solution

- A. Incorrect because the assumption of normality requires that the residuals be normally distributed. This does not mean that the dependent and independent variables must be normally distributed.
- B. Incorrect because for simple linear regression we assume that the observations (Y and X pairs) are uncorrelated with one another, meaning they are independent. If there is correlation between observations (that is, autocorrelation), they are not independent and the residuals will be correlated. As the pairs are uncorrelated, a transformation is not needed because independence (uncorrelated pairs of observations) is an assumption of simple linear regression.

C. **Correct** because if the relationship between the independent variable and the dependent variable is not linear, we can often transform one or both of these variables to convert this relation to a linear form, which then allows the use of simple linear regression.

Answer 20 of 90

Answer

Solution

- A. **Correct** because for a continuous positively skewed unimodal distribution, the mode is less than the median, which is less than the mean.
- B. Incorrect because for a continuous positively skewed unimodal distribution, the mode is less than the median, which is less than the mean. For the continuous negatively skewed unimodal distribution, the mean is less than the median, which is less than the mode.
- C. Incorrect because for a continuous positively skewed unimodal distribution, the mode is less than the median, which is less than the mean.

Answer 21 of 90

Answer

Solution

- A. Incorrect because the term Big Data has been in use since the late 1990s and refers to the vast amount of data being generated by industry, governments, individuals, and electronic devices. The term fintech is much broader.
- B. Incorrect because automated trading, not fintech, refers to executing investment decisions through computer algorithms or automated trading applications. The term fintech is much broader.
- C. **Correct** because in its broadest sense, the term 'fintech' generally refers to technology-driven innovation occurring in the financial services industry. For the purposes of this reading, **fintech** refers to technological innovation in the design and delivery of financial services and products. Note, however, that in common usage, fintech can also refer to companies (often new, startup companies) involved in developing the new technologies and their applications, as well as the business sector that comprises such companies.

Answer 22 of 90

Answer

Solution

- A. Incorrect because it is the F -statistic calculated as MSR / MSE ; $25 / 10 = 2.5$. For a simple linear regression, the F -Statistic is MSR divided by MSE, where the mean square regression (MSR) is the same as the sum of squares regression [SSR] and the mean square error (MSE) is calculated as the sum of squares error (SSE) adjusted by its degrees of freedom, $SSE / (n - 2)$; $280 / (30 - 2) = 10$.
- B. **Correct** because it is the standard error of the estimate calculated as the square root of the mean square error; $(10)^{.5} = 3.2$. The mean square error (MSE) is calculated as $SSE / (n - 2)$; $280 / (30 - 2) = 10.0$, where SSE is the sum of squares error.
- C. Incorrect because it is the mean square error (MSE) calculated as $SSE / (n - 2)$, where SSE is the sum of squares error; $280 / (30 - 2) = 10.0$.

Answer 23 of 90

Answer

Solution

- A. **Correct** because in bootstrap, we repeatedly draw samples from the original sample, and each resample is of the same size as the original sample. Note that each item drawn is replaced for the next draw (i.e., the identical element is put back into the group so that it can be drawn more than once). Assuming we are looking to find the standard error of sample mean, we take many resamples and then compute the mean of each resample.
- B. Incorrect because in cluster sampling, elements are selected from the population

not an original sample. The population is divided into clusters, each of which is essentially a mini-representation of the entire populations. Then certain clusters are chosen as a whole using simple random sampling.

C. Incorrect because in convenience sampling, elements are selected from the

population not from an original sample. An element is selected from the population based on whether or not it is accessible to a researcher or on how easy it is for a researcher to access the element. Because the samples are selected conveniently, they are not necessarily representative of the entire population, and hence the level of the sampling accuracy could be limited.

Answer 24 of 90

Answer

Solution

A. Incorrect because interest rates can be considered opportunity costs. The real risk-free interest rate is the single-period interest rate for a completely risk-free security if no inflation were expected. In economic theory, the real risk-free rate reflects the time preferences of individuals for current versus future real consumption. The sum of the real risk-free interest rate and the inflation premium is the nominal risk-free interest rate. As the inflation premium is 2%, the 1% real risk-free rate is not the opportunity cost of buying the US T-bill.

B. Incorrect because interest rates can be considered opportunity costs. The inflation premium compensates investors for expected inflation. The real risk-free interest rate is the single-period interest rate for a completely risk-free security if no inflation were expected. In economic theory, the real risk-free rate reflects the time preferences of individuals for current versus future real consumption. The sum of the real risk-free interest rate and the inflation premium is the nominal risk-free interest rate. As the real risk-free rate is 1%, the 2% inflation premium is not the opportunity cost of buying the US T-bill.

C. **Correct** because interest rates can be considered opportunity costs. The real risk-free interest rate is the single-period interest rate for a completely risk-free security if no inflation were expected. In economic theory, the real risk-free rate reflects the time preferences of individuals for current versus future real consumption. The sum of the real risk-free interest rate and the inflation premium is the nominal risk-free interest rate. Many countries have governmental short-term debt whose interest rate can be considered to represent the nominal risk-free interest rate in that country. The interest rate on a 90-day US Treasury bill (T-bill), for example, represents the nominal risk-free interest rate over that time horizon. Therefore, the opportunity cost of this investment is $1\% + 2\% = 3\%$.

Answer 25 of 90

Answer

Solution

A. Incorrect because it is the money-weighted return when the dividend is reinvested. Calculator solution: $CF_0 = -100$, $CF_1 = 0$, $CF_2 = 109$, compute $IRR = 4.403\% \approx 4.4\%$.

The answer is also the geometric mean return; $[(1 + 0.09)(1 + 0.00)]^{0.5} - 1 = 0.04403 \approx 4.4\%$.

B. Incorrect because it is the arithmetic mean return; $(0.09 + 0.00)/2 = 0.045 = 4.5\%$. The arithmetic or mean return is denoted by R_a and given by the following equation for asset i , where R_t is the return in period t and T is the total number of periods: $R_a = (R_1 + R_2 + \dots + R_T) / T$. The returns in Year 1 and Year 2 are calculated as $(100 - 100 + 9)/100 = 0.09$ and $(100 - 100 + 0)/100 = 0.00$, respectively.

C. **Correct** because it is the money-weighted return when the dividend is not reinvested. Calculator solution: $CF_0 = -100$, $CF_1 = 9$, $CF_2 = 100$, compute $IRR = 4.601\% \approx 4.6\%$.

Answer 26 of 90

Answer

Solution

- A. **Correct** because in regression analysis, we can use an F -distributed test statistic to test whether the slopes in a regression are equal to zero, with the slopes designated as b_i , against the alternative hypothesis that at least one slope is not equal to zero for simple linear regression, these hypotheses simplify to $H_0: b_1 = 0$. $H_a: b_1 \neq 0$.
- B. Incorrect because in regression analysis, we can use an F -distributed test statistic to test whether the slopes in a regression are equal to zero, with the slopes designated as b_i , against the alternative hypothesis that at least one slope is not equal to zero for simple linear regression, these hypotheses simplify to $H_0: b_1 = 0$. $H_a: b_1 \neq 0$.
- C. Incorrect because in regression analysis, we can use an F -distributed test statistic to test whether the slopes in a regression are equal to zero, with the slopes designated as b_i , against the alternative hypothesis that at least one slope is not equal to zero for simple linear regression, these hypotheses simplify to $H_0: b_1 = 0$. $H_a: b_1 \neq 0$.

Answer 27 of 90

Answer

Solution

- A. Incorrect because a tree-map is a graphical tool to display categorical data. It consists of a set of colored rectangles to represent distinct groups, and the area of each rectangle is proportional to the value of the corresponding group. Additional dimensions of categorical data can be displayed by nested rectangles.
- B. **Correct** because probabilities for different scenarios and different outcomes are best represented using a tree diagram.
- C. Incorrect because the probability estimates are for two discrete random variables, not one continuous random variable. A probability function specifies the probability that the random variable takes on a specific value. For continuous random variables, the probability function is called the probability density function (pdf), or just the density.

Answer 28 of 90

Answer

Solution

- A. Incorrect because tokenization is the process of representing ownership rights to physical assets on a blockchain or distributed ledger. It is not able to detect shifts in an analyst's sentiment.
- B. Incorrect because data curation refers to the process of ensuring data quality and accuracy through a data cleaning exercise. This process consists of reviewing all data to detect and uncover data errors—bad or inaccurate data—and making adjustments for missing data when appropriate. While the data used for natural language processing may need to be curated, it is not, in of itself, able to detect shifts in an analyst's sentiment.
- C. **Correct** because NLP [natural language processing] may be used to monitor analyst commentary to aid investment decision making. Since analysts tend not to change their buy, hold, and sell recommendations for a company frequently, they may instead offer nuanced commentary without making a change in their investment recommendation. NLP can, therefore, be used to detect, monitor, and tag shifts in sentiment, potentially ahead of an analyst's recommendation change.

Answer 29 of 90

Answer

Solution

- B. Incorrect because a tree-map is useful for displaying a frequency distribution, not correlation. In addition to bar charts and grouped bar charts, another graphical tool for displaying categorical data is a **tree-map**. It consists of a set of colored rectangles to represent distinct groups, and the area of each rectangle is proportional to the value of the corresponding group. The tree-map can represent data with additional dimensions by displaying a set of nested rectangles.
- C. **Correct** because scatter plots are a very useful tool for the sensible interpretation of a correlation coefficient. A **scatter plot** is a type of graph for visualizing the joint variation in two numerical variables. It is a useful tool for displaying and understanding potential relationships between the variables.
- D. Incorrect because a clustered bar chart is useful for displaying a frequency distribution, not correlation. In the case of two categorical variables, we need an enhanced version of the bar chart, called a **grouped bar chart** (also known as a **clustered bar chart**), to show joint frequencies.

Answer 30 of 90

Solution

- A. Incorrect because Option 3 (annuity due with 20 payments of \$13,000 each) has the highest present value of \$137,847. The PV of Option 1 is \$136,000.

If a candidate incorrectly calculated the present value of Option 3 as an ordinary annuity, the lump sum of \$136,000 would have the highest present value.

Incorrect calculator solution for Option 3: End mode; $N = 20$; $I/Y = 8$; $PMT = -13,000$; compute $PV = 127,636$.

- B. Incorrect because Option 3 (annuity due with 20 payments of \$13,000 each) has the highest present value of \$137,847. The PV of Option 2 is \$135,093.

If a candidate ignored the discount rate, the option with the 30 payments of \$12,000 each would have the highest value. That is, Option 2's 30 payments of \$12,000 for a total of \$360,000 is larger than Option 3's 20 payments of \$13,000 for a total of \$260,000 and the lump sum payment of \$136,000.

- C. **Correct** because Option 3 (annuity due with 20 payments of \$13,000 each) has the highest present value of the annuities and the \$136,000 lump sum.

Calculator solution for Option 2: End mode; $N = 30$; $I/Y = 8$; $PMT = -12,000$; compute $PV = 135,093$.

Calculator solution for Option 3: Begin mode; $N = 20$; $I/Y = 8$; $PMT = -13,000$; compute $PV = 137,847$.

Answer 31 of 90

Answer

Solution

- A. **Correct** because mean-variance analysis generally considers risk symmetrically in the sense that standard deviation captures variability both above and below the mean. An alternative approach evaluates only downside risk. We discuss one such approach, safety-first rules, as it provides an excellent illustration of the application of normal distribution theory to practical investment problems. Safety-first rules focus on shortfall risk, the risk that portfolio value will fall below some minimum acceptable level over some time horizon. Roy's safety-first criterion states that the optimal portfolio minimizes the probability that portfolio return, R_p , falls below the threshold level, R_L .
- B. Incorrect because the safety-first ratio uses standard deviation, not semideviation,

as a risk measure. The safety-first optimal portfolio maximizes the safety-first ratio (SFRatio): $\text{SFRatio} = [E(R_p) - R_L] / \sigma_P$. The quantity $E(R_p) - R_L$ is the distance from the mean return to the shortfall level. Dividing this distance by σ_P gives the distance in units of standard deviation. Although not used in Roy's safety-first criterion, semivariance,

semideviation, and related dispersion measures [also] focus on downside risk. Semivariance is defined as the average squared deviation below the mean. Semideviation (sometimes called semistandard deviation) is the positive square root of semivariance.

C. Incorrect because Roy's safety-first criterion assumes that portfolio returns are normally distributed. If returns are normally distributed, the safety-first optimal portfolio maximizes the safety-first ratio (SFRatio): $SFRatio = [E(R_p) - R_L] / \sigma_P$. The normal distribution, however, is less suitable as a model for asset prices than as a model for returns. A normal random variable has no lower limit. This characteristic has several implications for investment applications. An asset price can drop only to 0, at which point the asset becomes worthless. As a result, practitioners generally do not use the normal distribution to model the distribution of asset prices.

Answer 32 of 90

Answer

Solution

A. Incorrect because probability sampling gives every member of the population an equal chance of being selected. Hence it can create a sample that is representative of the population. In contrast, non-probability sampling depends on factors other than probability considerations, such as a sampler's judgment or the convenience to access data. Consequently there is a significant risk that non-probability sampling might generate a non-representative sample. In general, all else being equal, probability sampling can yield more accuracy and reliability compared with non-probability sampling.

B. Incorrect because probability sampling gives every member of the population an equal change of being selected. Hence it can create a sample that is representative of the population. In contrast, non-probability sampling depends on factors other than probability considerations, such as a sampler's judgment or the convenience to access data. Consequently there is a significant risk that non-probability sampling might generate a non-representative sample. In general, all else being equal, probability sampling can yield more accuracy and reliability compared with non-probability sampling.

C. **Correct** because probability sampling gives every member of the population an equal change of being selected. Hence it can create a sample that is representative of the population. In contrast, non-probability sampling depends on factors other than probability considerations, such as a sampler's judgment or the convenience to access data. Consequently there is a significant risk that non-probability sampling might generate a non-representative sample. In general, all else being equal, probability sampling can yield more accuracy and reliability compared with non-probability sampling.

Answer 33 of 90

Answer

Solution

A. Incorrect because the t -statistic for a paired comparisons test has $n - 1$ degrees of freedom, not $n - 2$ degrees of freedom, where n is the number of pairs of observations. When $n = 30$, the number of degrees of freedom is $30 - 1 = 29$, not $30 - 2 = 28$.

B. **Correct** because the t -statistic for a paired comparisons test has $n - 1$ degrees of freedom, where n is the number of pairs of observations. When $n = 30$, the number of degrees of freedom is $30 - 1 = 29$.

C. Incorrect because this is the number of degrees of freedom of the t -statistic for a hypothesis test concerning the equality of the population means of two independent samples, which is $n_1 + n_2 - 2$. When both n_1 and n_2 is 30, the number of degrees of freedom of the t -statistic for a hypothesis test concerning the equality of the population means of two independent samples is $30 + 30 - 2 = 58$.

Answer 34 of 90

Solution

- A. Incorrect because it is the difference between the third quartile and the second quartile, $Q_3 - Q_2 = 93 - 62 = 31$. The **interquartile range (IQR)** is the difference between the third quartile and the first quartile.
- B. **Correct** because the **interquartile range (IQR)** is the difference between the third quartile and the first quartile, or $IQR = Q_3 - Q_1 = 93 - 11 = 82$. **Quartiles** divide the distribution into quarters.
- C. Incorrect because it is the range of the four numbers given, and not the interquartile range. $Q_4 - Q_1 = 359 - 11 = 348$.

Answer 35 of 90

Answer

Solution

- A. Incorrect because the p -value is the smallest level of significance at which the null hypothesis can be rejected.
- B. **Correct** because the **power of a test** is the probability of **Correctly** rejecting the null—that is, the probability of rejecting the null when it is false.
- C. Incorrect because the significance level of a test is the probability of rejecting a true null hypothesis (i.e., the probability of incorrectly rejecting a null hypothesis).

Answer 36 of 90

Answer

Solution

- A. Incorrect because it is the money-weighted rate of return of the investment. The money-weighted return and its calculation are similar to the **internal rate of return**. The internal rate of return is the discount rate at which the sum of present values of these cash flows will equal zero. Calculator solution: $[CF_0] = \text{investment in the first share} = -100$, $[CF_1] = \text{investment in the second share} + \text{dividend from first share} = -110 + 10 = -100$, $[CF_2] = \text{proceeds of selling two shares} = 230$, $[IRR] [CPT] = 0.09687 \approx 9.7\%$.
- B. **Correct** because the **time-weighted rate of return** measures the compound rate of growth of \$1 initially invested in the portfolio over a stated measurement period.... We find this time-weighted return by taking the geometric mean of the two holding period returns. A **holding period return** is the return earned from holding an asset for a single specified period of time.....This return can be generalized and shown as a mathematical expression in which P is the price and I is the income: $R = [(P_t - P_0) + I_t]/P_0$. The subscript indicates the time of the price or income, ($t = 0$), is the beginning of the period and ($t = 1$) is the end of the period. Thus, the holding period return between time 0 and 1 is $HPR_0 = [(110 - 100) + 10]/100 = 0.2 = 20\%$, and the holding period return between time 1 and 2 is $HPR_1 = [(230 - 220) + 0]/220 = 0.04545 \approx 4.5\%$, where $220 = 110 \times 2$ is the price (value) of the two shares held by the investor at time 1. The time-weighted rate of return is then $[(1 + HPR_0) \times (1 + HPR_1)]^{0.5} - 1 = [(1 + 0.2) \times (1 + 0.04545)]^{0.5} - 1 = 0.12006 \approx 12.0\%$.
- C. Incorrect because it simply calculates the arithmetic average of the two holding period returns; $(20\% + 4.545\%)/2 = 12.273\% \approx 12.3\%$.

Answer 37 of 90

Answer

Solution

A. Incorrect because the second decile includes observations that are above the 10th percentile and at or below the 20th percentile. The 19th observation is in the second group of ten observations, but it is not in the second decile because there are only 75 observations in the sample.

B. **Correct** because the 19th observation is located at the 25th percentile;

$L_y = (1 + n) \left(\frac{y}{100} \right); 19 = (1 + 75) \left(\frac{y}{100} \right); y = \left(\frac{19 \times 100}{1 + 75} \right) = 25.0$, which is in the second quintile. The second quintile includes observations that are above the 20th percentile and at or below the 40th percentile.

C. Incorrect because the second quartile includes observations that are above the 25th percentile and at or below the 50th percentile. Because the 19th observation is at the 25th percentile, it is in the first quartile.

Answer 38 of 90

Answer

Solution

A. Incorrect because for a given dataset with different observations, the harmonic mean has the smallest value among harmonic, geometric, and arithmetic means. Unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

B. **Correct** because the arithmetic mean of a given dataset with different observations is higher than the harmonic and geometric means. Unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

C. Incorrect because for a given dataset with different observations, the geometric mean is larger than the harmonic mean but smaller than the arithmetic mean. Unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

Answer 39 of 90

Answer

Solution

A. Incorrect because sampling error is the difference between the observed value of a statistic and the quantity it is intended to estimate.

B. Incorrect because sampling error is the difference between the observed value of a statistic and the quantity it is intended to estimate.

C. **Correct** because sampling error is the difference between the observed value of a statistic and the quantity it is intended to estimate.

Answer 40 of 90

Answer

Solution

A. Incorrect because the central limit theorem allows us to make quite precise probability statements about the population mean by using the sample mean, whatever the distribution of the population (so long as it has finite variance), because the sample mean follows an approximate normal distribution for large-size samples. Therefore, having a normally distributed population is not a requirement for the central limit theorem to hold.

B. **Correct** because the central limit theorem states that the variance of the distribution of the sample mean is σ^2/n . The positive square root of variance is standard deviation.

The standard deviation of a sample statistic is known as the standard error of the statistic. The sample mean, in addition to being an efficient estimator, is also a consistent estimator of the population mean: As sample size n goes to infinity, its standard error,

$\sigma/\sqrt{n}$, goes to 0 and its sampling distribution becomes concentrated right over the value of population mean, μ .

C. Incorrect because the central limit theorem states that the sum (and mean) [not the product] of a large number of independent random variables is approximately normally distributed.

Answer 41 of 90

Answer

Solution

A. **Correct** because the **variance** of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value: $\sigma^2(X) = E[X - E(X)]^2$. Since each scenario is equally likely (probability = $1/3$), $E(X) = (2.0 + 2.2 + 2.4)/3 = 2.2$, so $\sigma^2(X) = [(2.0 - 2.2)^2 + (2.2 - 2.2)^2 + (2.4 - 2.2)^2]/3 = [0.04 + 0.04]/3 = 0.08/3 = 0.0267 \approx 0.03$ [in \$²].

B. Incorrect because it is the standard deviation, not the variance. The **variance** of a random variable is the expected value (the probability-weighted average) of squared deviations from the random variable's expected value: $\sigma^2(X) = E[X - E(X)]^2$. Since each scenario is equally likely (probability = $1/3$), $E(X) = (2.0 + 2.2 + 2.4)/3 = 2.2$, so $\sigma^2(X) = [(2.0 - 2.2)^2 + (2.2 - 2.2)^2 + (2.4 - 2.2)^2]/3 = [0.04 + 0.04]/3 = 0.08/3 = 0.0267$ [in \$²]. **Standard deviation** is the positive square root of variance. Therefore, $\sigma(X) = (0.0267)^{0.5} = 0.1633 \approx 0.16$ [in \$].

C. Incorrect because it is the difference between each pair of EPS values, not the variance: $\$2.40 - \$2.20 = \$2.20 - \$2.00 = \$0.20$.

Answer 42 of 90

Answer

Solution

A. **Correct** because the 50th percentile is the median, which is the average of the two middle items; $(£0.50 + £2.50)/2 = £1.50$. In an odd-numbered sample of n items, the median occupies the $(n + 1)/2$ position. In an even-numbered sample, we define the median as the mean of the values of items occupying the $n/2$ and $(n + 2)/2$ positions (the two middle items). Calculating the median may also be more complex; to do so, we need to order the observations from smallest to largest, determine whether the sample size is even or odd and, on that basis, apply one of two calculations. Alternatively, the 50th percentile when L_y is not a whole number or integer, L_y lies between the two closest integer numbers (one above and one below), and we use **linear interpolation** between those two places to determine P_y . That is, $L_y = (n + 1)(y/100) = (4 + 1)(50/100) = 2.5$. Hence, 2 is the closest integer below the calculated location and 3 is the closest integer above the calculated location. Using linear interpolation, $P_{50} = £0.50 + (£2.50 - £0.50) \times (2.5 - 2) = £1.50$.

B. Incorrect because it is the location of the median; $n/2 = 4/2$. Alternatively, it is also the average EPS; $(-£0.50 + £0.50 + £2.50 + £5.50)/4 = £2.00$.

C. Incorrect because it is the position (or location) of the 50th percentile instead of the 50th percentile: $(4 + 1) \times 50/100 = 2.5$. The formula for the position (or location) of a percentile in an array with n entries sorted in ascending order is $L_y = (n + 1)(y/100)$ where y is the percentage point at which we are dividing the distribution and L_y is the location (L) of the percentile (P_y) in the array sorted in ascending order. The value of L_y may or may not be a whole number. When L_y is not a whole number or integer, L_y lies between the two closest integer numbers (one above and one below), and we use **linear interpolation** between those two places to determine P_y .

Answer 43 of 90

Answer

Solution

A. Incorrect because unlike the coefficient of determination and the F -statistic, which are relative measures of fit, the standard error of the estimate is an absolute measure of the

distance of the observed dependent variable from the regression line.

B. Incorrect because it is the coefficient of determination, and not the standard error

of estimate. The **coefficient of determination**, also referred to as the R -squared or R^2 , is the percentage of the variation of the dependent variable that is explained by the independent variable.

C. **Correct** because the **standard error of the estimate** is a measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression.

Answer 44 of 90

Answer

Solution

A. Incorrect because n (5) is used in the calculation instead of $n - 1$: $(14/5)^{0.5}$, resulting in $1.67\% \approx 1.7\%$.

B. **Correct** because the formula for target downside deviation is $[\sum(X_i - B)^2 / (n - 1)]^{0.5}$, where X_i is the return for the period, B is the target return and n is the total number of sample observations. Moreover, the summation is taken over only those observations (X_i) that are less than or equal to the target B .

	Year Return (%)	Deviations from the target	Squared deviations
1	6	0	0
2	7	0	0
3	3	-2	4
4	2	-3	9
5	4	-1	1

C. Since the sum of squared deviations is 14, the target downside deviation = $[14/(5 - 1)]^{0.5}$, resulting in $1.87\% \approx 1.9\%$.

D. Incorrect because all the returns are used in the calculation, not just the returns below the target. In other words, the standard deviation from the target return is calculated. Since the sum of all squared deviations is $14 + 5 = 19$, the target downside deviation is calculated as: $[19/(5 - 1)]^{0.5}$, resulting in $2.18\% \approx 2.2\%$.

This answer is also closest if n is assumed to be the number of observations used in the correct calculation (3): $(14/3)^{0.5}$, resulting in $2.16\% \approx 2.2\%$.

Answer 45 of 90

Answer

Solution

A. Incorrect because the Sharpe ratio is the average return in excess of the risk-free rate divided by the standard deviation of returns. This ratio measures the excess return earned per unit of standard deviation of return. Therefore, the Sharpe ratio does not quantify the amount of risk per unit of mean return.

B. Incorrect because the standard deviation measures risk, but it does not quantify the amount of risk per unit of mean return. Variance is defined as the average of the squared deviations around the mean. Standard deviation is the positive square root of the variance.

C. **Correct** because the coefficient of variation, CV, is the ratio of the standard deviation of a set of observations to their mean value. When the observations are returns, for example, the coefficient of variation measures the amount of risk (standard deviation) per unit of mean return.

Answer 46 of 90

Answer

Solution

A. Incorrect because it confuses the location of the second quartile, which is calculated as $L_y = (7 + 1) \times 50/100 = 4$, with the value of the second quartile, which is the return of the 4th fund after placing the funds' returns in ascending order. 4% also corresponds to the return of the 4th fund (the middle fund) when the array has not been sorted in ascending order, which a candidate might interpret as the 50th percentile (the median).

B. **Correct** because the formula for the position of a percentile in an array with n entries sorted in ascending order is $L_y = (n + 1) \times y/100$, where y is the percentage point at which we are dividing the distribution and L_y is the location (L) of the percentile (P_y) in the array sorted in ascending order. With seven entries, the location of the second quartile, or 50th percentile, is: $L_y = (7 + 1) \times 50/100 = 4$. When placing the funds' returns in ascending order (3%; 3%; 4%; 5%; 7%; 8%; 12%), the return of the 4th fund is 5%.

Alternatively, candidates might realize that the second quartile or 50th percentile is the median. The median is the value of the middle item of a set of items that has been sorted into ascending or descending order. In an odd-numbered sample of n items, the median occupies the $(n + 1)/2$ position. Hence, the median return is 5%.

C. Incorrect because the mean is mistaken as the 50th percentile or second quartile; mean return = $(12\% + 8\% + 7\% + 5\% + 4\% + 3\% + 3\%)/7 = 6\%$.

Answer 47 of 90

Answer

Solution

A. Incorrect because correlation ranges from -1 and $+1$ for two random variables, X and Y .

B. Incorrect because correlation may also be an unreliable measure when outliers are present in one or both of the variables. As we have seen, outliers are small numbers of observations at either extreme (small or large) of a sample. The correlation may be quite sensitive to outliers.

C. **Correct** because the correlation coefficient expresses the strength of the linear relationship between the two random variables.

Answer 48 of 90

Answer

Solution

A. **Correct** because the correlation between two random variables, R_i and R_j , is defined as $\rho(R_i, R_j) = \text{Cov}(R_i, R_j) / [\sigma(R_i)\sigma(R_j)]$, where Cov denotes the covariance and σ the standard deviation. Since the standard deviation of each asset occurs in the denominator of the correlation formula, it is clear that, all else being equal, an increase in the variance (hence standard deviation) of either variable will decrease the correlation.

B. Incorrect because the correlation between two random variables, R_i and R_j , is defined as $\rho(R_i, R_j) = \text{Cov}(R_i, R_j) / [\sigma(R_i)\sigma(R_j)]$, where Cov denotes the covariance and σ the standard deviation. Hence an increase in variance will decrease the correlation, not keep it the same. Candidates may select this answer choice since the covariance of correlated variables has remained the same, or if they assume that correlation is independent of variance.

C. Incorrect because the correlation between two random variables, R_i and R_j , is defined as $\rho(R_i, R_j) = \text{Cov}(R_i, R_j) / [\sigma(R_i)\sigma(R_j)]$, where Cov denotes the covariance and σ the standard deviation. Hence an increase in variance will decrease the correlation, not increase it. Candidates may select this answer choice if they think that more variance means more covariance and hence more correlation, or if they swap numerator and

denominator of the correlation formula; $[\sigma(R_i)\sigma(R_j)]/\text{Cov}(R_i, R_j)$.

Answer 49 of 90

Answer

Solution

- A. **Correct** because the correlation coefficient is a measure of the linear association between two variables; it would not be appropriate to use the correlation coefficient to measure the non-linear relationship between variables.
- B. Incorrect because the correlation coefficient is a measure of the linear association between two variables. It does not measure non-linear relationships between variables.
- C. Incorrect because the correlation coefficient is a measure of the linear association between two variables. It does not measure non-linear relationships between variables.

Answer 50 of 90

Answer

Solution

- A. **Correct** because the equation to estimate the standard error of the sample mean effectively computes the sample standard deviation of the different means generated across all resamples. Hence the mean of each resample is required. However, neither the mean, nor the standard deviation, of the original sample are required.
- B. Incorrect because the equation to estimate the standard error of the sample mean effectively computes the sample standard deviation of the different means generated across all resamples. Hence the mean of each resample is required. However, neither the mean, nor the standard deviation, of the original sample are required.
- C. Incorrect because the equation to estimate the standard error of the sample mean effectively computes the sample standard deviation of the different means generated across all resamples. Hence the mean of each resample is required. However, neither the mean, nor the standard deviation, of the original sample are required.

Answer 51 of 90

Answer

Solution

- A. **Correct** because the expected value $E(X) = \sum_{i=1}^n P(X_i)X_i = (0.20 \times 35) + (0.30 \times 50) + (0.50 \times 80) = 62$. The variance $\sigma^2(X) = E\{[X - E(X)]^2\} = \sum_{i=1}^n P(X_i)[X - E(X)]^2 = 0.20 \times (35 - 62)^2 + 0.30 \times (50 - 62)^2 + 0.50 \times (80 - 62)^2 = 351$. **Standard deviation** is the positive square root of variance: $\sigma = 351^{1/2} \approx 18.73$.
- B. Incorrect because it uses the simple average of the outcomes instead of the expected value in the calculation. The simple average $X_{\text{bar}} = (35 + 50 + 80)/3 = 55$. Hence the variance becomes $\sigma^2(X) = \sum_{i=1}^n P(X_i)[X - X_{\text{bar}}]^2 = 0.20 \times (35 - 55)^2 + 0.30 \times (50 - 55)^2 + 0.50 \times (80 - 55)^2 = 400$. Then, the standard deviation is equal to $\sigma = 400^{1/2} = 20.00$.
- C. Incorrect because it ignores that this is a random variable and just calculates the sample standard deviation instead for a sample with 3 observations. The sample mean $X_{\text{bar}} = (35 + 50 + 80)/3 = 55$. Hence the sample variance is equal to $\sigma^2 = (\sum_{i=1}^n [X - X_{\text{bar}}]^2)/(n - 1) = [(35 - 55)^2 + (50 - 55)^2 + (80 - 55)^2]/(3 - 1) = 525$. Then, the sample standard deviation is equal to $\sigma = 525^{1/2} \approx 22.91$.

Answer 52 of 90

Answer

Solution

- A. Incorrect because it wrongly multiplies the expected dividend per share given the favorable scenario (\$1.90) by the probability of the scenario (0.60):

$$\text{Probability of favorable scenario} \times E(\text{Dividend} \mid \text{Favorable scenario})$$

$$= 0.60 \times [(0.80 \times \$2.00) + (0.20 \times \$1.50)]$$

$$= 0.60 \times \$1.90$$

$$= \$1.14.$$

- B. Incorrect because it is the unconditional expected dividend per share based on the total probability rule for expected value:

$$E(\text{Dividend} \mid \text{Favorable scenario}) = (0.80 \times \$2.00) + (0.20 \times \$1.50) = \$1.90.$$

$$E(\text{Dividend} \mid \text{Unfavorable scenario}) = (0.30 \times \$0.75) + (0.70 \times \$0.50) = \$0.575.$$

$$E(\text{Dividend}) = (0.60 \times \$1.90) + (0.40 \times \$0.575) = \$1.37.$$

- C. **Correct** because the expected value of a random variable X given an event or scenario S is denoted $E(X \mid S)$. Suppose the random variable X can take on any one of n distinct outcomes $X_1, X_2, \dots, X_n$ (these outcomes form a set of mutually exclusive and exhaustive events). The expected value of X conditional on S is the first outcome, X_1 , times the probability of the first outcome given S , $P(X_1 \mid S)$, plus the second outcome, X_2 , times the probability of the second outcome given S , $P(X_2 \mid S)$, and so forth. In our case, $S = \text{Favorable scenario}$, $X_1 = \text{Dividend of } \2.50 , $X_2 = \text{Dividend of } \1.50 , $P(X_1 \mid S) = 0.80$, and $P(X_2 \mid S) = 0.20$. Thus, the expected dividend given the favorable scenario $= (0.80 \times \$2.00) + (0.20 \times \$1.50) = \$1.90$.

Answer 53 of 90

Answer

Solution

- A. Incorrect because it takes the natural logarithm of Term Deposit 1's stated annual rate instead of its effective annual rate (EAR). Calculation: $\ln(1.04) = 0.039221 = 3.92\%$.

- B. **Correct** because the investor will be indifferent if the EAR for both term deposits is the same. Therefore, we need to find the stated annual rate with continuous compounding that corresponds to the EAR of the quarterly compounded term deposit. Calculations: EAR of Term Deposit 1 $= (1 + 0.04/4)^4 - 1 = 0.040604$. Hence, EAR of Term Deposit 2 $= 0.040604 = e^r - 1$, leading to a stated annual rate for Term Deposit 2 of $r = \ln(1.040604) = 0.039801 = 3.98\%$.

- C. Incorrect because it is the EAR of Term Deposit 1, not the stated annual rate of Term Deposit 2. EAR of Term Deposit 1 $= (1 + 0.04/4)^4 - 1 = 0.040604 = 4.06\%$. This answer is also closest to the calculations $e^{0.04} - 1 = 0.040811 = 4.08\%$ or $e^{0.040604} - 1 = 0.041440 = 4.14\%$.

Answer 54 of 90

Answer

Solution

- A. **Correct** because the harmonic mean is used to determine the average price paid per

$$\frac{3}{\frac{1}{14} + \frac{1}{12} + \frac{1}{17}} = 14.05.$$

share when using cost averaging;

The weighted mean formula could also be used, where the weights would be the proportion of the total number of shares purchased. However, in order to use this method a fixed investment amount would need to be created.

- B. **Incorrect** because it is the average price per share, not the cost per share; $(14 + 12 + 17)/3 = 14.33$. Using the arithmetic mean assumes equal weighting, which is not appropriate when the investment amount is fixed and the share price is variable.

- C. **Incorrect** because the price per share is incorrectly used as the weight in the weighted mean formula;

$$14\left(\frac{14}{14+12+17}\right) + 12\left(\frac{12}{14+12+17}\right) + 17\left(\frac{17}{14+12+17}\right) = 14.63.$$

Answer 55 of 90

Answer

Solution

- A. **Incorrect** because in the log-lin model, the dependent variable is in logarithmic form and the independent variable is not.

- B. **Correct** because the lin-log model is similar to the log-lin model, but only the independent variable is in logarithmic form.

- C. **Incorrect** because the log-log model is one in which both the dependent variable and the independent variable are linear in their logarithmic forms. In the lin-log model only the independent variable is in logarithmic form.

Answer 56 of 90

Answer

Solution

- A. **Incorrect** because the covariance formula is misapplied as $\text{Cov}(R_A, R_B) = \sum P(R_{Aij}, R_{Bij})(R_{Aij} - E[R_A]) + \sum P(R_{Aij}, R_{Bij})(R_{Bij} - E[R_B])$, resulting in $0.2(20 - 14) + 0.4(15 - 14) + 0.4(10 - 14) + 0.2(15 - 9) + 0.4(10 - 9) + 0.4(5 - 9) = 1.2 + 0.4 + (-1.6) + 1.2 + 0.4 + (-1.6) = 0$.

- B. **Incorrect** because the covariance formula is misapplied as $\text{Cov}(R_A, R_B) = \sum P(R_{Aij}, R_{Bij})(R_{Aij} - R_{Bij})$ resulting in $0.2(20 - 15) + 0.4(15 - 10) + 0.4(10 - 5) = 0.2(5) + 0.4(5) + 0.4(5) = 1 + 2 + 2 = 5$. This answer also represents the difference between the expected returns of X and Y: $14 - 9 = 5$.

- C. **Correct** because the formula for calculating the covariance between random variables R_A and R_B is $\text{Cov}(R_A, R_B) = \sum \sum P(R_{Aij}, R_{Bij})(R_{Aij} - E[R_A])(R_{Bij} - E[R_B])$.

The expected return (given) for each company is:

$$E[R_X] = 0.2(20) + 0.4(15) + 0.4(10) = 4 + 6 + 4 = 14,$$

$$E[R_Y] = 0.2(15) + 0.4(10) + 0.4(5) = 3 + 4 + 2 = 9.$$

$$\text{Hence, } \text{Cov}(R_X, R_Y) = 0.2(20 - 14)(15 - 9) + 0.4(15 - 14)(10 - 9) + 0.4(10 - 14)(5 - 9) = 0.2(6)(6) + 0.4(1)(1) + 0.4(-4)(-4) = 7.2 + 0.4 + 6.4 = 14.$$

Answer 57 of 90

Answer

Solution

- A. **Correct** because the mean absolute deviation of 1.5% is less than the sample standard deviation of 1.83%. The mean absolute deviation, MAD, is calculated as: $\frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$, where the sample mean, $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$. As the sample mean is: $\bar{X} = \frac{(-2-1+1+2)}{4} = 0$, the calculation of MAD is: $\frac{|-2-0|+|-1-0|+|1-0|+|2-0|}{4} = \frac{2+1+1+2}{4} = 1.5000\%$, while the sample standard deviation of observations, X_i , is $s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$, here: $\sqrt{\frac{(-2-0)^2+(-1-0)^2+(1-0)^2+(2-0)^2}{4-1}} = \sqrt{\frac{4+1+1+4}{3}} = \sqrt{3.3333} = 1.8257\%$.
- B. **Incorrect** because the mean absolute deviation, MAD, is less than standard deviation of returns in the sample. The mean absolute deviation, MAD, is calculated as: $\frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$, where the sample mean, $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$, which results in MAD = 1.50% while sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}} = 1.83\%$ therefore MAD is less than standard deviation of sample returns.
- C. **Incorrect** because the mean absolute deviation, MAD, is less than standard deviation of returns in the sample. The mean absolute deviation, MAD, is calculated as: $\frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$, where the sample mean, $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$, which results in MAD = 1.50% while sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}} = 1.83\%$ therefore MAD is less than standard deviation of sample returns.

Answer 58 of 90

Answer

Solution

- A. **Correct** because the money-weighted return and its calculation are similar to the internal rate of return and the yield to maturity. Just like the internal rate of return, amounts invested are cash outflows from the investor's perspective and amounts returned or withdrawn by the investor, or the money that remains at the end of an investment cycle, is a cash inflow for the investor. For the stock investment this is: $\$20 = \$3/(1+r) + (\$20 + \$1)/(1+r)^2$, yielding $r = 10.24\% \approx 10\%$. Calculator solution: $CF_0 = -20$; $CF_1 = 3$; $CF_2 = 21$; IRR CPT = 10.24.
- B. **Incorrect** because it is time-weighted, not money-weighted, rate of return. For the stock investment, the holding period returns (HPR) for the two periods are $HPR_1 = (\$12 - \$20 + \$3)/\$20 = -0.25 = -25\%$ and $HPR_2 = (\$20 - \$12 + \$1)/\$12 = 0.75 = 75\%$. Hence, TWR $= \sqrt{(0.75 \times 1.75)} - 1 = 0.1456 = 14.56\% \approx 15\%$.
- C. **Incorrect** because it confuses the money-weighted rate of return with the 'money returned', i.e. the net cash flow: $\$20 + \$3 + \$1 - \$20 = \$4$, which, on an initial investment of $\$20$, corresponds to a 20% return; $\$4/\$20 = 0.2 = 20\%$.

Answer 59 of 90

Solution

- A. **Incorrect** because the money-weighted return and its calculation are similar to the **internal rate of return** and the yield to maturity. Just like the internal rate of return, amounts invested are cash outflows from the investor's perspective and amounts returned or withdrawn by the investor, or the money that remains at the end of an investment cycle, is a cash

inflow for the investor. That is, cash withdrawals and investments are included in the money-weighted return calculation; however, cash withdrawals and investments are ignored in the calculation of the arithmetic and geometric mean returns.

- B. **Correct** because the money-weighted return is an accurate measure of what the investor actually earned on the money invested.
- C. **Incorrect** because although the money-weighted return is an accurate measure of what the investor actually earned on the money invested, it is limited in its applicability to other situations. For example, it does not allow for return comparison between different individuals or different investment opportunities. Two investors in the *same* mutual fund may have different money-weighted returns because they invested different amounts in different years.

Answer 60 of 90

Answer

Solution

A. **Incorrect** because an investment measure that is not sensitive to the additions and withdrawals of funds is the time-weighted rate of return. The time-weighted rate of return measures the compound rate of growth of \$1 initially invested in the portfolio over a stated measurement period. For the evaluation of portfolios of publicly traded securities, the time-weighted rate of return is the preferred performance measure as it neutralizes the effect of cash withdrawals or additions to the portfolio, which are generally outside of the control of the portfolio manager.

B. **Incorrect** because the simplest way to compute the return is to take a simple arithmetic average of all holding period returns. Moreover, the arithmetic mean return assumes that the amount invested at the beginning of each period is the same. Thus, the arithmetic mean return does not account for cash additions to the portfolio and it is not affected by them.

C. **Correct** because, if funds are added to a portfolio, the money-weighted rate of return puts a greater weight on the time period after the funds were added. (As a corollary, if funds are taken out of a portfolio, the money-weighted rate of return puts a greater weight on the time period before the funds were removed.) So, if a client gives an investment manager more funds to invest at an unfavorable time, the manager's money-weighted rate of return will tend to be depressed.

Answer 61 of 90

Answer

Solution

A. **Incorrect** because "uncorrelated" is misinterpreted as having a correlation value of -1 (i.e., perfect negative correlation is used instead of zero) when calculating the portfolio standard deviation;

$$\sigma_p = \sqrt{(0.5)^2(3)^2 + (0.5)^2(3)^2 + 2(0.5)(0.5)(3)(3)(-1)} = \sqrt{0} = 0\%$$

B. **Correct** because the portfolio standard deviation is 2.1%;

$$\sqrt{(0.5)^2(3)^2 + (0.5)^2(3)^2 + 2(0.5)(0.5)(3)(3)(0)} = \sqrt{4.5} \text{ or } 2.12\%, \text{ using the formula}$$

$$\sqrt{\sigma_p^2} = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\text{Cov}(R_1, R_2)}, \text{ where the } \text{Cov}(R_1, R_2) = \rho(R_1, R_2)\sigma(R_1)\sigma(R_2),$$

which is equal to zero because the funds are uncorrelated; $\rho(R_1, R_2) = 0$.

C. **Incorrect** because it is the weighted average of the two securities' standard deviation and calculates the portfolio standard deviation as: $\sigma_p = (0.5)(3\%) + (0.5)(3\%) = 3.0\%$.

Answer 62 of 90

Answer

Solution

A. **Correct** because the present value of the future lump sum payment is $PV = FV_n(1$

$+ r) \cdot N = \$500,000(1 + 0.04)^{-15} = \$277,632.25$. The 10 annual payments form an annuity due (since the payments start today) whose present value equals the present value of an ordinary annuity with 9 annual payments plus the first payment, i.e. $PV = A + A[1 - 1/(1 + r)^9]/r = A(1 + [1 - 1/(1 + 0.04)^9]/0.04) = 8.4353(A)$. Setting the PV of the cash outflows (the annuity) equal to the PV of the cash inflows (the return in 15 years), we can solve for the annual payment amount; $A = \$277,632.25/8.4353 \approx \$32,913$. Calculator solution: BGN; N = 10; I/Y = 4; PV = 277,632.25; solve for PMT = 32,913.

B. Incorrect because it mistakes the required payments as an ordinary annuity, rather than an annuity due, calculating their present value as: $PV = A[1 - 1/(1 + r)^9]/r = A([1 - 1/(1 + 0.04)^9]/0.04) = 8.1109(A)$. The present value of the return at $t = 15$ is $PV = FV_N(1 + r)^{-N} = \$500,000(1 + 0.04)^{-15} = \$277,632.25$. Setting these two values equal and solving for the annual payment thus yields: $A = \$277,632.25/8.1109 \approx \$34,230$. Calculator solution: END; N = 10; I/Y = 4; PV = 277,632.25; solve for PMT = 34,230.

C. Incorrect because it assumes the \$500,000 return is paid at $t = 10$ instead of $t = 15$. The present value of this payment is therefore $PV = FV_N(1 + r)^{-N} = \$500,000(1 + 0.04)^{-10} = \$337,782.08$. The 10 annual payments form an annuity due (since the payments start today) whose present value equals the present value of an ordinary annuity with 9 annual payments plus the first payment, i.e. $PV = A + A[1 - 1/(1 + r)^9]/r = A(1 + [1 - 1/(1 + 0.04)^9]/0.04) = 8.4353(A)$. Setting the PV of the cash outflows (the annuity) equal to the PV of the cash inflows (the return in 15 years), we can solve for the annual payment amount; $A = \$337,782/8.4353 \approx \$40,044$. Calculator solution: BGN; N = 10; I/Y = 4; FV = 500,000; solve for PMT = 40,044.

Answer 63 of 90

Answer

Solution

- A. Incorrect because the power of the test is equal to one minus the probability of Type II error. The p -value is a measure of Type I error and is synonymous with distracter B.
- B. Incorrect because the power of the test is equal to one minus the probability of Type II error. The p -value is a measure of Type I error and is synonymous with distracter B.
- C. **Correct** because the power of a test is the probability of **Correctly** rejecting the null—that is, the probability of rejecting the null when it is false. Failing to reject the null hypothesis when it is false is a Type II error. So the power of the test is equal to one minus the probability of Type II error.

Answer 64 of 90

Answer

Solution

- A. Incorrect because the prediction interval is equal to the predicted value of the dependent variable plus/minus the critical t -value times the standard error of the forecast. The larger the sample size (n) in the regression estimation, the smaller the standard error of the forecast. Meanwhile, when the sample size is larger, the critical t -value will be smaller. Both will lead to a narrower prediction interval if holding other things constant.
- B. Incorrect because the prediction interval is equal to the predicted value of the dependent variable plus/minus the critical t -value times the standard error of the forecast. When the level of significance increases, the critical t -value will decrease, which will lead to a narrower prediction interval if holding other things constant.
- C. **Correct** because the prediction interval is equal to the predicted value of the dependent variable plus/minus the critical t -value times the standard error of the forecast. The better the fit of the regression model, the smaller the standard error of the estimate (s_e) and, therefore, the smaller standard error of the forecast. When the standard error of the estimate increases, the standard error of the forecast will increase, which will lead to a wider prediction interval if holding other things constant.

Answer 65 of 90

Answer

Solution

A. Incorrect because the annuity is taken as an ordinary annuity rather than an annuity due, leading to a PV 10 years from today of $PV_{10} = \$50,000 \times [1 - 1/(1.03)^{10}]/0.03 = \$50,000 \times 3.717098 = \$185,855$. The PV of the annuity today would equal $PV_0 = \$185,855/(1.03)^{10} = \$138,294$.

Calculator solution: END mode; N = 4; I/Y = 3%; PMT = 50,000; solve for PV = 185,855. Discounted back 10 years: N = 10; I/Y = 3%; FV = 185,855; solve for PV = 138,294.

This is also the result if the PV of the annuity due is discounted back 11 years instead of 10 years: $\$191,431/(1.03)^{11} = \$138,294$.

Calculator solution: BEGIN mode; N = 4; I/Y = 3%; PMT = 50,000; solve for PV = 191,431. Discounted back 11 years: N = 11; I/Y = 3%; FV = 191,431; solve for PV = 138,294.

B. **Correct** because the present value (PV) of the annuity due 10 years from today equals $PV_{10} = \$50,000 + \$50,000 \times [1 - 1/(1.03)^9]/0.03 = \$50,000 + \$50,000 \times 2.828611 = \$191,431$. The PV of the annuity today equals $PV_0 = \$191,431/(1.03)^{10} = \$142,442$. Calculator solution: BEGIN mode; N = 4; I/Y = 3%; PMT = 50,000; solve for PV = 191,431. Discounted back 10 years: N = 10; I/Y = 3%; FV = 191,431; solve for PV = 142,442.

Alternatively, the annuity can be treated as an ordinary annuity, with a PV 9 years from today of $PV_9 = \$50,000 \times [1 - 1/(1.03)^9]/0.03 = \$50,000 \times 3.717098 = \$185,855$. The PV of the annuity today equals $PV_0 = \$185,855/(1.03)^9 = \$142,442$.

Calculator solution: END mode; N = 4; I/Y = 3%; PMT = 50,000; solve for PV = 185,855. Discounted back 9 years: N = 9; I/Y = 3%; FV = 185,855; solve for PV = 142,442.

C. Incorrect because the PV of the annuity due is discounted back 9 years instead of 10 years: $\$191,431/(1.03)^9 = \$146,716$.

Calculator solution: BEGIN mode; N = 4; I/Y = 3%; PMT = 50,000; solve for PV = 191,431. Discounted back 9 years: N = 9; I/Y = 3%; FV = 191,431; solve for PV = 146,716.

This answer is also closest to the PV of the entire required amount: $\$200,000/(1.03)^{10} = \$148,819$.

Answer 66 of 90

Answer

Solution

A. Incorrect because it assumes the ordinary annuity starts in Year 5 instead of Year 4. The present value of an ordinary annuity with 7 payments of \$10,000 at a 6% discount rate is calculated as follows:

$$\begin{aligned} PV &= A[1 - 1/(1 + r)^N]/r \\ PV_4 &= \$10,000 \times [1 - 1/(1 + 0.06)^7]/0.06 \\ PV_4 &= \$55,823.81 \end{aligned}$$

Then the PV of the investment is incorrectly discounted from Year 5 instead of Year 4 such that:

$$\begin{aligned} PV_0 &= FV_5/(1 + r)^5 \\ PV_0 &= \$55,823.81 \times (1 + 0.06)^{-5} \\ PV_0 &= \$41,714.80 \approx \$41,715. \end{aligned}$$

Calculator solution: (1) END mode; N = 7; I = 6; PMT = -10,000; FV = 0; solve for PV = 55,823.81.

(2) END mode; N = 5; I = 6; PMT = 0; FV = -55,823.81; solve for PV = 41,714.80.

B. **Correct** because the present value in Year 4 of an ordinary annuity with 7 payments of \$10,000 at a 6% discount rate is calculated as follows:

$$\begin{aligned} PV &= A[1 - 1/(1 + r)^N]/r \\ PV_4 &= \$10,000 \times [1 - 1/(1 + 0.06)^7]/0.06 \\ PV_4 &= \$55,823.81 \end{aligned}$$

Then, using a time line, the PV of the annuity in today's dollars is

$$\begin{aligned}PV_0 &= FV_4(1 + r)^{-N} \\PV_0 &= \$55,823.81 \times (1 + 0.06)^{-4} \\PV_0 &= \$44,217.69 \approx \$44,218.\end{aligned}$$

Calculator solution: (1) END mode; N = 7; I = 6; PMT = -10,000; FV = 0; solve for PV = 55,823.81.

END mode; N = 4; I = 6; PMT = 0; FV = ->55,823.81; solve for PV = 44,217.69.

- C. Incorrect because it is an intermediate step in the calculation and it represents the value of the annuity in 4 years. The present value in Year 4 of an ordinary annuity with 7 payments of \$10,000 at a 6% discount rate is calculated as follows:

$$\begin{aligned}PV &= A[1 - 1/(1 + r)^n]/r \\PV_4 &= \$10,000 \times [1 - 1/(1 + 0.06)^7]/0.06 \\PV_4 &= \$55,823.81 \approx \$55,824.\end{aligned}$$

Calculator solution: (1) END mode; N = 7; I = 6; PMT = -10,000; FV = 0; solve for PV = 55,823.81.

Answer 67 of 90

Answer

Solution

A. Incorrect because we look at the sum of squared deviations of the observations from the mean to capture [the variation of the dependent variable]. Our goal is to understand what explains the variation of Y . The variation of Y is often referred to as the sum of squares total (SST), or the total sum of squares. Therefore, the difference between an observation and the mean of the dependent variable is not the observation's residual.

B. **Correct** because the residual for the i^{th} observation, e_i , is how much the observed value of Y_i differs from the estimated [value] using the regression line. Further, the residual refers to the fitted linear relation based on the sample.

C. Incorrect because the error term refers to the true underlying population relationship, whereas the residual refers to the fitted linear relation based on the sample. Further, the error term, or simply the error, represents the difference between the observed value of Y and that expected from the true underlying population relation between Y and X .

Answer 68 of 90

Answer

Solution

A. Incorrect because the **liquidity premium** compensates investors for the risk of loss relative to an investment's fair value if the investment needs to be converted to cash quickly. US T-bills, for example, do not bear a liquidity premium because large amounts can be bought and sold without affecting their market price.

B. Incorrect because the **maturity premium** compensates investors for the increased sensitivity of the market value of debt to a change in market interest rates as maturity is extended. For example, the difference between the interest rate on longer-maturity, liquid Treasury debt and that on short-term Treasury debt reflects a positive maturity premium for the longer-term debt.

C. **Correct** because the real risk-free rate reflects the time preferences of individuals for current versus future real consumption.

Answer 69 of 90

Answer

Solution

- A. **Correct** because the target downside deviation = $[\sum(X_i - B)^2 / (n - 1)]^{0.5}$, where X_i are the periodic returns below the target return, B is the target return, and n is the total number of periods. Since the sample has a standard deviation of 2.7%, it will have values below and above its mean of 1.0%. Since the target downside deviation ignores the deviations above the mean, it will be less than the standard deviation.
- B. **Incorrect** because the target downside deviation = $[\sum(X_i - B)^2 / (n - 1)]^{0.5}$, where X_i are the periodic returns below the target return, B is the target return, and n is the total number of periods. Since the sample has a standard deviation of 2.7%, it will have values below and above its mean of 1.0%. Since the target downside deviation ignores the deviations above the mean, it will be less than the standard deviation.
- C. **Incorrect** because the target downside deviation = $[\sum(X_i - B)^2 / (n - 1)]^{0.5}$, where X_i are the periodic returns below the target return, B is the target return, and n is the total number of periods. Since the sample has a standard deviation of 2.7%, it will have values below and above its mean of 1.0%. Since the target downside deviation ignores the deviations above the mean, it will be less than the standard deviation.

Answer 70 of 90

Answer

Solution

- A. **Correct** because the trimmed mean is computed by excluding a stated small percentage of the lowest and highest values and then computing an arithmetic mean of the remaining values. For example, a 5% trimmed mean discards the lowest 2.5% and the highest 2.5% of values and computes the mean of the remaining 95% of values.
- B. **Incorrect** because the harmonic mean is the value obtained by summing the reciprocals of the observations—terms of the form $1/X_i$ —then averaging that sum by dividing it by the number of observations n , and, finally, taking the reciprocal of the average. The harmonic mean may be viewed as a special type of weighted mean in which an observation's weight is inversely proportional to its magnitude. The harmonic mean does not exclude or replace any outliers.
- C. **Incorrect** because the winsorized mean is calculated by assigning a stated percentage of the lowest values equal to one specified low value and a stated percentage of the highest values equal to one specified high value, and then it computes a mean from the restated data. For example, a 95% winsorized mean sets the bottom 2.5% of values equal to the value at or below which 2.5% of all the values lie (as will be seen shortly, this is called the '2.5th percentile' value) and the top 2.5% of values equal to the value at or below which 97.5% of all the values lie (the '97.5th percentile' value).

Answer 71 of 90

Answer

Solution

- A. **Incorrect** because the stock price will be distributed in a lognormal manner, not in a normal or uniform manner.
- B. **Incorrect** because the stock price will be distributed in a lognormal manner, not in a normal or uniform manner.
- C. **Correct** because the relationship between normal and lognormal distributions is if a stock's continuously compounded return is normally distributed, then future stock price is necessarily lognormally distributed.

Answer 72 of 90

Answer

Solution

- A. Incorrect because the procedure does not involve the systematic selection of a sample from the population. With systematic sampling, we select every k th member until we have a sample of the desired size. The sample that results from this procedure should be approximately random.
- B. **Correct** because the sampling procedure does not give every member of the population an equal chance of being selected. It is based on the analyst's convenience. Non-probability sampling depends on factors other than probability considerations, such as a sampler's judgment or the convenience to access data.
- C. Incorrect because cluster sampling requires the division or classification of the population into subpopulation groups, called clusters. In this method, the population is divided into clusters, each of which is essentially a mini-representation of the entire populations. Then certain clusters are chosen as a whole using simple random sampling. If all the members in each sampled cluster are sampled, this sample plan is referred to as one-stage cluster sampling. If a subsample is randomly selected from each selected cluster, then the plan is referred as two-stage cluster sampling. Choosing only company names starting with the letter P is not two-stage cluster sampling.

Answer 73 of 90

Answer

Solution

- A. Incorrect because it fails to account for any compounding; $(3\% + 5\%) = 8\%$;
 $90,000(1.08) = 97,200$.
- B. Incorrect because it fails to account for quarterly compounding; $90,000(1.03)(1.05) = 97,335$.
- C. **Correct** because the returns are compounded quarterly;
 $90,000 \left(1 + \frac{0.03}{4}\right)^4 \left(1 + \frac{0.05}{4}\right)^4 = 97,455$.

Answer 74 of 90

Answer

Solution

- A. Incorrect because the two most noteworthy observations about the lognormal distribution are that it is bounded below by 0 and it is skewed to the right.
- B. **Correct** because the two most noteworthy observations about the lognormal distribution are that it is bounded below by 0 and it is skewed to the right (it has a long right tail), i.e. it is asymmetrical.
- C. Incorrect because like the normal distribution, the lognormal distribution is completely described by two parameters. Unlike the other distributions, a lognormal distribution is defined in terms of the parameters of a different distribution. The two parameters of a lognormal distribution are the mean and standard deviation (or variance) of its associated normal distribution. Remember, we must keep track of two sets of means

and standard deviations (or variances): the mean and standard deviation (or variance) of the associated normal distribution (these are the parameters) and the mean and standard deviation (or variance) of the lognormal variable itself. The mean of the lognormal distribution is not equal to the mean of the associated normal distribution.

Answer 75 of 90

Answer

Solution

- A. Incorrect because a scatter plot (or scattergram) represents the two variables in two dimensions. The variation of Y (the dependent variable) is often referred to as the sum of squares total (SST), or the total sum of squares.
- B. **Correct** because the variation of Y (the dependent variable) is often referred to as the sum of squares total (SST), or the total sum of squares.
- C. Incorrect because Y (the dependent variable) is often referred to as the sum of squares total (SST), or the total sum of squares.

Answer 76 of 90

Answer

Solution

- A. **Correct** because using the equation $PV = FVN(1 + r/m)^{-Nm}$

where

m = number of compounding periods per year

r = quoted annual interest rate

N = number of years

we compute $PV = \$10,000,000 \times (1 + 0.05/2)^{-15 \times 2} = \$4,767,426.85 \approx \$4,767,427$.

In applying the equation, we use the periodic rate (in this case, the semi-annual rate) and the appropriate number of periods with semi-annual compounding.

Alternative solution using a financial calculator in END mode:

$N = 30$; $I/Y = 0.025$; $PMT = 0$; $FV = 10,000,000$; $CPT PV = \$4,767,426.852 \approx \$4,767,427$

- B. Incorrect because when computing present value, it uses 5% instead of EAR to get $\$10,000,000 \times (1.05)^{-15} = \$4,810,170.98 \approx \$4,810,171$.
- C. Incorrect because in the present value equation, it incorrectly accounts for compounding (multiplies by 2 instead of dividing) and does not use the appropriate number of periods for semi-annual compounding (divides by 2 instead of multiplying by 2); to incorrectly compute $PV = 10,000,000 \times (1 + 0.05 \times 2)^{-15/2} = \$4,892,770.68 \approx \$4,892,771$.

Answer 77 of 90

Answer

Solution

- A. **Correct** because one of the four key assumptions we need to make to be able to draw valid conclusions from a simple linear regression model is that regression residuals are normally distributed.
- B. Incorrect because one of the four underlying assumptions for simple linear regression is Independence: The observations, pairs of Y s and X s, are independent of one another. This implies the regression residuals are uncorrelated [and do not have high correlation] across observations.
- C. Incorrect because one of the four underlying assumptions for simple linear

regression is Homoskedasticity: The variance of the regression residuals is the same [and not different] for all observations.



Answer 78 of 90

Answer

Solution

- A. **Correct** because when the observations are returns, the coefficient of variation measures the amount of risk (standard deviation) per unit of mean return.
- B. Incorrect because it describes the Sharpe ratio, not the coefficient of variation. The Sharpe ratio measures the reward, in terms of mean excess return, per unit of risk, where mean excess return is defined as the average rate of return in excess of the risk-free rate.
- C. Incorrect because it describes the mean absolute deviation (MAD), not the coefficient of variation. The MAD is defined as the mean of the absolute values of deviations from the sample mean.

Answer 79 of 90

Answer

Solution

- A. **Correct** because when the unknown population variances are equal, a t -test based on independent random samples is given by $t = (X_1 - X_2 - \mu_1 + \mu_2) / (s_p^2 / n_1 + s_p^2 / n_2)^{1/2}$ where s_p^2 is a pooled estimator of the common variance. Hence, a change in hypothesized difference $\mu_1 - \mu_2$ will change the value of the test statistic.
- B. Incorrect because the number of degrees of freedom is $n_1 + n_2 - 2$ and it is independent of the hypothesized difference $\mu_1 - \mu_2$.
- C. Incorrect because $s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2) / (n_1 + n_2 - 2)$ is a pooled estimator of the common variance. It is independent of the hypothesized difference $\mu_1 - \mu_2$.

Answer 80 of 90

Solution

- A. **Correct** because with n as the sample size, the Spearman rank correlation is given by:
- $$r_s = 1 - (6 \sum d^2) / (n(n^2 - 1)) = 1 - 6(12) / (4(4^2 - 1)) = 1 - 6/5 = -1/5 = -0.2, \text{ where}$$
- the sum of squared differences in ranks $\sum d^2 = (2 - 1)^2 + (3 - 2)^2 + (4 - 3)^2 + (1 - 4)^2 = 1 + 1 + 1 + 9 = 12$.
- B. Incorrect because this answer is calculated by erroneously omitting the multiple (of the sum of squared deviations) of 6, giving a rank correlation of $1 - 1/5 = 4/5 = 0.8$.
- C. Incorrect because instead of using the sum of squared differences in ranks, this answer erroneously uses the sum of unsquared differences in ranks (which, by definition, sums to zero), giving a rank correlation of 1.

Answer 81 of 90

Answer

Solution

- A. Incorrect because variation in the supply of wheat is a dependent variable not limited to values of 0 or 1. Suppose we want to examine whether a company's quarterly earnings announcements influence its monthly stock returns. In this case, we could use an **indicator variable**, or dummy variable, that takes on only the values 0 or 1 as the independent variable.
- B. **Correct** because variation in the demand for corn is being used to explain the variation in the supply of wheat. Therefore the variation in the supply of wheat is the dependent variable, or explained variable. We refer to the variable whose variation is being explained as the **dependent variable**, or the explained variable; it is typically denoted by Y .
- C. Incorrect because the variation in the demand for corn is being used to explain the variation in the supply of wheat. Therefore the variation in the demand for corn is the independent variable. We refer to the variable whose variation is being used to explain the

variation of the dependent variable as the **independent variable**, or the explanatory variable; it is typically denoted by X .

Answer 82 of 90

Answer

Solution

- A. **Correct** because a return distribution with negative skew has frequent small gains and a few extreme losses.
- B. Incorrect because a return distribution with positive skew has frequent small losses and a few extreme gains. A return distribution with negative skew has frequent small gains and a few extreme losses.
- C. Incorrect because a return distribution with positive skew has frequent small losses and a few extreme gains. A return distribution with negative skew has frequent small gains and a few extreme losses.

Answer 83 of 90

Answer

Solution

- A. Incorrect because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.
- B. **Correct** because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.
- C. Incorrect because when we look at the residuals of a model, what we would like to see is that the residuals are random. The residuals should not exhibit a pattern when plotted against the independent variable.

Answer 84 of 90

Answer

Solution

- A. **Correct** because, in cluster sampling, the population is divided into clusters, each of which is essentially a mini-representation of the entire populations. Then certain clusters are chosen as a whole using simple random sampling. Cluster sampling is commonly used for market surveys, and the most popular version identifies clusters based on geographic parameters.
- B. Incorrect because judgmental, not cluster, sampling involves selectively handpicking elements from the population based on a researcher's knowledge and professional judgment. For example, when auditing financial statements, seasoned auditors can apply their sound judgment to select accounts or transactions that can provide sufficient audit coverage.
- C. Incorrect because creating a bond portfolio to mirror the performance of a specified index is an application of stratified sampling. Bond indexing is one area in which stratified sampling, rather than cluster sampling, is frequently applied. **Indexing** is an investment strategy in which an investor constructs a portfolio to mirror the performance of a specified index.

Answer 85 of 90

Answer

Solution

- A. Incorrect because it is the underlying binomial distribution that exhibits skewness when $p \neq 0.5$, not the sampling distribution of the sample mean. Since the sample size is large, the distribution of the sample mean will be approximately normal according to the central limit theorem, and therefore symmetric.
- B. **Correct** because, according to the central limit theorem, the sampling distribution of the sample mean will be approximately normal when the sample size n is large. The normal distribution has a skewness of 0 (it is symmetric). Since the binomial distribution has a

mean of np and finite variance of $np(1 - p)$, where n is the number of trials and p is the probability of success, the central limit theorem holds.

C. Incorrect because it is the underlying binomial distribution that exhibits skewness when $p \neq 0.5$, not the sampling distribution of the sample mean. Since the sample size is large, the distribution of the sample mean will be approximately normal according to the central limit theorem, and therefore symmetric.

Answer 86 of 90

Answer

Solution

- A. **Correct.** A negatively skewed distribution appears as if the left tail has been pulled away from the mean. The average magnitude of negative deviations from the mean is larger than the average magnitude of positive deviations.
- B. Incorrect. Kurtosis refers to relative peakedness of a distribution; leptokurtosis means more peaked than normal.
- C. Incorrect. A negatively skewed distribution appears as if the left tail has been pulled away from the mean.

Answer 87 of 90

Answer

Solution

- A. Incorrect. Tests related to differences between means concern a parameter. Nonparametric tests are used when the data do not concern a parameter or the data do not meet distributional assumptions.
- B. **Correct.** A nonparametric test is used under three circumstances: 1) when the data do not meet distributional assumptions, 2) when the data are given in ranks, and 3) when the hypothesis does not concern a parameter.
- C. Incorrect. When the data meet distributional assumptions, a parametric test is used instead of a nonparametric test.

Answer 88 of 90

Answer

Solution

- A. **Correct** because the time-weighted rate of return is the preferred performance measure as it neutralizes the effect of cash withdrawals or additions to the portfolio, which are generally outside of the control of the portfolio manager.
- B. Incorrect because the arithmetic return is biased upward. For the evaluation of portfolios of publicly traded securities, the time-weighted rate of return is the preferred performance measure.
- C. Incorrect because investment managers find time-weighted returns more meaningful. If a client gives an investment manager more funds to invest at an unfavorable time, the manager's money-weighted rate of return will tend to be depressed. If a client adds funds at a favorable time, the money-weighted return will tend to be elevated. The time-weighted rate of return removes these effects.

Answer 89 of 90

Answer

Solution

- A. Incorrect because, in cluster sampling, the population is divided into clusters, each of which is essentially a mini-representation of the entire population. Then certain clusters are chosen as a whole using simple random sampling. The sampling method used by the analyst is not cluster sampling because, first, each sector is not a mini-representation of all public firms; second, certain sectors are not chosen as a whole.
- B. Incorrect because, in simple random sampling, a public firm would be randomly selected without the firms being grouped into sectors first. A simple random sample is a subset of a larger population created in such a way that each element of the population has an equal probability of being selected to the subset.
- C. **Correct** because, in stratified random sampling, the population is divided into subpopulations (strata) based on one or more classification criteria. Simple random samples are then drawn from each stratum in sizes proportional to the relative size of each stratum in the population.

Answer 90 of 90

Answer

Solution

- A. Incorrect because if we mistakenly reject the null hypothesis, we can only be making a Type I error.
- B. Incorrect because rejecting a false null hypothesis is a correct decision and therefore not a Type II error.
- C. **Correct** because, when we make a decision in a hypothesis test, we run the risk of making either a Type I or a Type II error. These are mutually exclusive errors: If we mistakenly reject the null hypothesis, we can only be making a Type I error; if we mistakenly fail to reject the null, we can only be making a Type II error.